

Experimental Evidence on the Learning Impact of Generative AI*

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Abstract

We study whether access to off-the-shelf generative AI changes how much students learn during a fixed-time assignment. In a randomized experiment with proctored, in-person sessions, undergraduates learn about an unfamiliar topic and write an essay with or without access to off-the-shelf AI, then complete unaided assessments immediately and one week later. AI access raises immediate test scores by 0.29 standard deviations. **These gains persist, at a reduced size, one week later. Estimates for overall essay quality are positive but imprecise in both sessions; dimension-level estimates one week later, when students write unaided, are positive in every dimension but precise only for writing style and clarity. AI access also changes the learning process:** students shift time away from writing and toward information seeking, and they report greater learning enjoyment.

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1 Introduction

Generative artificial intelligence (AI) is now widely used in higher education. ChatGPT has over 900 million weekly users, college-age users send nearly half of its messages, and more than a quarter of those messages relate to learning (Chatterji et al., 2025; OpenAI, 2025). Despite this widespread use, evidence on whether AI helps students learn is scarce.

Generative AI can change how much students learn through two margins. The first is time allocation: students may spend less time by automating some academic tasks when AI is available (Kazemitabaar et al., 2023; Kestin et al., 2025; Kreijkes et al., 2026), and whether that helps or harms learning depends on what students do with the time they save. The second margin is how much students learn from a fixed amount of study time (henceforth, “learning productivity”). Holding study time fixed, AI may raise learning by acting as an on-demand tutor that explains difficult concepts and provides personalized feedback (Nickow et al., 2024), or lower it by offloading the cognitive effort that builds durable knowledge (Chi and Wylie, 2014; Risko and Gilbert, 2016).

In this paper, we focus on the second margin and study how off-the-shelf AI affects learning productivity in a common class of academic tasks: time-bounded assignments in which students must learn unfamiliar material, identify relevant information, and demonstrate understanding through writing and tests. Our randomized experiment with 211 undergraduates unfolds over two in-person sessions approximately one week apart. In the first session, students have the same amount of time to learn about a technical topic and write an analytical essay. We randomly assign students to an *AI-allowed* or an *AI-forbidden* condition. The AI-allowed group may use any generative AI tool along with traditional resources, while the AI-forbidden group may use only traditional resources. We monitor AI usage and compliance through direct proctor observation, ChatGPT conversation logs, and self-reports. In the second session, all students complete another test and write an essay on the same topic, this time without AI or any other resources.

We measure learning along two dimensions. Knowledge tests target factual and conceptual understanding: we ask students to recall key information and identify core mechanisms. Open-ended essays target higher-order skills: we ask students to analyze relationships between concepts and support their arguments with specific evidence. Human and AI graders evaluate each essay along several dimensions, such as writing style and accuracy of content. Students complete all tests and the Session Two essay without AI or other resources, so these assessments measure what they learned. The Session One essay, which

AI can help write, measures the work they submit. We also measure characteristics of the essay text: the fraction flagged as AI-generated by a state-of-the-art detector, length, readability, lexical diversity, and textual similarity among student essays.

We first document that our random assignment generated substantial variation in AI usage. About 70 percent of students in the AI-allowed condition used generative AI during the first session, compared with near-zero usage in the control group. Users value this access: 88 percent report that AI is helpful for the learning task. ChatGPT conversation logs reveal that students use AI most often to explain concepts (58 percent of treated students). Drafting responses (32 percent), editing responses (21 percent), and summarizing readings (18 percent) are the next most common substantive uses.

We turn next to our main result: AI access improves learning outcomes, and the gain persists after the tool is removed. Students with AI access score 7.4 percentage points (pp) higher on the immediate knowledge test on a baseline of 56.3 percent. Standardized by the control-group standard deviation (SD), this is equivalent to a 0.29 SD gain in test scores. One week later, AI-allowed students score 3.7 pp higher on the Session Two test on a baseline of 50.0 percent, or 0.20 SD.

The essay estimates point in the same direction as the test-score estimates but are less conclusive. In Session One, the share of treated students' text flagged as AI-generated rises by 9.8 pp, yet their essays are no more homogeneous than control students', contrary to the convergence documented in workplace settings (Brynjolfsson et al., 2025). Averaging human and AI graders, overall quality rises by 0.19 SD, though the estimate is imprecise. This gain blends AI-produced text with any learning; the Session Two essays, written one week later without AI, separate the two. There, the AI-detection effect disappears. Consistent with the test-score results, overall quality is 0.20 SD higher, and all five dimensions show positive gains.

To shed light on why AI access improves learning, we look at time use and the learning experience. While both groups spend on average the same amount of time learning, AI access shifts the composition of learning time away from producing text: treated students report spending 4.7 pp less of their learning time on writing activities and 4.4 pp more on reading and searching for information, relative to control-group shares of 53.4 and 44.5 percent. AI access also makes learning more enjoyable: treated students report 13 percent higher enjoyment ratings.

We organize these results with a model in which students choose effort in research and writing, both activities produce knowledge, and writing also produces the essay. Used as

a tutor, AI raises the return to research effort. Used to write the essay, it removes part of the incentive to write and, with it, some of the learning that writing would have produced. The net effect on knowledge depends on whether the tutoring gain exceeds the losses from delegation, and students who value learning more are less likely to delegate. The model also implies that outcomes should differ by how students use AI: AI-written text raises the submitted essay but does not carry over to unaided work, whereas knowledge acquired through tutoring does.

Consistent with the prediction of the model, we provide descriptive evidence that outcomes differ with how students use AI. Using treated students’ ChatGPT conversation logs, we classify 51 percent of AI users as “augmentation” users (who work *with* AI) and 31 percent as “automation” users (who use AI to do the work *for* them); the rest mix the two or fall in neither category. This distinction shows up in both usage and output: augmentation users prompt AI mainly as a personalized tutor, while automation users prompt it to draft their essays for them and have AI-detection rates more than twice as high. Automation users submit higher-rated essays in Session One, but this advantage is absent from their unaided essays one week later. Augmentation users show positive test-score differences relative to the control group in both sessions, and their point estimates exceed automation users’ on every Session Two outcome.

Students choose whether and how to use AI, and those choices depend on what they believe AI does to their learning. Are those beliefs accurate? We ask students to estimate how much AI access changed (or would have changed) their test scores. Both treated and control students expect positive effects, but only the treated group’s prediction comes close to our estimate; the control group overestimates it. Across subgroups, perceived gains are only weakly related to actual gains. To understand why students expect these effects, we elicit an open-ended account of how AI affects student learning in general and code each response as a causal graph from AI use to learning. Most students name both a channel through which AI helps learning and one through which it harms it, condition the effect on how AI is used, and spontaneously organize the channels around the augmentation-versus-automation distinction along which our use-type results differ.

Our results contribute to an emerging literature on the learning impacts of AI (e.g., [Bastani et al., 2025](#); [Franco et al., 2026](#); [Strömberg et al., 2026](#)). We benchmark our estimates against the experimental literature in the main text, but note three challenges for interpreting existing evidence. First, many studies evaluate AI systems that are scaffolded,

restricted, or otherwise customized beyond a standard off-the-shelf chatbot.¹ These designs inform the particular system studied but speak less directly to the chatbots students use in practice. Second, some studies compare AI with active alternatives rather than a business-as-usual counterfactual, so the estimates reflect AI’s value relative to those alternatives rather than its absolute effect on learning.² Third, in many designs students choose how much time to spend on the task, so the estimated effects combine the two margins: time allocation and learning productivity. Our design addresses all three: treated students access off-the-shelf AI tools, control students learn the same material without any AI access, and both groups face the same time limit and spend the same time on average, so our estimates capture the learning-productivity margin. Beyond these design features, we contribute by measuring what students retain one week later, after AI is removed; distinguishing augmentation from automation uses; tracing how AI shifts time use and the learning experience; assessing multiple learning dimensions rather than test scores alone; and comparing students’ beliefs about AI’s effects with those we estimate.

We also contribute to the literature on digital technologies and student learning. Prior research has examined laptops and personal computers (Malamud and Pop-Eleches, 2011; Carter et al., 2017; Cristia et al., 2017); online courses (Figlio et al., 2013; Bettinger et al., 2017); internet access (Vigdor et al., 2014; Dettling et al., 2018); and interactive classroom technologies such as clickers and whiteboards (Caldwell, 2007; Lewin et al., 2008). Expanding access to general-purpose technologies tends to yield limited, mixed, or negative effects on academic performance. Computer-assisted learning systems—which tailor content to students’ levels and provide immediate feedback—produce larger and more consistent gains (see Bulman and Fairlie, 2016; Escueta et al., 2020, for reviews). Generative AI combines features of both: it is a general-purpose tool that can give personalized explanations and respond to students’ questions in real time, but it can also do the work for students. Our

¹For example, Bastani et al. (2025) test GPT-4 with guardrails that provide teacher-designed hints rather than direct solutions in Turkish high-school math classes; Xu et al. (2025) add metacognitive prompts to a generative-AI learning environment for Chinese college students; Kim et al. (2025) restrict AI to tutoring after students attempt each problem on a math practice platform; De Simone et al. (2025) study teacher-guided sessions with Microsoft Copilot in after-school English classes in Nigeria; Poulidis et al. (2025) compare a system-regulated AI that auto-delivers tips at key moments with on-demand AI in a 12-week chess training program; and Kumar et al. (2025) expose online adults practicing math problems to explanations pre-generated by GPT-4 under a hidden tutoring prompt, rather than to the chatbot itself.

²For example, Kestin et al. (2025) compare AI with in-class active learning in a college physics course, LearnLM Team (2024) with static pre-written hints and human tutoring across a range of tutoring scenarios, Kreijkes et al. (2026) with note-taking on reading-comprehension tasks in secondary schools, and Chung et al. (2026) compare an AI tutor that adapts the order and difficulty of practice problems to each student with a fixed-sequence version of the same tutor in a high-school Python course.

experiment studies what students learn when both capabilities are available and students choose how to use them.

Finally, we contribute to the literature on the productivity effects of AI. Prior studies document substantial productivity gains from AI access in professional writing, software development, consulting, customer support, legal work, team-based problem solving, radiology, and taxi driving (Noy and Zhang, 2023; Peng et al., 2023; Choi et al., 2024; Jia et al., 2024; Brynjolfsson et al., 2025; Dell’Acqua et al., 2025; Kanazawa et al., 2025; Baird et al., 2026; Cruces et al., 2026; Cui et al., 2026; Dell’Acqua et al., 2026; Goldsmith-Pinkham et al., 2026). This literature focuses on task performance: how well workers complete tasks when given access to AI. Productivity gains can also accrue through skill accumulation, that is, through building durable human capital. The few studies that speak directly to learning in work settings provide mixed results. Brynjolfsson et al. (2025) show that AI facilitates worker learning among customer-support agents, Autor et al. (2026) find that an AI drafting assistant improves the unassisted performance of senior patent lawyers and not of junior ones, and Shen and Tamkin (2026) find that software engineers who learned an unfamiliar programming library with an AI assistant scored substantially worse on a subsequent unassisted test. We provide experimental evidence that AI access raises learning productivity in an academic context: given the same amount of time, students accumulate more durable knowledge. This result identifies a channel through which AI may raise long-run productivity beyond its immediate effects on task performance.

2 Experimental Design

The experiment consists of two in-person sessions conducted approximately one week apart. Survey instruments (including tests) are available in the [Supplementary Materials](#).

2.1 Session One

Overview. The first session employs a between-subjects design with random assignment to either an AI-allowed or AI-forbidden condition. We schedule eight time slots with two parallel labs per slot. We randomly assign each student who signed up for a time slot to a specific computer in one of two labs. For each time slot, we then randomly assign one lab to the AI-allowed condition and the other to the AI-forbidden condition. Each workstation has a privacy divider to minimize distractions and prevent students from viewing other screens (see Appendix Figure A1). We administer all components of the session through

Qualtrics, a survey platform that tracks each student’s progress, enforces section time limits, and records how long students spend on each component.

Session Structure. The first session follows a fixed 60-minute structure with the following components (Figure 1, Panel A).

Welcome and Instructions. Each session begins with a staff member reading a condition-specific welcome script and asking students to put their phones away for the duration of the session. We tell students that their primary task involves learning about a prespecified topic and that they will demonstrate their understanding through a series of assessments. To ensure comprehension of the experimental procedures, students complete a series of comprehension checks, with laboratory staff providing clarification if needed. Students correctly answer an average of 4.76 out of 5 comprehension questions, and the median student answers all questions correctly.

Topic Assignment and Baseline Assessment. We select three topics rich in factual content, about which most students have limited prior knowledge: blockchain technology, carbon capture systems, and CRISPR gene editing. Students are randomly assigned to one of these three topics and cannot change their assignment. Students then complete a five-question multiple-choice test to assess their baseline knowledge of the assigned topic. In all tests, the order of questions and answer options within each question is randomized. Students may not use notes, the internet, or AI tools during tests.

Learning Phase. Following the baseline test, students enter the learning phase. During this phase, they have up to 35 minutes to learn about their assigned topic. We instruct them to “use the same learning approach you would typically follow for a college assignment” and emphasize that their performance affects their payoffs, as described below.

During this phase, students also write an approximately 500-word analytical essay demonstrating their understanding of the topic. Each essay prompt asks students to apply critical thinking skills by analyzing relationships between concepts, comparing elements, and supporting their analysis with specific evidence (see Appendix B.2).³ For each topic, we develop two prompts; each student is randomly assigned one for Session One and the other for Session Two. We provide all students with a link to an introductory reading about their assigned topic, created to be of comparable length and reading difficulty across

³For example, a student assigned to carbon capture may be asked to “analyze the three main barriers to scaling carbon capture technologies (technical limitations, economic costs, and policy challenges),” identify which barrier is most critical, and support their analysis with specific examples.

the three topics.⁴

Students can allocate their time freely across reading, searching, drafting, editing, or other activities such as surfing the web. Students can finish before the 35-minute maximum but cannot exceed it. The interface automatically advances at 35 minutes.

Post-Learning Survey. After the learning phase, students complete a brief survey that measures self-assessed current knowledge of the topic (0–10 scale); time allocation during the learning phase; subjective task assessment (enjoyment, feeling skilled or effective); whether they had previously done similar tasks in their current academic year; and an attention check.

Post-Learning Test. Students next complete five multiple-choice questions to test factual knowledge and conceptual understanding of the assigned topic. These questions differ from the baseline test. Students complete this assessment without access to external resources, the internet, or AI tools regardless of treatment condition.

Waiting Room. Students who complete all tasks before the 60-minute session ends are directed to a virtual waiting room. The “Finish Session 1” button is enabled only after 50 minutes have elapsed from the session start time. We tell students that we record their responses only after they click this button. This prevents early departures from disrupting others or creating social pressure effects. During this waiting period, students may browse the internet freely.

Treatment. We randomly vary students’ access to generative AI tools during the learning phase. The treatment has two components: explicit instructions about whether AI use is permitted, and a logged-in ChatGPT account for students in the AI-allowed condition.

Students in the *AI-allowed* condition receive explicit instructions that permit the use of any generative AI tools during the learning phase. The instructions are delivered orally through the proctor-read welcome script and in writing through the survey interface (Appendix Figure A2, Panel A). The script states: “[I]f you typically use generative AI tools such as ChatGPT, feel free to use them here as well. Using ChatGPT or other generative AI is completely allowed. We provide you with a ChatGPT account, already logged in on one of the tabs, so you don’t need to use your personal account.” To ease access, each computer has three browser tabs already open: a dedicated ChatGPT (GPT-4o) account,

⁴Texts range from 1,253 to 1,399 words (Appendix Table B1). At an average silent reading speed of approximately 250 words per minute for college students (Carver, 1992; Brysbaert, 2019), the estimated reading time is 5.0 to 5.6 minutes per text. The full text of the learning materials is available in the [Supplementary Materials](#).

Wikipedia, and the Middlebury College Library website.

Students in the *AI-forbidden* condition receive explicit instructions prohibiting use of generative AI tools. The instructions are delivered orally through the proctor-read welcome script and in writing through the survey interface (Appendix Figure A2, Panel B). The script states: “You are not allowed to use generative AI tools such as ChatGPT, Claude, or other AI assistants.” Each computer has three browser tabs already open: Google, Wikipedia, and the Middlebury College Library website.

Compliance Monitoring. We enforce treatment compliance through multiple mechanisms. First, two proctors per lab directly observe students throughout the session using classroom seating charts to record any unauthorized resource use (Appendix Figure A3). Proctors note instances of students accessing AI tools in the AI-forbidden condition or violating other experimental protocols such as accessing external resources during the tests. Second, the exit survey elicits self-reported AI usage during the learning phase, including which specific AI tools were used (ChatGPT, Claude, Gemini, etc.). Third, the platform periodically captures students’ writing interfaces to detect sudden appearances of large text blocks that may indicate copying from external sources.

Compensation and Incentives. Students receive both attendance compensation and performance-based incentives. For attendance, students receive \$5 for completing Session One alone but \$50 for completing both sessions. For performance, students earn lottery tickets: one ticket per correct response on each test and one per point on their rounded quality score. At the end of the experiment, we conduct a drawing in which 30 lottery tickets are randomly selected, with each winning ticket worth \$100. To further support attendance, we send email and text reminders on the day of each scheduled session and let students schedule the second session flexibly, rather than requiring exactly seven days between sessions.

2.2 Session Two

Session Two takes place approximately one week after Session One.⁵ This session measures knowledge retention when all students—regardless of Session One treatment assignment—

⁵When scheduling, we encouraged students to sign up for a Session Two slot approximately one week after their Session One slot, but allowed flexibility to accommodate scheduling constraints. Among students who attended Session Two, the average gap between sessions was 6.98 days. Most returned exactly seven days after Session One (61.0 percent), and 80.0 percent returned within six to eight days.

work without access to AI or external resources.

Session Structure. The session includes two assessments presented in randomized order (Figure 1, Panel B): a 10-item multiple-choice knowledge test, and a 20-minute analytical essay that uses the complementary prompt developed for each topic.

Session Two concludes with a questionnaire on four topics. First, beliefs about AI’s impact on their own and others’ test scores. Second, self-reported AI skills (Likert scale), frequency of use across academic tasks, and when students began using AI. Third, retrospective Session One behavior, including which tools they used and for what purposes (explaining concepts, summarizing, drafting, proofreading, editing) and whether they studied the topic between sessions. Fourth, an open-ended question about generative AI’s impact on learning.⁶

3 Data and Research Design

3.1 Sample and Recruitment

The experiment took place at Middlebury College during Spring 2025. At the time of the experiment, AI adoption at Middlebury exceeded 80 percent (Contractor and Reyes, 2026), so our treatment varies AI access among students already familiar with the tool. We recruited students through campus-wide email announcements, posted flyers, student group chats, and a dedicated recruitment website. To minimize selection concerns, we framed the study as “a study on learning” without mentioning AI in any recruitment materials.

⁶We note four main deviations from the experimental protocol that occurred during early time slots. First, the initial Session One verbal welcome script did not mention AI permissions or restrictions; students learned their treatment condition only through written platform instructions. We introduced verbal acknowledgment of treatment conditions starting with the second Session One time slot. Second, the initial Session One included a timer that malfunctioned, displaying available time after the 35-minute limit had elapsed. We removed the timer for subsequent sessions. Third, the first Session Two time slot lacked a waiting room, requiring students to manually track time before leaving. We introduced a virtual waiting room for all following sessions. Fourth, a technical glitch during one Session Two time slot prevented students from completing the essay on the platform. We added extra time, asked affected students to write their essays in Microsoft Word, and entered those essays into the dataset manually. All analyses include session fixed effects to account for these protocol variations. In Appendix Figure A4, we show robustness of the test-score effects to excluding each time slot in turn.

3.2 Summary Statistics and Balance

Table 1 reports summary statistics for three groups: the 256 students who completed the sign-up survey and scheduled a Session One slot (column 1), the 211 (82.4 percent) who attended Session One (column 2), and the 205 (80.1 percent) who attended both sessions (column 3).⁷ We focus on column 2. The typical student is 20.1 years old, 35.1 percent are male, 49.8 percent white, 26.5 percent international, and just over half (54.0 percent) attended a public high school (Panel A).⁸ The mean GPA is 3.68 and the mean SAT score is 1386 (Panel B).⁹ The majority of students (97.2 percent) returned for Session Two (Panel C). Students self-reported little baseline knowledge of their assigned topic (1.5 on a 0–10 scale) and, consistent with this self-assessment, answered only 30.3 percent of baseline multiple-choice questions correctly (Panel D).

The sample is balanced on most observable characteristics across treatment groups. Table 2 compares mean characteristics of students in the AI-forbidden versus AI-allowed conditions (columns 1 and 2). Only the coefficient on age is statistically significant at the 10 percent level. Measures of academic ability—strong predictors of learning outcomes—are well balanced: mean GPA (3.69 versus 3.67) and SAT scores (1392 versus 1383) are nearly identical across groups. To address any residual imbalance, we use the double-lasso procedure (Belloni et al., 2014), as described below. We find no differential attendance in Session One (82.5 versus 82.3 percent) and no differential attrition between sessions.

3.3 Regression Model

We estimate linear models of the form

$$Y_i = \alpha + \beta \text{AI-Allowed}_i + \mathbf{X}_i' \gamma + \varepsilon_i, \quad (1)$$

where Y_i is an outcome for student i , AI-Allowed_i is an indicator that equals one if the student was randomly assigned to the AI-allowed condition, $\mathbf{X}_i$ is a vector of baseline control variables to improve precision, and ε_i is an error term. The vector $\mathbf{X}_i$ contains fixed effects

⁷Our sample is relatively large by the standards of in-person laboratory experiments in economics. For example, Camerer et al. (2016) replicate 18 between-subject laboratory experiments published in the *American Economic Review* and the *Quarterly Journal of Economics* between 2011 and 2014. Among these, the median sample size is 120, and only two exceed ours.

⁸Table A2 compares the sample with the Middlebury student body. The sample overrepresents women and first-year students and matches the student body on college GPA.

⁹We asked students for both SAT and ACT scores. Most students took the SAT. When students provided ACT but not SAT scores, we use concordance tables to convert ACT scores to SAT equivalents.

for the randomization strata (the eight Session One time slots) and additional controls selected through the double-lasso procedure (Belloni et al., 2014), which selects controls that predict either the outcome or the treatment assignment from a pool of potential covariates.¹⁰ We report heteroskedasticity-robust standard errors.

The coefficient of interest, β , measures the causal effect of AI access on a given outcome. This intent-to-treat (ITT) estimate captures the effect of being *allowed* to use AI during the learning phase, regardless of whether students actually used AI. This may be the relevant parameter for policy evaluation: institutions can choose whether to permit AI use, but they cannot force students to use it. To estimate the effect of actual AI use rather than assignment, we complement our ITT estimates with a treatment-on-the-treated (TOT) analysis. We instrument actual AI use with the randomly assigned treatment status (AI-allowed) and estimate by two-stage least squares (2SLS). These estimates identify the local average treatment effect (LATE) for compliers—students whose treatment assignment influenced their AI use.

3.4 Main Outcomes

We estimate equation (1) separately for several categories of outcomes:

AI Usage. As a first-stage measure, we examine whether students used generative AI during the learning phase. Our primary measure is an indicator that equals one if the ChatGPT account provided to the student during Session One contained at least one prompt. We measure intensity through the number of prompts sent and the number of distinct conversations (i.e., separate chat threads) initiated. We complement these with self-reported measures of whether students used any generative AI tool, ChatGPT specifically, or alternative AI models (Claude, Gemini, Copilot, DeepSeek, or Llama).

¹⁰The pool of potential controls includes: (1) demographic characteristics—age, gender, race/ethnicity indicators (Black, Latino, Asian, white), international student status, and public/private high school attendance; (2) academic background—cohort fixed effects, declared major status, field-of-study indicators (natural sciences, social sciences, humanities/arts), self-reported study hours per week, college GPA and SAT score (each entered continuously and as decile bins), high school GPA (as four letter-grade bands), and an indicator for taking a standardized admissions test; (3) baseline measures—pre-learning self-assessed knowledge and pre-learning test score; (4) experimental design features—topic fixed effects, writing prompt version, Session Two assessment order, and lab location; and (5) missing-data indicators. For students who did not take a standardized admissions test, we impute SAT scores from baseline covariates to assign them to bins, as described in Appendix B.5.

Test-Based Learning Outcomes. We measure knowledge acquisition and retention using self-reported and objective measures. Self-assessed knowledge captures students’ subjective evaluation of their understanding on a 0–10 scale, with 0 indicating “I know nothing about this topic” and 10 indicating “I am an expert.”¹¹ The fraction correct is the proportion of multiple-choice questions answered correctly (out of five questions in Session One, and ten in Session Two). We construct standardized test scores by normalizing raw scores to have mean zero and standard deviation (SD) one in the control group (separately for each session). We also define binary indicators for scoring above multiple thresholds to trace treatment effects across the score distribution.

Essay-Based Learning Outcomes. Essays measure higher-order skills such as analytical reasoning, synthesis, and argumentation. We recruited 311 graders through Prolific, restricting eligibility to individuals holding a master’s or PhD degree. Each essay was independently evaluated on a 0–10 scale by three or four graders blind to treatment condition and student identity; each grader scored five essays. Graders provided an overall quality rating and scores on five dimensions: accuracy of content, use of evidence and examples, relevance to the prompt, organization and structure, and writing style and clarity (see Appendix B.3 for details). We also constructed a quality index as the average of the five dimension scores, each on the 0–10 scale; we standardize it by the control-group standard deviation when we report it in SD units. As specified in our pre-analysis plan, the overall quality rating and this average index are our primary essay-based outcomes, and the individual dimensions are secondary outcomes. We complement human grading with AI grading: a large language model (LLM) evaluates each essay on the same dimensions and rubric (see Appendix B.4 for the procedure and prompts).¹² For our main regressions, we use the average of human and AI grader scores for each dimension and show robustness to using only the human or only the AI ratings.

Essay Writing Characteristics. Beyond quality, we measure three characteristics of each essay that describe how AI access changes the writing students submit. First, we

¹¹This measure may capture dimensions of learning that multiple-choice questions cannot test, though we acknowledge that it requires accurate self-knowledge and is subject to the usual biases of subjective questions, such as social desirability bias.

¹²Two considerations motivate our use of AI grading: a growing literature validates LLMs as evaluators of text quality (Chiang and Lee, 2023; Liu et al., 2023; Zheng et al., 2023), and our learning topics (blockchain, carbon capture, CRISPR) are technical subjects on which human graders recruited from a general population may have limited expertise.

use Pangram, a state-of-the-art AI content detector (Emi and Spero, 2024; Masrour et al., 2025; Thai et al., 2026), to measure the fraction of text classified as AI-generated.¹³ We also use Pangram to measure the fraction flagged as plagiarized.¹⁴ Second, we measure writing style through three indices: a length index (number of tokens, words, and sentences), a readability index (sentence length, syllables per word, the Flesch-Kincaid grade level, and the Flesch Reading Ease score, with difficulty measures reversed so that higher values indicate easier-to-read text), and a lexical diversity index (the ratio of unique to total words and the share of words used only once). Each index averages its components after standardizing them to mean zero and SD one in the control group. Third, to examine whether AI access homogenizes student writing, we follow Brynjolfsson et al. (2025) in computing sentence embeddings. We measure within-group textual similarity as the average pairwise cosine similarity between a student’s essay and all other essays in the same treatment group, topic, and prompt cell, and reading material similarity as the cosine similarity between the essay and the provided reading material.¹⁵

4 Effects of AI Access on Learning

4.1 First Stage: AI Adoption and Usage Patterns

We begin with the first-stage estimates of how random assignment to AI access translated into AI use. Table 3 reports estimates from equation (1) using several measures of AI usage as outcomes. Panel A presents measures recorded in treated students’ ChatGPT accounts. Panel B presents self-reported measures. Figure 2 illustrates these first-stage effects.

The experimental manipulation generated substantial variation in AI usage. Assignment to the AI-allowed group increased AI usage by $\hat{\beta} = 70.1$ pp as measured by activity in the provided ChatGPT account ($p < 0.001$). This effect represents a large increase from the near-zero baseline in the control group. Self-reported measures yield similar though slightly attenuated effects: any generative AI use increased by $\hat{\beta} = 60.9$ pp ($p < 0.001$). This adoption was almost entirely ChatGPT: a 60.2 pp increase in ChatGPT use ($p < 0.001$), versus

¹³Jabarian and Imas (2025) evaluate four AI text detectors and find that Pangram achieves near-zero false positive and false negative rates, even when AI-generated text is modified using “humanizer” tools.

¹⁴Pangram’s plagiarism checker matches sentences against the open web—webpages, books, news articles, and other publicly indexable sources (Masrour, 2025).

¹⁵As a validation exercise, we confirm that the embeddings capture content variation: average within-topic cosine similarity is 0.74, compared to 0.15 across topics, and essays responding to the same prompt are more similar to each other (0.77) than essays on the same topic but a different prompt (0.71).

just a 2.9 pp increase in the use of other AI models ($p = 0.182$).¹⁶ Consistent with this strong first stage, 88 percent of AI users reported that AI was somewhat or very helpful for the learning task.¹⁷

Treated students use AI for a mix of purposes—most often to explain concepts, but also to draft and edit text. Figure 3 reports the fraction of treated students who used AI for each of six purposes, measured through self-reports (Panel A) and ChatGPT conversation logs (Panel B), which we classify using LLMs (see Appendix B.6). In the logs, explaining concepts is the most common use (58 percent of treated students), followed among the substantive categories by drafting responses (32 percent), editing responses (21 percent), and summarizing readings (18 percent). The self-reports give a similar picture except for writing: explaining concepts is again the most common use (46 percent), but students report drafting (17 percent) and editing (14 percent) less often than the logs show.¹⁸

4.2 Effects on Test Scores

AI access improves students’ test performance but not their self-assessed knowledge (Table 4, columns 1–3). Both treated and control students perform better after the learning phase, consistent with skill accumulation. Self-assessed knowledge in the control group rises sharply, from 1.56 to 4.79 on the 0–10 scale ($p < 0.001$). The rise is nearly identical among treated students, leaving an ITT of $\hat{\beta} = -0.09$ points, which is statistically indistinguishable from zero. Test performance, by contrast, responds to AI access. Control students’ fraction of questions answered correctly rises from 30.6 percent to 56.3 percent ($p < 0.001$); AI access adds a further $\hat{\beta} = 7.4$ pp, a 13 percent improvement over the control mean. In standardized units, this is a $\hat{\beta} = 0.29$ SD effect (column 2, $p = 0.016$). The corresponding TOT estimate is 10.6 pp, or 0.42 SD ($p = 0.017$, column 3).¹⁹

¹⁶The near-exclusive use of ChatGPT is consistent with surveys of adoption among college students (Hirabayashi et al., 2024; Stöhr et al., 2024; Contractor and Reyes, 2026) and the U.S. working-age population (Bick et al., 2026).

¹⁷Appendix Table A1 reports correlates of AI take-up among treated students. Prior AI use is the strongest and most robust predictor. Self-assessed proficiency also predicts take-up in bivariate specifications, while early adoption does not. White students are less likely to use the provided account in the bivariate specification. We find no significant association between take-up and either AI’s perceived benefits or GPA. Men are directionally more likely to adopt than women, but the difference is imprecise.

¹⁸The high rate of “other” uses in the revealed data largely reflects conversational turns (e.g., “yes,” “sure”), synonym lookups, and off-topic questions—prompts that students would not consider a distinct use of AI when responding to a survey.

¹⁹One concern in interpreting these results as pure learning is that AI access increases the likelihood of rule violations on the Session One test (Appendix Table A3). Proctor-observed violations rise by $\hat{\beta} = 5.7$ pp ($p = 0.077$), while self-reported violations rise by $\hat{\beta} = 9.6$ pp ($p = 0.003$). A back-of-the-envelope

The learning gains from AI access persists one week later (Table 4, columns 4–6).²⁰ Both treated and control students perform worse in Session Two than in Session One, consistent with skill depreciation. Self-assessed knowledge among control students falls from 4.77 to 3.83 on the 0–10 scale ($p < 0.001$). The decline is smaller among treated students, with an ITT of $\hat{\beta} = 0.19$ points, though this difference is small and not statistically distinguishable from zero. Test performance shows a similar fade-out pattern. Among control students, the fraction of questions answered correctly falls from 56.9 percent in Session One to 50.0 percent in Session Two ($p = 0.012$). Treated students decline as well but remain ahead of the control group. The resulting ITT of $\hat{\beta} = 3.7$ pp ($p = 0.082$) is 50 percent of the 7.4 pp Session One effect (though given the standard errors, the two estimates are not statistically distinguishable, $p = 0.261$). Standardized, this corresponds to a $\hat{\beta} = 0.20$ SD effect.²¹ AI-assisted learning therefore produces knowledge gains that are still present one week later.

The point estimates place the learning gains from AI access in the middle of the performance distribution in both sessions (Figure 4, Panel A). In Session One, treated students are 8.9 pp more likely to score at least 40 percent correct ($p = 0.060$) and 12.9 pp more likely to score at least 60 percent correct ($p = 0.045$). AI access has minimal effects on reaching the 80 percent threshold or achieving perfect scores. In Session Two, the largest threshold estimate is an increase of $\hat{\beta} = 11.4$ pp at the 60 percent threshold ($p = 0.074$); the other threshold estimates are smaller. Panels B and C of Figure 4 show the underlying score distributions: AI access shifts the middle of the performance distribution rightward while leaving the tails largely unchanged. This pattern is consistent with previous work showing that AI’s gains tend to concentrate in the lower half of the performance distribution (Noy and Zhang, 2023; Choi et al., 2024; Doshi and Hauser, 2024; Brynjolfsson et al., 2025; Kanazawa et al., 2025; Dell’Acqua et al., 2026).

calculation for each measure multiplies its treatment effect by the test-score gap between violators and non-violators (22.1 pp for proctor-observed violations and 16.4 pp for self-reported ones). The implied contribution of violations to the ITT is 1.3 pp under the proctor measure and 1.6 pp under the self-report measure, or 17 and 21 percent of the estimate. Excluding students who violated the rules as captured by proctors, by self-reports, or by both reduces the learning effect to 0.22 SD, 0.22 SD, and 0.26 SD, respectively. We note that no student violated the rules in Session Two, by either proctor observation or self-report, and thus those estimates are likely a better reflection of learning.

²⁰Students were not informed that they would complete additional assessments in Session Two. We find no evidence that students prepared between sessions: only 5 percent of students reported looking up extra information about the topic, with no significant difference between treatment and control groups ($p = 0.192$), and no student reported studying it.

²¹The test-score effects are robust to leaving out any one test item (Appendix Figure A5).

Benchmarking the Test-Score Effect. The effect of AI access on test scores is large by the standards of the educational-intervention literature. Kraft (2020) reports a median effect of 0.10 SD among 747 RCTs; our ITT of $\hat{\beta} = 0.29$ SD is substantially larger. Two in-sample benchmarks make this magnitude concrete. A single learning session raises AI-forbidden students’ test scores by 1.02 SD above baseline, and AI access adds roughly another quarter of that gain. Among control students, a one-SD increase in college GPA predicts about 0.22 SD higher test performance, so the effect of AI access exceeds the gain associated with a one-SD increase in GPA. Our estimate is similar to the 0.29 SD pooled gain from structured human tutoring programs (Nickow et al., 2024), with the advantage that AI access is not constrained by the supply of qualified tutors.

Figure 5 places our estimates alongside 24 others from 14 randomized experiments (see Appendix B.7 for the studies and inclusion criteria). Aggregating estimates within each study and excluding our own, the random-effects grand mean across the 14 studies is 0.19 SD (95 percent confidence interval 0.08 to 0.30). The gains often come from designs that cast AI as a coach: teacher-guided classroom practice with a chatbot raises test scores by 0.21–0.26 SD (De Simone et al., 2025; LearnLM Team, 2026), an AI tutor grounded in the course text raises exam performance by 0.34 SD (Fischer et al., 2025), and guided AI study outperforms unguided access (Hou et al., 2026; Gallegos, 2026). Students in our sample receive an unrestricted configuration, yet their gains are positive in both sessions. Section 5.3 relates the persistence of these gains to how students use AI.

4.3 Effects on Essay-Based Outcomes

We examine how AI access affects students’ writing in three steps. First, we check whether AI-generated text enters treated students’ essays. Second, we ask whether AI homogenizes the writing students submit. Third, we ask whether AI access improves essay quality. Because Session One essays are written during the learning phase—when treated students have AI access—these estimates reflect three channels: direct AI output, changes in how students approach the writing task, and any learning accumulated during the session itself. Session Two essays, which students write one week later without AI, isolate the learning channel.

AI Detection. AI use is detectable in treated students’ essays (Figure 6 and Table 5, Panel A). The fraction of text classified as AI-generated rises by $\hat{\beta} = 9.8$ pp in the AI-

allowed group ($p = 0.052$), a 76 percent increase over the control mean of 12.8 percent.²² The corresponding TOT estimate is 14.2 pp ($p = 0.049$). Despite this large increase in the fraction of AI-generated text, we find little evidence of stylistic changes in student essays: the effects on the length, readability, and lexical diversity indices are small and statistically insignificant (Appendix Table A4 and Appendix Figure A7). By Session Two, AI traces disappear, consistent with students writing without AI access (Table 5, Panel A). The treatment effect collapses from $\hat{\beta} = 9.8$ pp to $\hat{\beta} = 0.2$ pp ($p = 0.933$). We find no evidence of copying from online sources: the fraction of text flagged by Pangram’s plagiarism checker is near zero in both groups, with no significant treatment effect in either session.

Homogenization. Despite concerns that AI homogenizes writing (Doshi and Hauser, 2024; Meincke et al., 2025; Moon et al., 2025, 2026), we see no such effect in our data (Table 5, Panel B). Treatment effects on cosine similarity—our measure of homogeneity—are near zero and statistically insignificant: within-group text similarity changes by $\hat{\beta} = 0.004$, a 0.5 percent change relative to the control mean of 0.773. Similarity to the provided reading material changes by $\hat{\beta} = -0.002$, a 0.3 percent change relative to the control mean of 0.745. Both similarity effects remain near zero in Session Two (columns 4–6). These null results contrast with the convergence patterns documented by Brynjolfsson et al. (2025) among customer-support agents and with the reductions in collective creative diversity observed in lab studies of AI-assisted writing and brainstorming (Doshi and Hauser, 2024; Meincke et al., 2025; Moon et al., 2025). One possible explanation is task structure: customer-service and ideation tasks have well-defined correct answers or converge on common AI-suggested ideas, whereas essay prompts are open-ended and admit multiple valid responses.

Essay Quality and Dimension-Level Ratings. Effects on overall essay quality are positive but imprecise in both sessions, and the dimension-level estimates are larger in Session Two, when students write from memory (Figure 6 and Table 5, Panel C). In Session One, overall quality shows a positive but imprecise effect: treated essays score $\hat{\beta} = 0.28$ points higher (or 0.19 SD, $p = 0.180$). All five dimensions show positive but imprecise effects (Appendix Table A5). In Session Two, overall quality rises by $\hat{\beta} = 0.31$

²²That 12.8 percent of control-group text is flagged as AI-generated likely reflects a combination of control students using AI, false positives, and students whose natural writing style resembles AI-generated text—a pattern consistent with recent evidence that exposure to LLMs shifts human communication patterns toward AI-like prose (Yakura et al., 2024).

points (or 0.20 SD), though this holistic rating is imprecisely estimated ($p = 0.123$). Effects on four of the five dimensions are larger than in Session One (all but organization and structure). Writing style and clarity shows the most precise gain ($\hat{\beta} = 0.34$ points, or 0.25 SD, $p = 0.058$). The gains in relevance to the prompt ($\hat{\beta} = 0.32$ points, or 0.21 SD, $p = 0.111$), factual accuracy ($\hat{\beta} = 0.26$ points, or 0.19 SD, $p = 0.183$), and evidence and examples ($\hat{\beta} = 0.32$ points, or 0.24 SD, $p = 0.111$) are of similar size but imprecise, and organization and structure shows the smallest effect ($\hat{\beta} = 0.20$ points, or 0.13 SD, $p = 0.334$).

Together with the test-score gains of Section 4.2, these estimates suggest that treated students learned more about the topic and express that learning in several ways: in the facts they recall and in how clearly they argue and explain the topic in their own writing, though the essay estimates are too imprecise to be conclusive.

5 How AI Changes the Learning Process

We understand why AI access increased learning, we first examine how AI access changes the way students work during the learning phase: the time they allocate across learning activities and how they experience the task. To organize these findings, we then present a model in which students produce two outputs: the essay they submit and the knowledge they acquire while producing it. The model accounts for the results and yields one further implication: the effects on the submitted essay and on knowledge differ with how students use AI. We provide descriptive evidence examining this.

5.1 Effects on How Students Learn

Time and Allocation Across Learning Activities. AI access does not impact time spent on the learning phase (Table 6, Panel A). We measure time in two ways: Qualtrics records the duration of the learning phase, and the post-learning survey provides a self-reported measure. Control students spend an average of 32.6 minutes by the Qualtrics measure and 34.3 minutes by self-report. Most students use the full 35 minutes (Appendix Figure A9). Treatment effects are small and not statistically significant: a 0.2-minute decrease by Qualtrics and a 0.5-minute decrease by self-report.

Although total learning time is unchanged, its composition shifts away from producing text (Table 6, Panel B). Treated students spend about 4.7 pp less of their learning time on writing activities (drafting, editing, note-taking, and organizing), down from a control-

group share of 53.4 percent ($p = 0.041$), and about 4.4 pp more on research activities (reading and searching), up from 44.5 percent ($p = 0.050$).²³ The dominant shift is away from generating text and toward reviewing and absorbing it, with editing the one writing activity that rises, which plausibly reflects students revising AI-generated text rather than drafting their own. This pattern is consistent with a broader finding in professional settings: AI shifts effort from task execution to task stewardship (Lee et al., 2025). For example, Mozannar et al. (2024) find that GitHub Copilot users spend more than half their coding time verifying and editing AI-generated suggestions rather than writing code.

The Learning Experience. AI access makes learning more enjoyable (Table 6, Panel C): enjoyment of the learning task rises by $\hat{\beta} = 0.70$ points on the 0–10 scale ($p = 0.028$), a 13 percent improvement over the control mean of 5.22. The effect on how skilled or effective students feel is small and statistically insignificant ($\hat{\beta} = 0.07$ points, $p = 0.795$). These findings align with Noy and Zhang (2023), who find that ChatGPT access increases task satisfaction by 0.40 SD and self-efficacy by 0.20 SD among knowledge workers (though the latter is not statistically significant). One interpretation is that access to a new technology changes how students (and workers) experience the task. For example, AI access may lower the cost of effort. Understanding a confusing concept takes less effort when a student can ask for an explanation, and producing text takes less effort when AI supplies a draft, so the same task involves less frustration for students.

5.2 Producing an Essay versus Acquiring Knowledge

The evidence so far shows that AI access raised test scores, shifted time from writing toward research, and made the task more enjoyable.

To organize this evidence, we build a model in which a student produces two outputs during the learning phase: the essay the student submits and the knowledge the student acquires during the learning phase. The goal of the model is to account for these results and to generate an additional testable implication about how outcomes vary with the way students use AI. Here, we provide an abbreviated version of the model and its implications and show the complete model in Appendix D.

²³The activity-level breakdown is consistent but individually imprecise: drafting falls the most as a share of time, and note-taking and organizing also decline, while editing and reading rise (Appendix Figure A8).

Summary of the Model. In the model, a student decides how much cognitive effort to spend on two tasks: “research” (which includes activities such as reading and searching) and “writing” (which includes drafting, editing, note-taking, and organizing). The assignment has two outputs: both activities produce knowledge, and writing also produces the essay the student submits. **The quality of that essay reflects both the student’s writing and the knowledge the student has acquired.**

The model rests on two key assumptions about what AI does. First, we assume that AI used as a personalized tutor raises the return to research effort: with AI, each unit of reading and searching produces more knowledge, because the student can ask for explanations and follow-up questions. Second, we assume that the student can delegate the essay to AI, which replaces the student’s own writing with an AI-written essay of a given quality. We say that a student delegates when the submitted essay is AI-written. **The AI-detection results show that this capability was used: the share of treated students’ Session One text flagged as AI-generated is well above the control group’s.** The two capabilities can coexist: a student can ask AI to explain the topic and also ask it to write the essay.

The student values knowledge and essay quality and finds effort costly. In our experiment, the lottery tickets paid for correct test answers and for essay quality contribute to these valuations, alongside the student’s own interest in the topic and any intrinsic motivation in writing a high-quality essay.

Under these assumptions, the effect of using AI on learning **combines three forces: a gain from tutoring and two losses from delegation.** Tutoring raises knowledge because each unit of research effort produces more of it and because the student supplies more research effort in response. **Delegation removes the essay-quality reward from the student’s own work, so the student supplies less effort in both activities and forgoes the knowledge that this effort would have produced. Delegation also lowers the gain from tutoring: the knowledge that tutoring produces no longer improves the submitted essay, so research effort responds less to the higher return.** For a student who uses both capabilities the net effect is ambiguous, and a better AI-written essay need not come with more knowledge. **The model without AI is a special case of the model with AI, in which the tutoring gain is zero and the student does not delegate.**

Model Implications. **The model can rationalize the three stylized facts documented so far. First, AI access raised test scores in both sessions. In the model, tutoring raises knowledge and delegation lowers it, so a positive average test-score effect implies that, averaged**

across treated students, the tutoring gain exceeded the two losses from delegation. Second, treated students shifted time from writing toward research. In the model, both capabilities lower the share of effort devoted to writing, tutoring by raising research effort and delegation by lowering writing effort by a larger proportion than research effort, although effort differs from time: a student can spend more minutes on an activity while engaging with it less. Third, treated students enjoyed the task more. In the model, each capability the student uses raises the utility the student obtains from the task, as would a lower cost of effort, which would in addition raise effort in both activities and knowledge with it. The enjoyment gain is consistent with all three channels and does not identify which produced it.

The model also generates a testable implication about the mode of use. An AI-written essay raises the submitted essay without adding knowledge, and Session Two essays were written without AI, so the essay advantage that AI-written text gives students who delegated in Session One should be absent from their unaided work one week later, whereas the knowledge that tutoring produces should persist and show up on the tests and in the unaided essay. To examine this implication, Section 5.3 classifies treated students by how they used AI, as augmentation or automation, and compares each group’s Session One essays, unaided Session Two essays, and test scores with the control group.

The model also shows why that comparison is descriptive. Appendix D derives the delegation decision: a student delegates when the AI-written essay is good enough relative to the essay the student would write, and the threshold rises with the value the student places on knowledge. A student who values knowledge more therefore delegates less and supplies more effort in both activities. The same incentive determines the use type and the learning gain, so differences across use types reflect incentives as well as any effect of the use type itself.

We use the model to organize the evidence rather than to estimate its parameters. Two features of the design bear on the delegation decision. Correct answers on the unaided tests earned lottery tickets, which raises the value of knowledge. Students could not leave before minute 50, which lowers the value of the time that delegating saves, although students who finished early could browse the internet while they waited. Both features favor tutoring relative to ordinary coursework, in which grades reward the essay rather than the knowledge behind it, and saved time has other uses. Random assignment of AI access therefore identifies its effect for students with an incentive to learn and a small return to saving time. It does not identify the effect under coursework incentives, and it does not identify

the effect of the use type.

5.3 Outcomes by Type of AI Use

Section 5.2 implies that outcomes should differ with how students use AI. We distinguish augmentation, in which AI works with the student, from automation, in which AI does the work for the student (Brynjolfsson and Mitchell, 2017; Acemoglu and Restrepo, 2019). In the model, tutoring is our shorthand for augmentation, and delegating the essay is one form of automation. The empirical categories are broader: they describe how students interacted with AI, and a student in either category may have used both capabilities. Because students choose the use type, the comparison is descriptive. Appendix Table A6 reports baseline characteristics and unadjusted outcomes by use type; automation users enter with higher baseline test scores than augmentation users, so the comparison across use types also reflects who chooses each one.

To identify augmentation versus automation empirically, we asked an LLM to read each ChatGPT conversation log and label it *Automation* if the AI does the work *for* the student, *Augmentation* if the AI works *with* the student, *Mixed* if AI does both, or *Other* if the conversation is off-topic (see Appendix B.6 for the full prompt). Of treated AI users, 51 percent are pure augmentation users, 31 percent are pure automation users, 8 percent are mixed, and the remaining 11 percent are classified as “other,” having engaged AI only off-topic.²⁴

For the outcome comparisons in Table 7, we use inclusive groups. A student is an automation user if any conversation is labeled Automation or Mixed, and an augmentation user if any conversation is labeled Augmentation or Mixed, so students with a mixed conversation, or with separate conversations of both types, appear in both groups. We compare each group with the full control group using the specification of equation (1). The coefficients are adjusted differences between self-selected AI users and control students rather than causal effects of the use type.

Three pieces of evidence validate this classification. First, the two groups prompt AI differently: automation users rely on AI more for drafting and editing, while augmentation users turn to AI more for explaining concepts (Appendix Figure A10). Second, automation users reallocate time during the learning phase: relative to augmentation users, they spend

²⁴Similar mixed usage patterns appear in observational data: Contractor and Reyes (2026) in survey responses from Middlebury undergraduates, Handa et al. (2025) and OpenAI (2025) in large-scale Claude and ChatGPT logs, and Ammari et al. (2025) in ChatGPT logs from undergraduates at another U.S. university.

about 16 percent less time on research activities (reading and searching; $p = 0.045$) and 8 percent less time overall ($p = 0.049$). Third, these behavioral differences show up in essay outputs: Pangram flags 53.6 percent of automation users’ text as AI-generated, versus 20.5 percent for augmentation users.

The clearest difference across use types is in essay quality (Table 7). In Session One, automation users’ essays are rated $\hat{\beta} = 0.57$ SD above the control group’s ($p = 0.020$) and augmentation users’ essays $\hat{\beta} = 0.07$ SD above. In Session Two, when students write without AI, the automation users’ difference is close to zero ($\hat{\beta} = 0.03$ SD on overall quality and 0.04 SD on the quality index), while the augmentation users’ difference is positive but imprecise ($\hat{\beta} = 0.19$ SD on overall quality, $p = 0.290$, and 0.24 SD on the index). The Session One essay advantage of automation users thus does not appear in the point estimates for their unaided essays.²⁵

The test-score comparisons show less of a difference between use types. In Session One, the adjusted differences relative to control students are 0.33 SD for automation users and 0.45 SD for augmentation users. One week later, both remain positive, at 0.15 SD for automation users ($p = 0.463$) and 0.24 SD for augmentation users ($p = 0.125$), and the two are not statistically distinguishable ($p = 0.667$). On every Session Two outcome, the augmentation users’ point estimate exceeds the automation users’, so the comparison rests on point estimates. These estimates do not establish that augmentation produces more learning, or more persistent learning, than automation.

The essay pattern is consistent with the model’s distinction between the submitted essay and acquired knowledge: AI-written text can raise the quality of the first essay without raising the quality of the unaided one, whereas knowledge raises both. It does not imply that automation users learned nothing. They can learn from research, from reading the generated text, and from tutoring-style prompts, and their test-score differences remain positive.

The essay pattern is also consistent with [Strömberg et al. \(2026\)](#), who find that AI raises homework scores but lowers exam performance, with the learning losses concentrated among students whose unusually short completion times and high homework scores indicate they outsource their work to AI—the field analogue of our automation users. [Franco et al.](#)

²⁵To test the change between sessions, we estimate the four use-type regressions jointly with standard errors clustered by student. The automation users’ essay-quality difference falls by 0.54 SD from Session One to Session Two ($p = 0.084$), the augmentation users’ difference does not change ($p = 0.604$), and the difference between the two changes is 0.65 SD ($p = 0.039$; 0.70 SD, $p = 0.038$, on the quality index). Neither group’s test-score difference changes between sessions ($p = 0.442$ for automation users and 0.315 for augmentation users).

(2026) measure delegation directly in ChatGPT logs, as the share of prompts that paste practice questions verbatim into the chatbot, and report, in descriptive analyses, that the heaviest copy-pasters score 16 percent below the control mean on the unaided exam. It matches a recurring pattern in the AI-and-learning literature: AI access can boost short-term output while reducing what students learn from the task (Bastani et al., 2025; Liu et al., 2026; Shen and Tamkin, 2026; Strömberg et al., 2026). This comparison cannot separate the effect of the use type from the incentives behind it (Section 5.2). Gallegos (2026) randomizes this margin directly: among undergraduates who all had access to a course-specific AI assistant, guidance on learning-oriented use shifted interactions toward tutor-style engagement and raised final-exam scores by 0.22 SD, with the gains concentrated in open-response questions, while a companion experiment that encouraged adoption alone raised usage but left exam performance unchanged. Guidance does not raise learning everywhere: Franco et al. (2026) also randomize brief written guidance and find that it reduces verbatim copy-pasting of practice questions without changing average exam scores.

6 Beliefs About AI’s Impact on Learning

Students’ beliefs about AI’s effect on learning may shape their adoption decisions. In observational data, students report that AI improves their learning (Stöhr et al., 2024; Ravšelj et al., 2025). Yet these beliefs may not track actual effects: students hold biased beliefs in other educational settings (Jensen, 2010; Wiswall and Zafar, 2015), and even experienced professionals misjudge AI’s productivity effects.²⁶

We test the accuracy of students’ beliefs—and how treatment shapes them—in two ways. First, after the Session Two test, students estimate (1) how many questions they answered correctly, (2) how many they would have answered under the opposite treatment condition, (3) how many questions other students in their group answered correctly, and (4) the same for students in the opposite group. From these we compute perceived treatment effects on own and others’ performance, in percentage points, and compare them with the actual estimates from Section 4.2, overall and across subgroups. Second, we analyze responses to an open-ended question about the effects of generative AI on student learning in college. We validate students’ open-ended responses against their AI usage patterns (Appendix C.1) and, following Andre et al. (2026), code each response as a causal graph

²⁶Becker et al. (2025) find that experienced open-source developers predicted AI tools would speed them up by 24 percent and, after the study, still believed AI had accelerated their work by roughly 20 percent. The actual measured effect was a 19 percent *slowdown*.

of the mechanisms it describes (Appendix C.2).

6.1 Beliefs About AI’s Effect on Test Scores

Control students overestimate the effect of AI on their test performance; treated students predict an effect close to the one we estimate (Figure 7, Panel A). **The figure and the numbers that follow use the adjusted estimates of Appendix Table A7.** Both groups give similar predictions of their own raw score (control 37.8 percent correct, treated 37.2 percent, both below the actual mean of 51.2 percent); what differs is what they attribute to AI. Control students predict an AI gain for themselves of 24.8 pp, **6.7 times** the actual effect of 3.7 pp. Treated students, who experienced AI firsthand, predict only 3.3 pp, close to the effect we estimate.²⁷ Predictions about others’ performance show the same pattern in muted form: control students predict 14.1 pp and treated 10.9 pp, both still inflated relative to the actual effect. Across subgroups, perceived and actual gains are only weakly related (Figure 7, Panel B; correlation = 0.14).

6.2 Students’ Mental Models of AI and Learning

The above beliefs capture students’ expected test-score gains from AI, but are silent about the mechanisms by which they think AI affects learning. To recover these mental models, we asked an open-ended question on the exit survey: “In your opinion, how does generative AI (e.g., ChatGPT) affect student learning in college? Please explain your reasoning.” Following Andre et al. (2026), we code each short narrative as a causal graph running from AI use to learning through the mechanisms the student names (see Appendix C.2 for the coding procedure and its reliability, and Appendix C.3 for additional results). Figure 8 aggregates students’ mental models into a single causal graph in which each node is one such mechanism (sized by how often students name it), and each link is green where they describe the mechanism as promoting learning and maroon where it harms it.

The augmentation-versus-automation distinction along which outcomes differ in Section 5.3 is also the structure most students use to reason about the effects of AI on learning. Among students naming any learning channel, 69 percent name both one through which AI helps and one through which it harms. The single most common element, present in 54.7 percent of coded responses, is that the effect depends on how AI is used; as one student put

²⁷The lower average perceived effect among treated students reflects two shifts in the distribution of beliefs relative to control students (Appendix Figure A11): more mass at zero (no perceived effect) and less mass at large positive values (five or more questions).

it, AI is “a double-edged sword, with each individual student’s use determining whether it is helpful or hurtful for their education.” The two most common mechanisms map onto the two sides of this distinction: AI explaining concepts and tutoring, as an augmentation channel (40 percent of coded responses), and AI shortcutting the work, as an automation channel (42 percent).²⁸ Taken together, students’ reasoning mirrors our estimates: they expect AI to raise their learning, their average estimate of the magnitude moves toward ours once they have access to AI, and they locate the gains where our estimates by use type are larger, in AI that augments their work rather than does it for them.

7 Conclusion

This paper provides causal evidence on how access to generative AI affects student learning. In a randomized experiment with undergraduate students, AI access produces learning gains, part of which persists one week later. AI access also reshapes how students approach learning a new topic: they spend less time drafting text and more time reading and searching for information, and report higher enjoyment. Taken together, these results provide proof of concept that off-the-shelf AI can improve learning, and outcomes differ across the ways students chose to use it.

We do not view our findings as implying that widespread AI adoption will raise learning overall. We find no effect on total learning time—likely because the lab setting offered few competing uses of time—so the learning gains hold time-on-task fixed. In ordinary academic work, students choose how long to spend on each task, and many use AI to save time. Whether AI raises or lowers total learning therefore depends on how students reallocate the time they save and on whether learning-productivity gains more than compensate for any reduction in time spent learning.

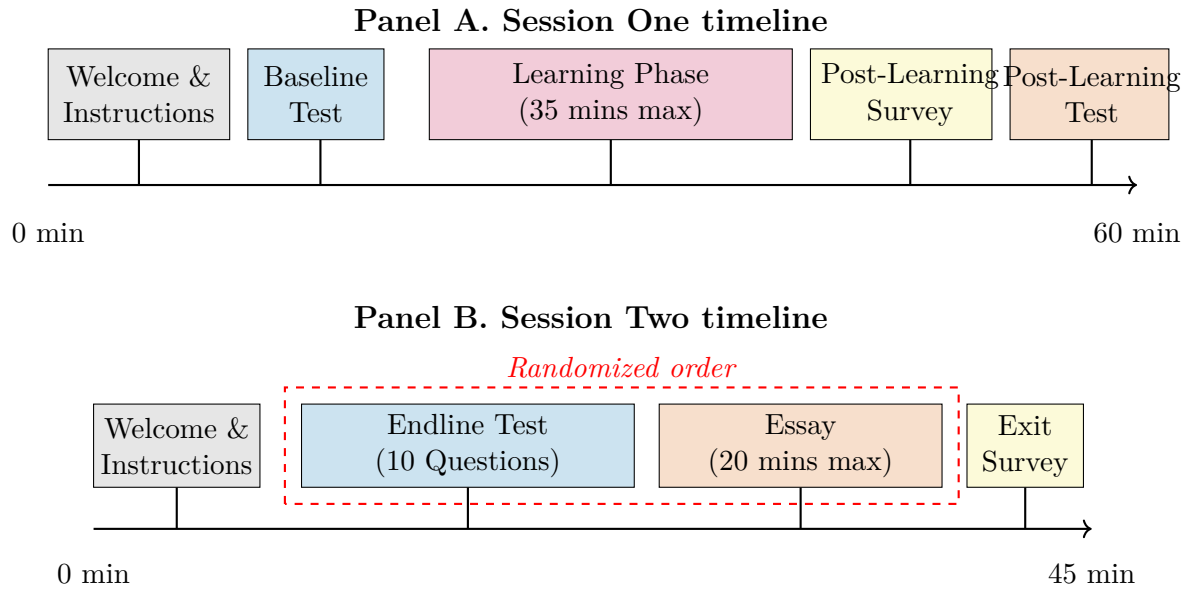
Our findings have implications for incentive design in higher education. In our model, using AI as a tutor raises knowledge, while delegating the essay to AI displaces the learning that writing would have produced. Students who value learning less are more likely to delegate, so how students use AI depends on the incentives they face. Before AI, as-

²⁸Appendix Table C1 reports how students’ narratives about AI and learning vary with their characteristics. Students with above-median GPA hold more sophisticated mental models: conditional on the other characteristics, they name 0.47 more mechanisms (column 1, $p = 0.020$), relative to a mean of 3.1 mechanisms. Among the AI-experience measures, self-assessed proficiency is the only one that predicts framing: conditional on demographics and frequency of use, students at or above the median in proficiency are 17.1 pp more likely to frame AI as augmentation ($p = 0.019$) and 12.0 pp less likely to frame it as automation ($p = 0.062$).

signments produced learning even when students cared mainly about the submitted work, because producing it required the activities through which students learn. Delegation to AI weakens that link, and features of colleges shape the incentive to delegate. For example, grade inflation weakens the link between effort and grades, pushes students to differentiate themselves through extracurriculars, and gives them reason to automate coursework with AI to free up time. Similarly, students who view college primarily as a signal rather than as human-capital accumulation have less reason to use AI as a learning tool, since learning matters less for their perceived returns to education. How higher-education incentives shape AI usage, and through it learning, is a question for future research.

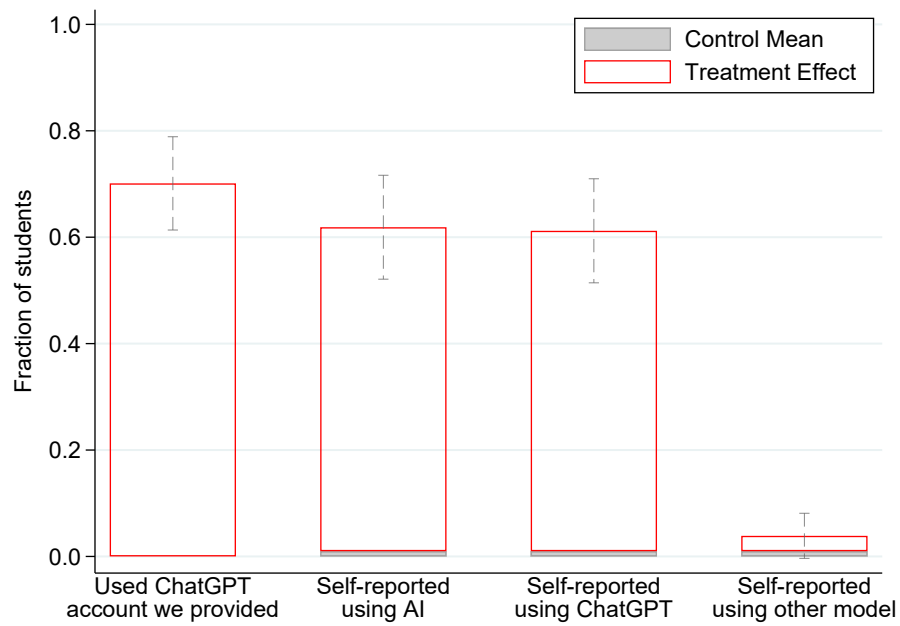
Figures and Tables

Figure 1: Experimental Sessions Timelines



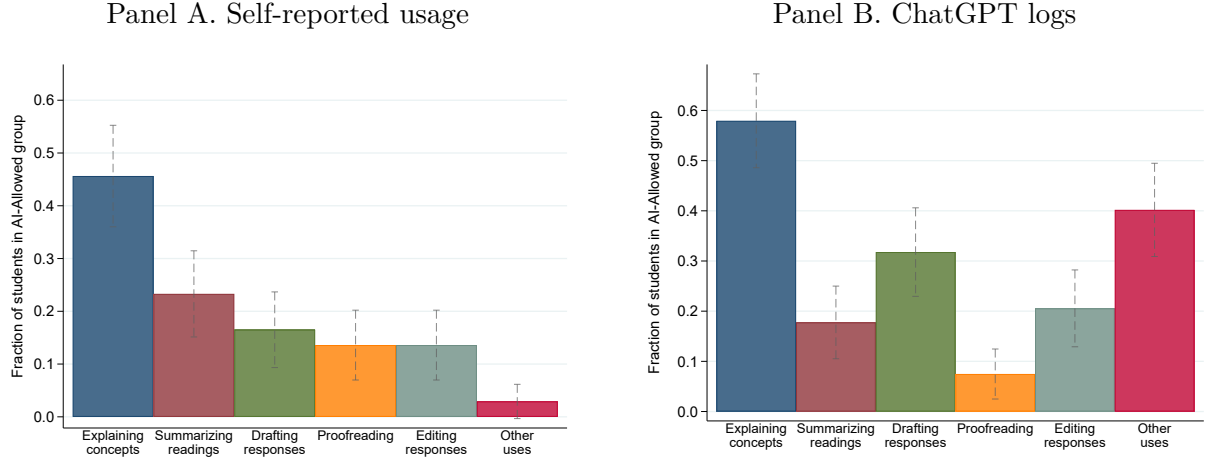
Notes: This figure shows the timeline of the two experimental sessions. Panel A shows the structure of Session One, during which students were randomly assigned to AI-allowed or AI-forbidden conditions. Panel B shows Session Two, conducted approximately one week later (mean of 6.98 days), during which all students completed tasks without AI access. The red dashed box in Panel B indicates that the order of the Endline Test and the Essay was randomized across students. The Baseline Test and Post-Learning Test in Session One each contained 5 multiple-choice questions. The Endline Test in Session Two contained 10 multiple-choice questions.

Figure 2: The Impact of AI Access on Generative AI Usage During the Learning Phase



Notes: This figure shows the impact of AI access on generative AI usage during the learning phase in Session One. Gray bars represent control group means, while red outlines show treatment effects. The first measure is constructed from activity in the ChatGPT account provided to each student. The remaining three measures are based on self-reported usage collected in the exit survey. Vertical bars represent 95 percent confidence intervals.

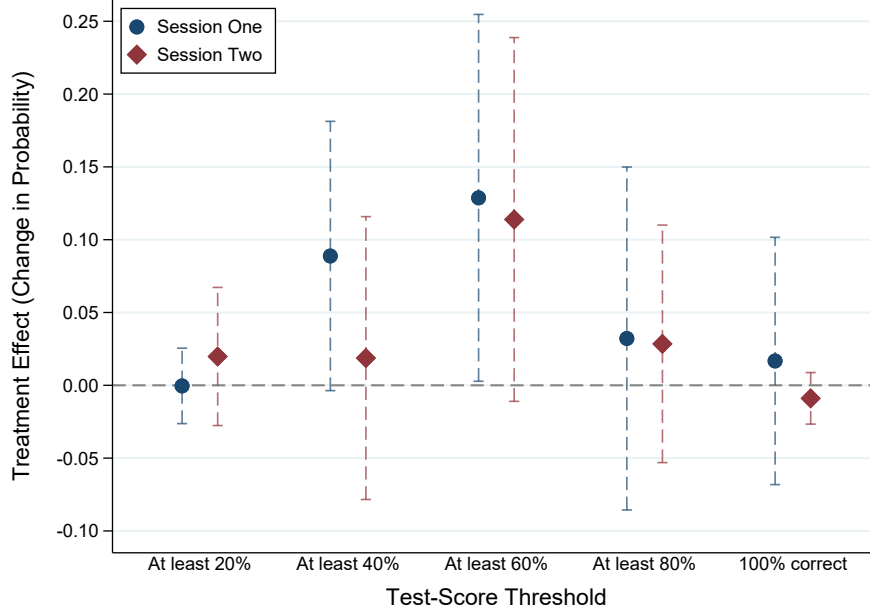
Figure 3: Types of Generative AI Use During the Learning Phase



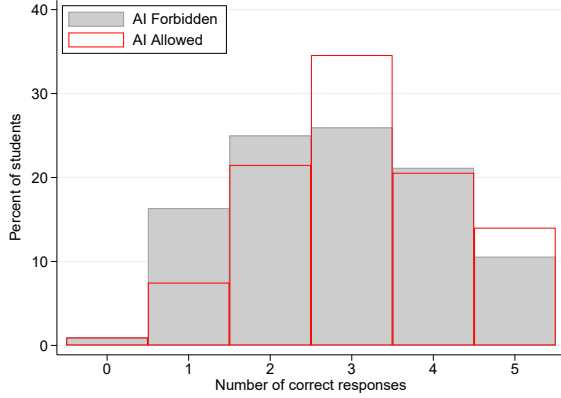
Notes: This figure shows AI usage types among treated students. Panel A presents self-reported usage types collected in the exit survey; students could select multiple categories. Panel B is constructed from the actual ChatGPT conversation logs. We first use LLMs to classify each student prompt into one of six categories: explaining concepts, summarizing readings, drafting responses, proofreading, editing responses, and other uses (see Appendix B.6 for the classification procedure). We then construct student-level indicators: for each category, the indicator equals one if any of the student's prompts fell into that category. Students who were assigned to the AI-allowed condition but did not use the provided ChatGPT account are coded as zero for all categories.

Figure 4: Distributional Effects of AI Access on Test Performance

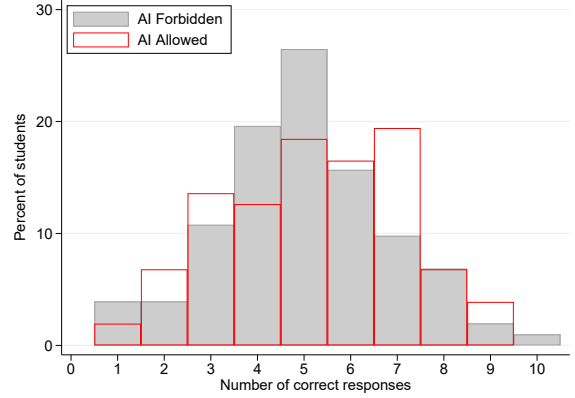
Panel A. Treatment effects by score threshold



Panel B. Session One (knowledge acquisition)

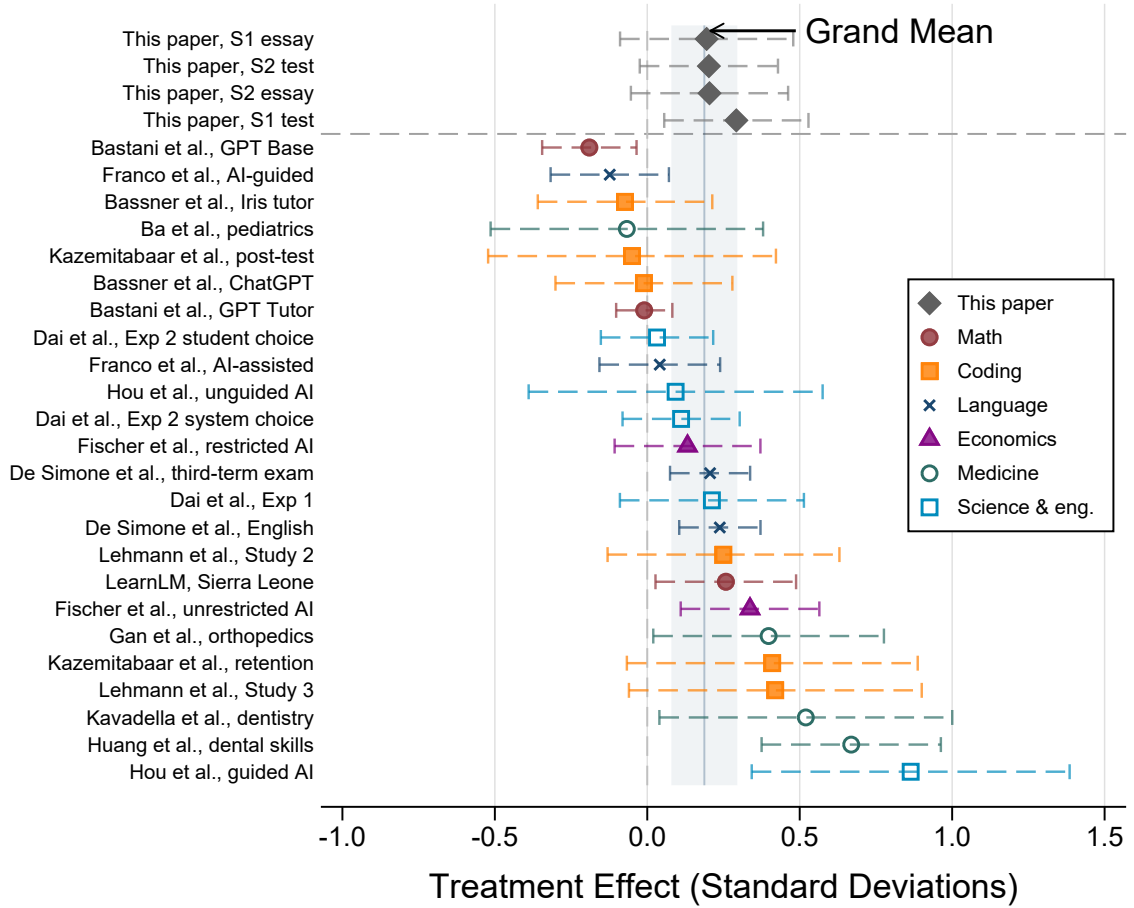


Panel C. Session Two (knowledge retention)



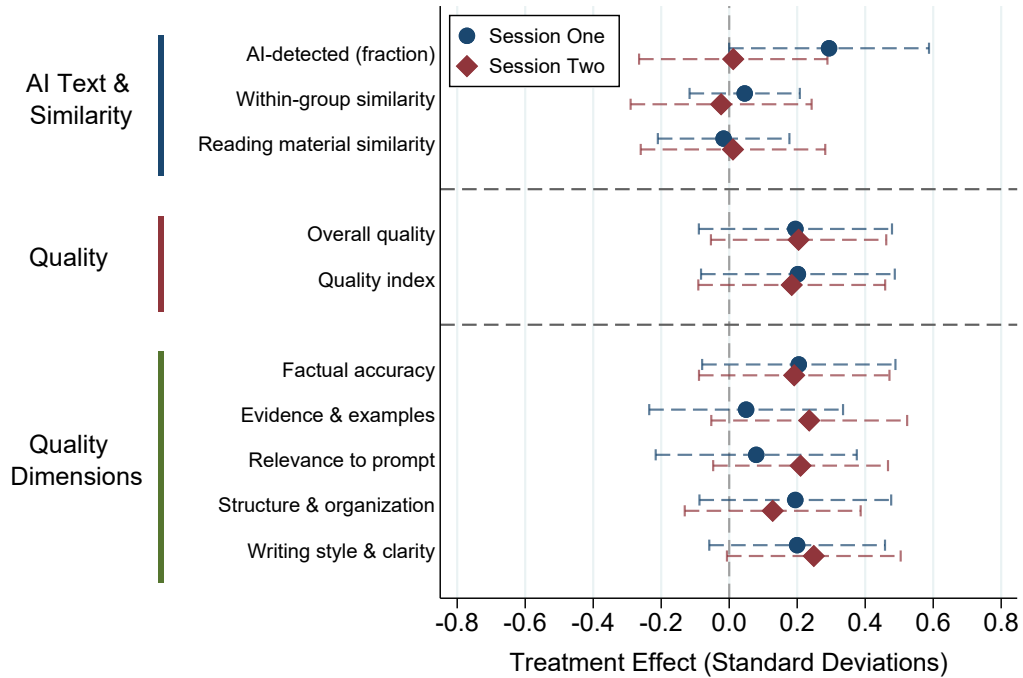
Notes: Panel A plots ITT estimates of the effect of AI access on indicators for scoring above each threshold, separately for Session One (navy circles) and Session Two (maroon diamonds). Each estimate comes from a regression that includes strata fixed effects and controls selected through a double-lasso procedure (Belloni et al., 2014). Vertical bars show 95 percent confidence intervals based on heteroskedasticity-robust standard errors. Panels B and C show the distribution of test scores by treatment assignment. The x -axis shows the number of correct responses; the y -axis shows the percentage of students achieving each score. The Session One test had 5 questions; the Session Two test had 10 questions.

Figure 5: Effect Sizes Across AI-and-Learning Experiments



Notes: This figure compares treatment effect sizes (in standard deviations) across randomized experiments that provide students with AI tools during a learning task and measure performance on unassisted assessments. Marker shapes indicate the outcome domain: diamonds for this paper, circles for math, squares for coding, crosses for language learning, triangles for economics, hollow circles for medicine, and hollow squares for science and engineering. Our estimates include both knowledge test scores and essay quality for Session One (immediate) and Session Two (retention), shown above the dashed separator line. Literature estimates come from the 14 randomized experiments described in Appendix B.7 and summarized in Appendix Table B3. Horizontal lines show 95 percent confidence intervals. The shaded band and vertical line show the random-effects grand mean across the other studies and its 95 percent confidence interval, estimated via the DerSimonian and Laird (1986) method after aggregating each study's estimates by inverse-variance weighting (our own estimates are excluded from the pool). The dashed vertical line marks zero.

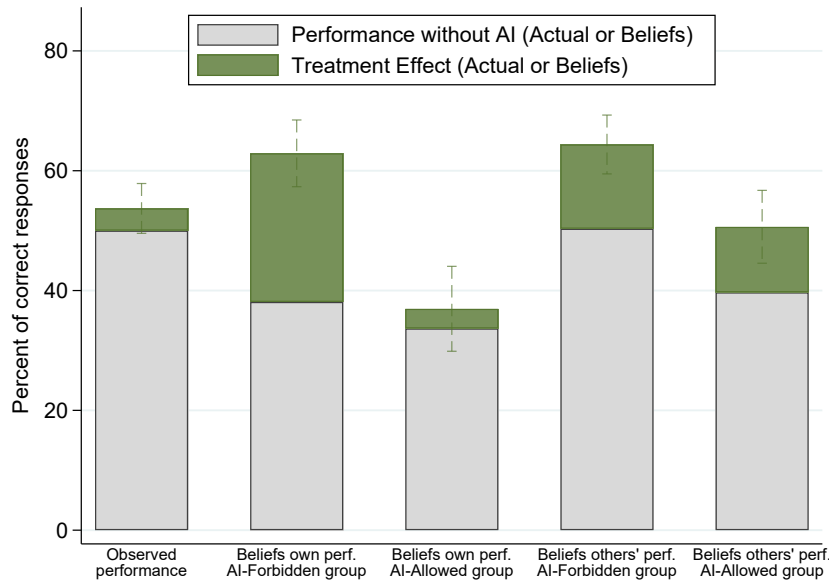
Figure 6: Effects of AI Access on Essay Outcomes



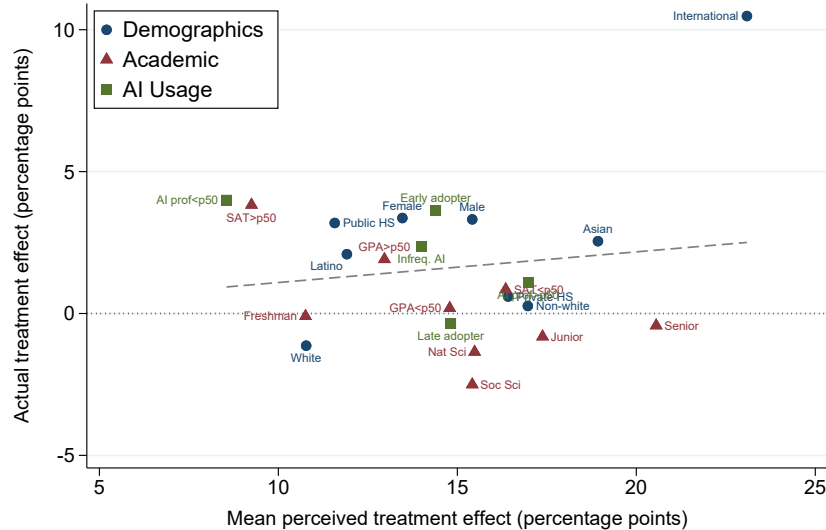
Notes: This figure presents treatment effects of AI access on AI detection and textual similarity (top panel), overall essay quality (middle panel), and individual quality dimensions (bottom panel), measured in standard deviations. The top-panel measures are the AI-detected fraction (the share of text classified as AI-generated by Pangram, standardized in the control group), within-group similarity (the average pairwise cosine similarity within treatment×topic×prompt cells), and reading material similarity (between the student essay and the provided reading material). The overall quality measures are overall quality (the grader’s holistic assessment) and the quality index (averaging the five dimensions below), each averaged across the human and AI graders. The quality dimensions are the five sub-components of a standardized rubric—accuracy of content, use of evidence and examples, relevance to the prompt, organization and structure, and writing style and clarity—each averaged across the human and AI graders. Circles represent Session One effects (essays written with or without AI access); diamonds represent Session Two effects (essays written one week later without AI access). Horizontal lines represent 95 percent confidence intervals. Point estimates underlying each coefficient in raw units are reported in Table 5 and Appendix Table A5. The five quality dimensions split by grader appear in Appendix Figure A6.

Figure 7: Actual and Perceived Treatment Effects on Test Performance

Panel A. Beliefs versus actual effects

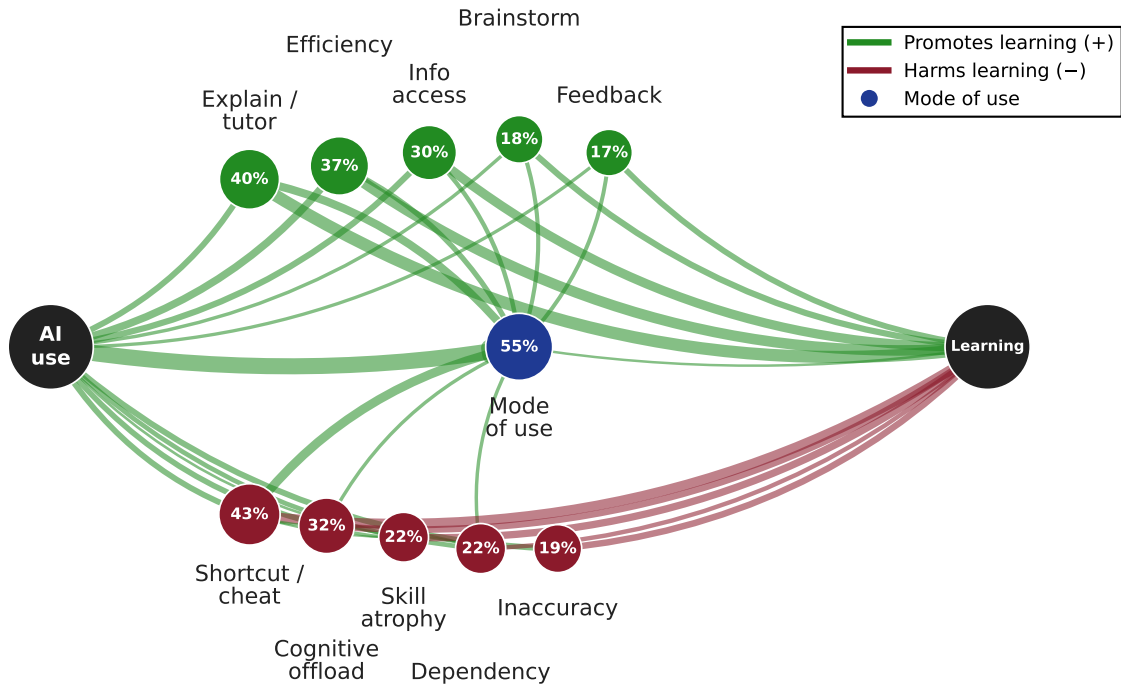


Panel B. Perceived versus actual effects by subgroup



Notes: This figure shows actual and perceived treatment effects of AI access on test performance. In Panel A, the first bar shows observed performance—gray is the control mean, and green is the estimated treatment effect. The remaining bars show students' beliefs about own and others' performance, separately by treatment group: gray is the believed score without AI and green the perceived AI gain. AI-forbidden bars are control means; AI-allowed bars add the double-lasso treatment effect of Appendix Table A7 to the control mean. Panel B plots the actual treatment effect against the mean perceived treatment effect (own performance) for each demographic, academic, and AI-usage subgroup with at least 40 observations; the dashed line is a linear fit. Vertical bars represent 95 percent confidence intervals, from the standard error of the mean for the AI-forbidden bars and of the treatment effect elsewhere.

Figure 8: The Average Narrative About AI’s Effect on Learning



Notes: This figure shows the aggregate signed causal graph across the 201 open-ended responses codable as a narrative, constructed following Andre et al. (2026). We code each student’s answer to the question “in your opinion, how does generative AI (e.g., ChatGPT) affect student learning in college? Please explain your reasoning” as a causal graph from AI use (left) to learning (right) through the mechanisms the student names; the figure overlays these graphs. Node size is proportional to the share of students who name a mechanism, and link width to the share drawing the link; we omit mechanisms below 10 percent and links below 2 percent. Green links denote mechanisms described as promoting learning, maroon links mechanisms that harm it, and navy the mode-of-use node. Appendix C.2 describes the coding and its reliability.

Table 1: Summary Statistics

	Signed-up (1)	Attended Session One (2)	Attended Both Sessions (3)
Panel A. Demographic characteristics			
Age	20.211	20.147	20.127
Male	0.359	0.351	0.351
White	0.496	0.498	0.507
International student	0.270	0.265	0.273
Public high school	0.539	0.540	0.537
Panel B. Academic background			
Took SAT/ACT	0.742	0.763	0.766
SAT score (conditional)	1387.593	1385.969	1386.571
SAT score (including predicted)	1379.537	1375.642	1375.953
College GPA	3.680	3.683	3.686
Freshman	0.359	0.374	0.385
Already declared major	0.783	0.785	0.778
Natural-science major	0.301	0.308	0.312
Social-science major	0.367	0.365	0.356
Humanities/Arts/Languages major	0.109	0.104	0.102
Study hours per week	16.918	16.654	16.620
Panel C. Attendance and comprehension			
Attended Session One	0.824	1.000	1.000
Attended Session Two	0.801	0.972	1.000
Attended Session Two (conditional S1)	0.972	0.972	1.000
Attended both sessions	0.801	0.972	1.000
Days between sessions	6.951	6.986	6.980
Comprehension score (0-5)	4.763	4.763	4.761
Panel D. Experimental conditions			
Assigned Blockchain topic	0.365	0.365	0.371
Assigned CRISPR topic	0.327	0.327	0.322
Assigned Carbon-capture topic	0.308	0.308	0.307
Baseline self-assessed knowledge (0-10)	1.517	1.517	1.512
Fraction correct (baseline)	0.303	0.303	0.308
Shared info between sessions	0.059	0.059	0.059
Looked up info about topic	0.049	0.049	0.049
Studied topic for Session Two	0.000	0.000	0.000
Number of students	256	211	205

Notes: This table presents summary statistics for three groups of students: students who completed the sign-up survey and scheduled a Session One slot (column 1), those who attended Session One (column 2), and those who attended both sessions (column 3). GPA is on the 0–4 scale. SAT score (conditional) is the mean among students with an SAT score or an ACT score converted to the SAT scale, and SAT score (including predicted) adds the imputed scores described in Appendix B.5.

Table 2: Balance of Baseline Characteristics by Treatment Assignment

	Treatment group:		Difference:	
	AI-forbidden (1)	AI-allowed (2)	$\hat{\beta}$ (3)	(SE) (4)
Panel A. Demographic characteristics				
Age	20.016	20.400	0.382	(0.227)*
Male	0.405	0.315	-0.091	(0.060)
White	0.524	0.469	-0.055	(0.063)
International student	0.286	0.254	-0.033	(0.055)
Public high school	0.540	0.538	-0.001	(0.063)
Panel B. Academic background				
Took SAT/ACT	0.778	0.708	-0.067	(0.054)
SAT score (conditional)	1392.371	1382.554	-7.862	(21.039)
SAT score (including predicted)	1377.853	1381.169	2.537	(18.796)
College GPA	3.694	3.666	-0.025	(0.038)
Freshman	0.357	0.362	0.007	(0.061)
Already declared major	0.825	0.742	-0.083	(0.052)
Natural-science major	0.302	0.300	-0.002	(0.058)
Social-science major	0.413	0.323	-0.093	(0.061)
Humanities/Arts/Languages major	0.111	0.108	-0.000	(0.039)
Study hours per week	16.864	16.969	0.205	(1.199)
Panel C. Attendance and comprehension				
Attended Session One	0.825	0.823	-0.006	(0.048)
Attended Session Two	0.810	0.792	-0.020	(0.050)
Attended Session Two (conditional S1)	0.981	0.963	-0.017	(0.022)
Attended both sessions	0.810	0.792	-0.020	(0.050)
Days between sessions	6.975	6.929	-0.030	(0.156)
Comprehension score (0-5)	4.760	4.766	0.001	(0.065)
Panel D. Experimental conditions				
Assigned Blockchain topic	0.375	0.355	-0.023	(0.066)
Assigned CRISPR topic	0.327	0.327	0.002	(0.064)
Assigned Carbon-capture topic	0.298	0.318	0.021	(0.062)
Baseline self-assessed knowledge (0-10)	1.558	1.477	-0.086	(0.251)
Fraction correct (baseline)	0.306	0.301	-0.007	(0.038)
<i>N</i> (Students)	126	130	256	

Notes: This table shows average student characteristics by treatment assignment. Students in the AI-allowed group could use generative AI tools during the learning phase; students in the AI-forbidden group could not. Panels A and B report characteristics measured before treatment assignment. Panel C reports session attendance and instruction comprehension, realized after assignment; Panel D reports the randomly assigned essay topics and baseline knowledge measures collected at the start of Session One, before the learning phase. Heteroskedasticity-robust standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Table 3: The First-Stage Impact of Generative AI Access on AI Usage

Outcome	Control mean (1)	Effect ($\hat{\beta}$) (2)
Panel A. Revealed measures		
Used ChatGPT account	0.000	0.701*** (0.045)
Number of prompts	0.000	3.972*** (0.427)
Number of conversations	0.000	0.798*** (0.060)
Panel B. Self-reported measures		
Used any AI	0.010	0.609*** (0.050)
Used ChatGPT	0.010	0.602*** (0.050)
Used other AI tool	0.010	0.029 (0.022)
<i>N</i>	104	211

Notes: This table reports first-stage effects of AI-access assignment on measures of AI usage. Each row reports the effect of being assigned to the AI-allowed group on the indicated measure. Panel A presents revealed measures: Used ChatGPT account equals one if at least one prompt was recorded in the ChatGPT account provided to the student; Number of prompts is the total number of prompts sent to the provided ChatGPT account during the session; Number of conversations counts the number of distinct ChatGPT conversations (i.e., separate chat threads) initiated during the session. Panel B presents self-reported measures collected in Session Two. All specifications include controls selected through a double-lasso procedure (Belloni et al., 2014). Heteroskedasticity-robust standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Table 4: The Impact of Generative AI on Test Scores

Outcome	Session One			Session Two		
	Control mean (1)	ITT (2)	TOT (3)	Control mean (4)	ITT (5)	TOT (6)
Self-assessed	4.788	−0.094 (0.206)	−0.137 (0.302)	3.833	0.193 (0.192)	0.271 (0.269)
Fraction correct	0.563	0.074** (0.031)	0.106** (0.044)	0.500	0.037* (0.021)	0.052* (0.030)
Test score (SD)	0.000	0.292** (0.121)	0.418** (0.174)	0.000	0.202* (0.116)	0.284* (0.165)
<i>N</i>	104	210	210	102	205	205

Notes: This table reports treatment effects of AI access on learning outcomes in Session One, immediately after the learning phase, and in Session Two, one week later. Students took both tests without access to any resources. The AI-allowed group had AI access only during the Session One learning phase. Each row reports the effect of being assigned to the AI-allowed group on the indicated outcome. Self-assessed knowledge is measured on a 0–10 scale, with 0 indicating “I know nothing about this topic” and 10 indicating “I am an expert.” Fraction correct is measured on a 0–1 scale. Test score (SD) is the fraction correct standardized to have mean zero and standard deviation one in the control group. Columns 1 and 4 report the control group mean. Columns 2 and 5 report intent-to-treat (ITT) estimates. Columns 3 and 6 report treatment-on-the-treated (TOT) estimates from two-stage least squares, instrumenting actual ChatGPT use with random treatment assignment. All specifications include controls selected through a double-lasso procedure (Belloni et al., 2014). Heteroskedasticity-robust standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Table 5: Effects of Generative AI Access on Essay Outcomes

Outcome	Session One (with AI)			Session Two (without AI)		
	Control mean (1)	ITT (2)	TOT (3)	Control mean (4)	ITT (5)	TOT (6)
Panel A. AI detection and plagiarism						
AI-detected (fraction)	0.128	0.098* (0.050)	0.142** (0.072)	0.039	0.002 (0.028)	0.003 (0.039)
Plagiarism (fraction)	0.000	0.000 (0.000)	0.000 (0.000)	0.000	0.000 (0.000)	0.000 (0.001)
<i>N</i>	104	210	210	102	204	204
Panel B. Homogeneity and similarity						
Within-group sim.	0.773	0.004 (0.007)	0.006 (0.010)	0.746	−0.003 (0.016)	−0.004 (0.022)
Reading material sim.	0.745	−0.002 (0.009)	−0.002 (0.013)	0.696	0.001 (0.016)	0.002 (0.023)
<i>N</i>	104	210	210	102	204	204
Panel C. Overall quality						
Overall quality (average)	6.190	0.277 (0.206)	0.401 (0.298)	5.076	0.312 (0.201)	0.443 (0.290)
Quality index (average)	6.450	0.246 (0.177)	0.353 (0.255)	5.297	0.240 (0.183)	0.343 (0.265)
<i>N</i>	104	210	210	97	198	198

Notes: This table reports treatment effects of AI access on essay outcomes: AI-detection and plagiarism measures, textual similarity, and overall quality. Each row reports the effect of being assigned to the AI-allowed group on the indicated outcome. Panel A reports AI detection and plagiarism measures: AI-detected (fraction) is the fraction of text classified as AI-generated by the Pangram AI content detector; Plagiarism (fraction) is the fraction of text flagged as plagiarized by Pangram’s plagiarism checker. Panel B reports homogeneity and similarity measures: within-group similarity is the average pairwise cosine similarity between a student’s essay embedding and all other essays in the same treatment $\times$ topic $\times$ prompt cell; reading material similarity is the cosine similarity between a student’s essay embedding and the embedding of the provided reading material. Both are computed using a sentence-embedding model (Reimers and Gurevych, 2019) fine-tuned from the MPNet architecture (Song et al., 2020). Panel C reports overall quality (the grader’s holistic assessment) and the quality index (the average of the five dimension scores: accuracy, evidence, relevance, organization, and writing style), each scored on a 0–10 scale and averaged across human and AI graders; human grades are averaged across each essay’s graders before estimation, and Appendix Table A5 reports the five dimensions separately. The number of observations is reported below each panel. Columns 1 and 4 report the control group mean. Columns 2 and 5 report intent-to-treat (ITT) estimates. Columns 3 and 6 report treatment-on-the-treated (TOT) estimates from two-stage least squares, instrumenting actual ChatGPT use with random treatment assignment. All specifications include controls selected through a double-lasso procedure (Belloni et al., 2014). Heteroskedasticity-robust standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Table 6: Time Use and Experience

Outcome	Control mean (1)	ITT (2)	TOT (3)
Panel A. Time spent learning			
Time spent learning (Qualtrics, minutes)	32.611	−0.205 (0.678)	−0.292 (0.968)
Time spent learning (self-reported, minutes)	34.312	−0.532 (0.669)	−0.769 (0.966)
Panel B. Time allocation			
Share on research activities (percent)	44.462	4.435** (2.245)	6.403** (3.244)
Share on writing activities (percent)	53.381	−4.728** (2.301)	−6.817** (3.304)
Share on other activities (percent)	2.157	0.911 (1.310)	1.316 (1.883)
Panel C. Learning experience			
Enjoyed learning (continuous)	5.221	0.696** (0.314)	1.005** (0.458)
Found effective (continuous)	5.000	0.073 (0.282)	0.103 (0.399)
<i>N</i>	104	210	210

Notes: This table reports treatment effects of AI access on total time spent learning (Panel A), how that time is allocated across activities (Panel B), and the learning experience (Panel C). Each row reports the effect of being assigned to the AI-allowed group on the indicated outcome. **The two time spent learning rows give the duration of the learning phase in minutes, recorded by Qualtrics and in the post-learning survey, respectively.** The share on research, writing, and other activities rows express each category’s self-reported minutes as a percentage of the student’s total self-reported learning time; these rows report effects in percentage points. Enjoyed learning and Found effective are 0–10 self-reports. All specifications include controls selected through a double-lasso procedure (Belloni et al., 2014). Heteroskedasticity-robust standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Table 7: Outcomes by Type of AI Use: Automation versus Augmentation

	Session One			Session Two		
	Test score (1)	Overall quality (2)	Quality index (3)	Test score (4)	Overall quality (5)	Quality index (6)
Panel A. All students						
Overall (TOT)	0.418** (0.174)	0.282 (0.210)	0.291 (0.210)	0.279* (0.165)	0.294 (0.195)	0.266 (0.204)
<i>N</i>	210	210	210	205	198	198
Panel B. By type of AI use						
Automation users	0.327 (0.228)	0.567** (0.240)	0.572** (0.238)	0.151 (0.205)	0.027 (0.232)	0.042 (0.248)
<i>N</i>	133	132	132	131	126	126
Augmentation users	0.446*** (0.161)	0.075 (0.181)	0.065 (0.183)	0.237 (0.154)	0.189 (0.178)	0.238 (0.181)
<i>N</i>	148	148	148	144	137	137

Notes: This table reports treatment effects on test scores and essay quality, separately for the full sample (Panel A) and by how students used AI (Panel B). Overall (TOT) is the treatment-on-the-treated estimate, instrumenting actual ChatGPT use with random assignment to the AI-allowed group. Treated students' conversation logs are classified into four mutually exclusive categories (see Appendix B.6): *Automation*, if the AI did the work *for* the student; *Augmentation*, if the AI worked *with* the student (e.g., explaining concepts or giving feedback on the student's own writing); *Mixed*, if both occurred within the same conversation; and *Other*, if the conversation was off-topic. Panel B reports effects for two subgroups, each pooled with the control group as the comparison: *Automation users* (Automation + Mixed) and *Augmentation users* (Augmentation + Mixed). A student belongs to a subgroup if any of their conversations falls in it, so students with a Mixed conversation—or with separate Automation and Augmentation conversations—appear in both rows; the two subgroups are therefore not mutually exclusive, and the coefficients need not average to the Panel A overall effect. Treated students whose conversations were classified as Other, or who did not use ChatGPT at all, appear only in Panel A. **All six outcomes are standardized by the control-group mean and standard deviation of the corresponding session, so the coefficients are in control-group SD units.** All specifications use controls selected by double-lasso on the full sample (Belloni et al., 2014), with strata fixed effects. Heteroskedasticity-robust standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

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Appendix

A Appendix Figures and Tables

Figure A1: Computer Lab Setup with Privacy Dividers



Notes: This figure shows the computer lab setup used during experimental sessions. Each workstation was equipped with privacy dividers to minimize distractions and prevent participants from viewing other screens.

Figure A2: Treatment Instructions Displayed to Participants

Panel A. AI-allowed condition

How to Approach the Learning Phase

How should I approach learning this topic?

- You are encouraged to use the same learning approach you would typically follow for a college assignment.
- For example, if you typically use generative AI tools such as ChatGPT, feel free to use them here as well. **Using ChatGPT or other generative AI is completely allowed.** We provide you with a ChatGPT account, already logged in one of the tabs, so you don't need to use your personal account.
- You are welcome to use standard online resources like [Google](#), [Wikipedia](#), [Middlebury College Library's website](#), or other websites to look up information.
- To get you started, in the next screen we will provide you with an introductory text about the topic.

Click "Next" to begin the learning phase.

Panel B. AI-forbidden condition

How to Approach the Learning Phase

How should I approach learning this topic?

- You are encouraged to use the same learning approach you would typically follow for a college assignment.
- You are **not allowed to use generative AI tools such as ChatGPT, Claude, or other AI assistants.**
- You are welcome to use standard online resources like [Google](#), [Wikipedia](#), [Middlebury College Library's website](#), or other websites to look up information.
- To get you started, once the learning phase begins, we will provide you with an introductory text about the topic.

Click "Next" to begin the learning phase.

Notes: This figure shows the treatment-specific instructions displayed to participants in the survey interface.

Figure A3: Laboratory Seating Charts

Panel A. Library computer lab (LIB 140)

LIB 140

Session # _____ # Students _____

Date _____ Proctor 1 _____

Treatment _____ Proctor 2 _____

The seating chart for LIB 140 is a rectangular grid. At the top center is a horizontal line labeled "WHITEBOARD". Below it, 20 numbered computer workstations are arranged in two groups of 10. The first group (1-10) is on the left, and the second group (11-20) is on the right. Each workstation is represented by a small icon of a computer monitor. At the bottom of the chart, there are three boxes: "Proctor 1" on the left, "Proctor 2" in the middle, and "Door" on the right.

Panel B. Language Building computer lab (SDL 122)

SDL 122

Session # _____ # Students _____

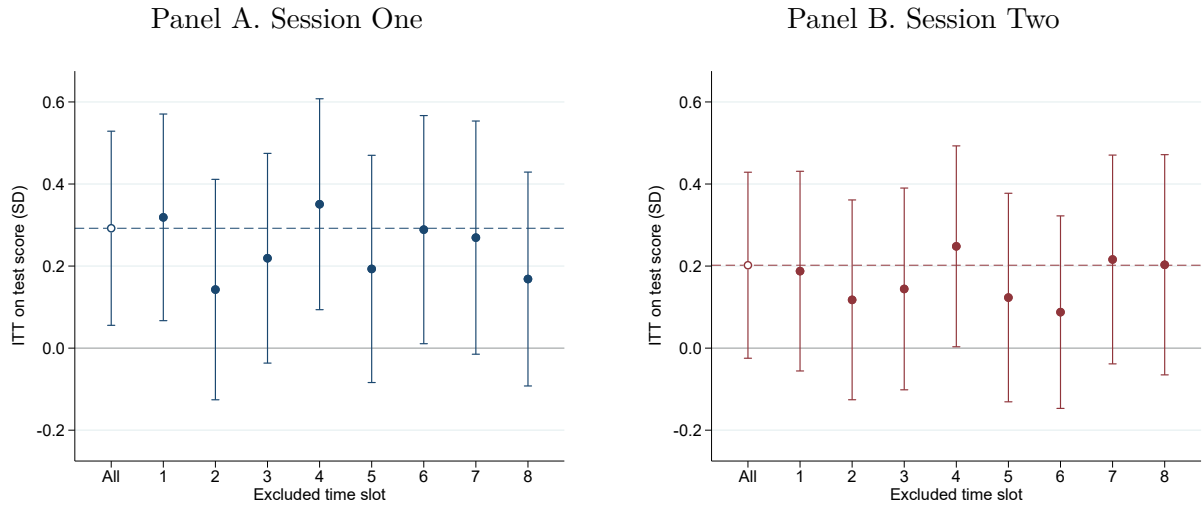
Date _____ Proctor 1 _____

Treatment _____ Proctor 2 _____

The seating chart for SDL 122 is a rectangular grid. At the top center is a horizontal line labeled "WHITEBOARD". Below it, 20 numbered computer workstations are arranged in two groups of 10. The first group (1-10) is on the left, and the second group (11-20) is on the right. Each workstation is represented by a small icon of a computer monitor. At the bottom of the chart, there are three boxes: "Proctor 1" on the left, "Proctor 2" in the middle, and "Door" on the right.

Notes: This figure shows the seating charts used by laboratory staff to monitor compliance during experimental sessions. Each numbered square represents a computer workstation. For each session time slot, one lab was randomly assigned to the AI-allowed condition and the other to the AI-forbidden condition. Proctors used these charts to record any instances of unauthorized resource use, with two proctors assigned to each laboratory. The physical separation of treatment conditions across different buildings helped prevent cross-contamination between experimental groups.

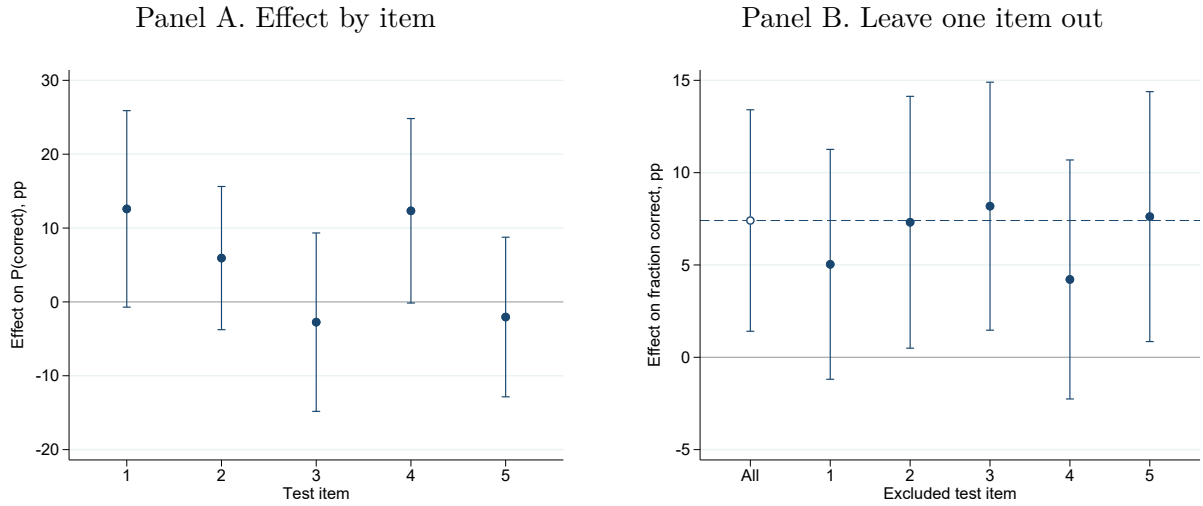
Figure A4: Test-Score Effects Excluding One Time Slot at a Time



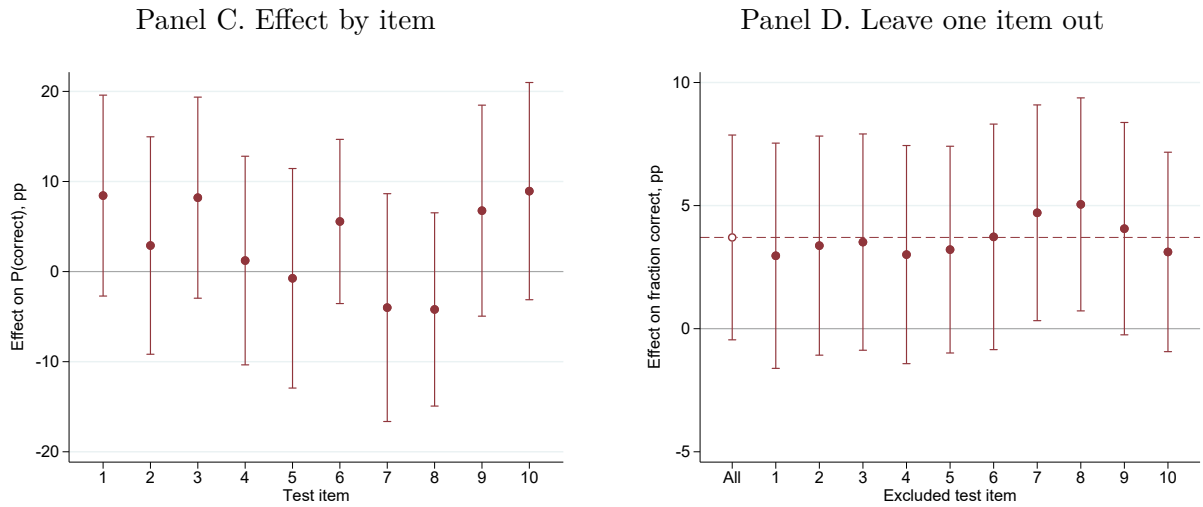
Notes: This figure re-estimates the test-score ITT for Session One (Panel A) and Session Two (Panel B) excluding one of the eight time slots at a time. The hollow marker keeps all slots and reproduces the estimates in Table 4; the dashed line marks that benchmark. Each estimate re-selects double-lasso controls on its subsample (Belloni et al., 2014) and includes strata fixed effects; bars are 95 percent confidence intervals from heteroskedasticity-robust standard errors. Excluding a slot leaves 173–190 students in Session One and 169–183 in Session Two.

Figure A5: Test-Score Effects by Item and Excluding One Item at a Time

Session One

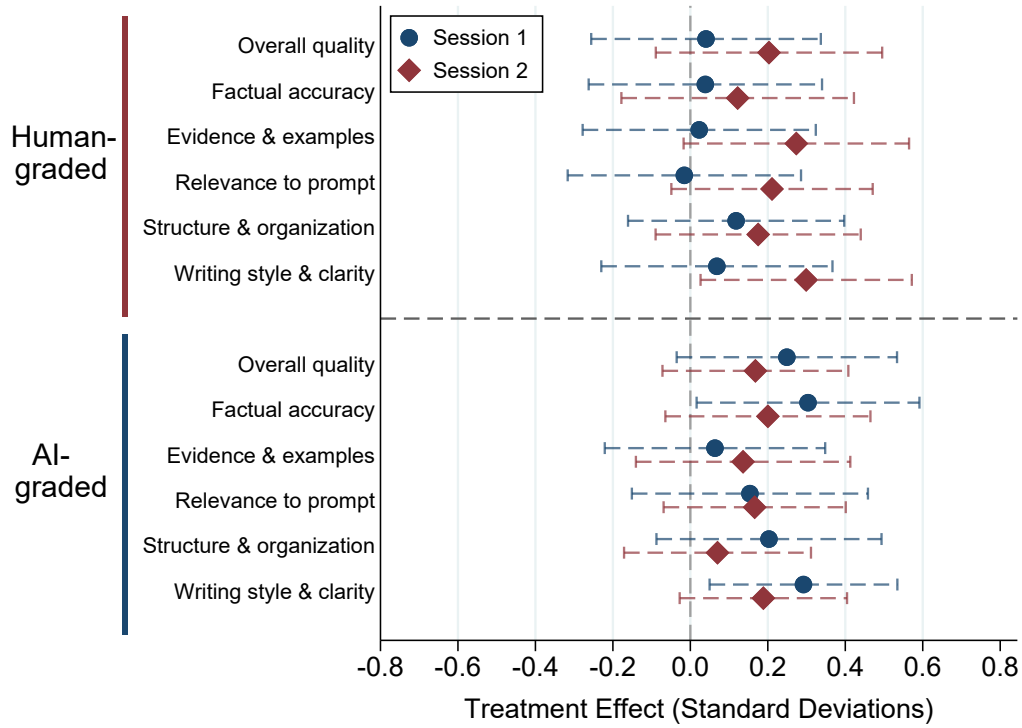


Session Two



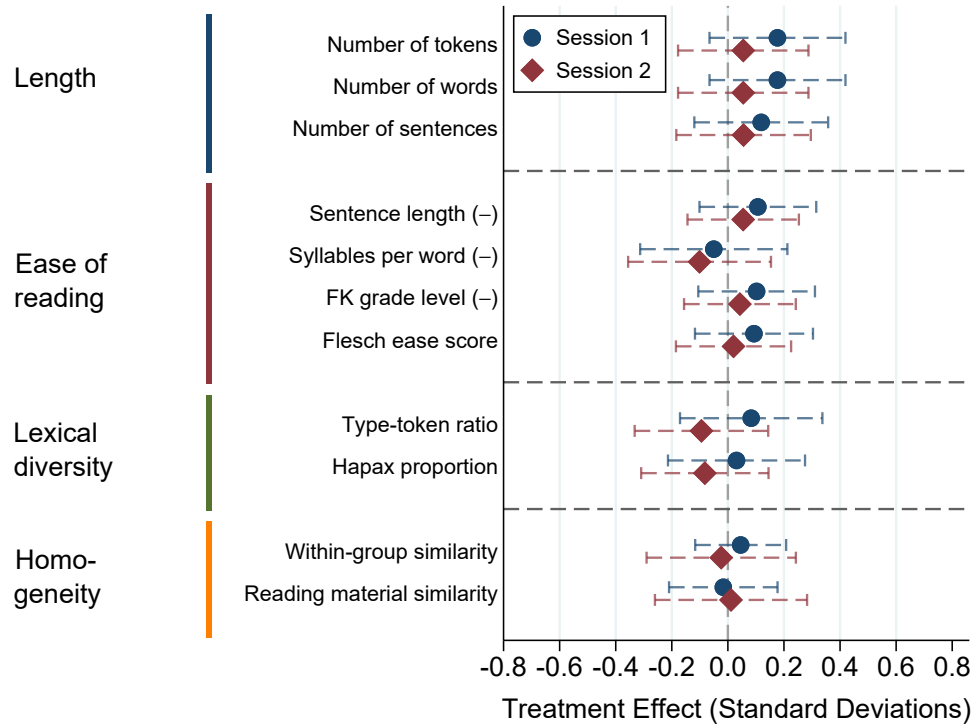
Notes: This figure decomposes the test-score effects of Table 4 by item. The Session One post-learning test has five items and the Session Two test has ten; item positions pool the three topics, each with its own fixed set of items. Panels A and C report the ITT on the probability of answering each item correctly, in percentage points. Panels B and D report the ITT on the fraction of items answered correctly when one item at a time is excluded; the hollow marker keeps all items and reproduces the percentage-point estimates in Table 4, and the dashed line marks that benchmark. Each estimate selects double-lasso controls for its own outcome (Belloni et al., 2014) and includes strata fixed effects; bars are 95 percent confidence intervals from heteroskedasticity-robust standard errors.

Figure A6: Effects of AI Access on Essay Quality, by Dimension



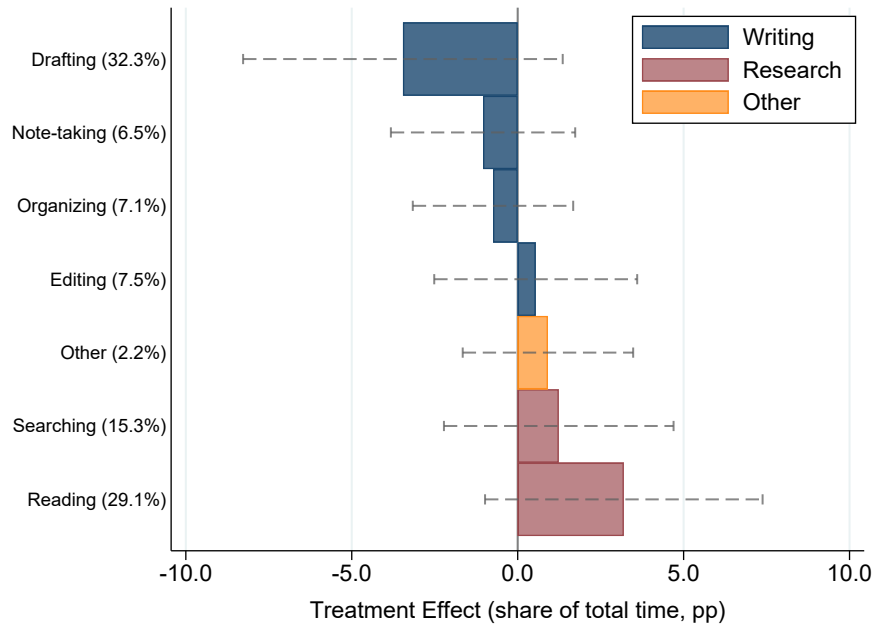
Notes: This figure presents treatment effects of AI access on individual essay quality dimensions, measured in standard deviations. The top panel shows human-graded dimensions: essays were evaluated by independent graders on five rubric dimensions plus an overall quality rating (factual accuracy, use of evidence and examples, relevance to the prompt, structure and organization, writing style and clarity, and overall quality). Human-graded regressions are at the student level, averaging each essay's scores across its graders. The bottom panel shows the same dimensions as scored by an AI grader (see Appendix B.4). Each dimension is scored on a scale from 0 to 10 and standardized to have mean zero and standard deviation one in the control group. Circles represent Session One effects (essays written with or without AI access); diamonds represent Session Two effects (essays written one week later without AI access). Horizontal lines represent 95 percent confidence intervals based on heteroskedasticity-robust standard errors.

Figure A7: Effects of AI Access on Essay Characteristics, by Dimension



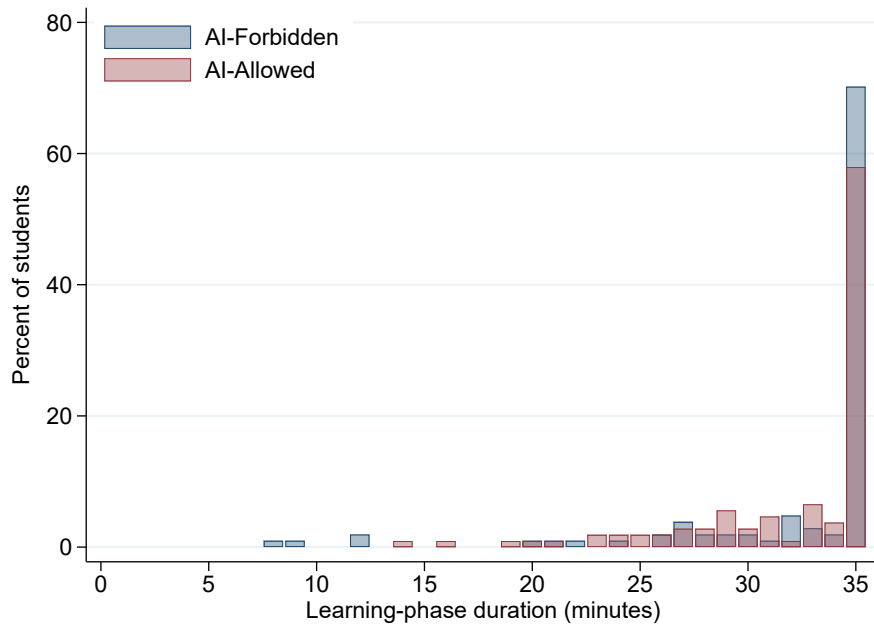
Notes: This figure presents treatment effects of AI access on individual essay characteristics, measured in standard deviations. Each variable is standardized to have mean zero and standard deviation one in the control group. Variables are grouped into four categories, indicated by colored vertical bars: length (number of tokens, words, and sentences), ease of readability (sentence length, syllables per word, Flesch-Kincaid grade level, and Flesch Reading Ease score, with difficulty measures reversed so that higher values indicate easier readability), lexical diversity (type-token ratio and hapax proportion), and homogeneity and similarity (within-group cosine similarity and reading material cosine similarity). Circles represent Session One effects; diamonds represent Session Two effects. Horizontal lines represent 95 percent confidence intervals based on heteroskedasticity-robust standard errors.

Figure A8: Mechanisms: Time Allocation Across Learning Activities



Notes: This figure presents treatment effects of AI access on self-reported time allocation across different learning activities, measured as each activity’s share of the student’s total self-reported learning time (in percentage points). Students reported how many minutes they spent on each activity during the 35-minute learning phase; each activity’s share is its minutes divided by the sum of minutes across all activities. The survey labels were: “Writing a first draft of your write-up” (Drafting), “Revising and editing your draft” (Editing), “Taking notes on key concepts and facts” (Note-taking), “Planning your writing task structure and arguments” (Organizing), “Finding information about the topic, searching for relevant sources” (Searching), “Reading and comprehending information about the topic” (Reading), and “Casually browsing the web, unrelated to the topic” and “Other activities” combined (Other). Navy bars represent writing activities, maroon bars represent research activities, and orange bars represent other activities. Bars represent the estimated treatment effect for each activity, with horizontal lines showing 95 percent confidence intervals. Numbers in parentheses indicate the control group mean share.

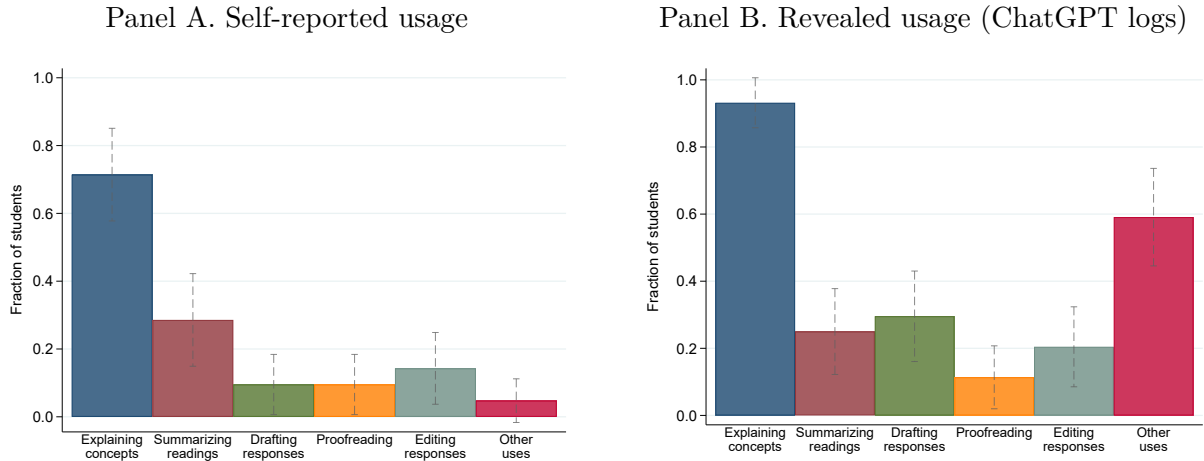
Figure A9: Distribution of Learning-Phase Duration by Treatment Arm



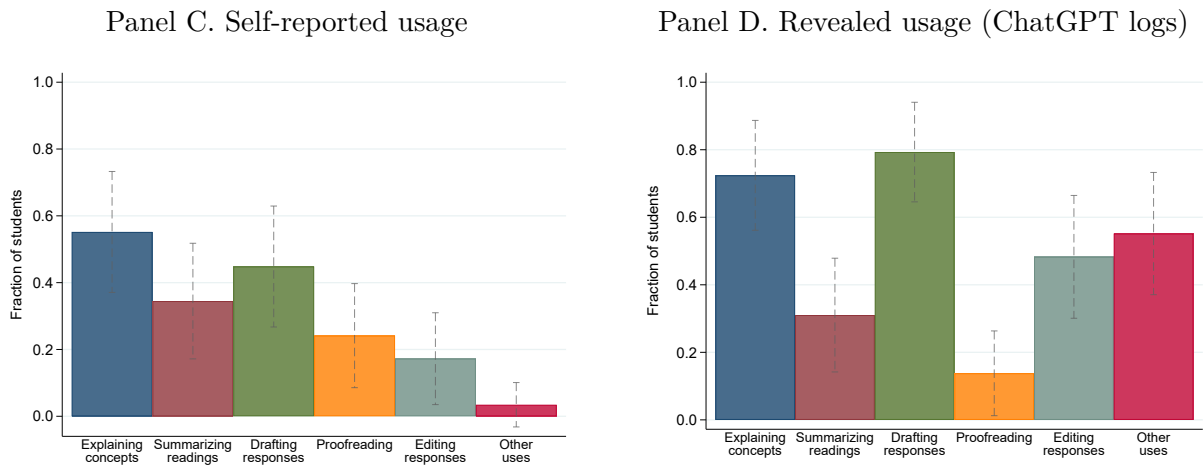
Notes: This figure plots the distribution of the Qualtrics-recorded duration of the Session One learning phase, by whole minute, separately for the AI-forbidden and AI-allowed conditions. The interface enforces a 35-minute maximum, and the bar at 35 collects every student who used the full time. The control mean and the treatment effect are reported in Table 6, Panel A.

Figure A10: Types of Generative AI Use by Automation and Augmentation Users

Augmentation users

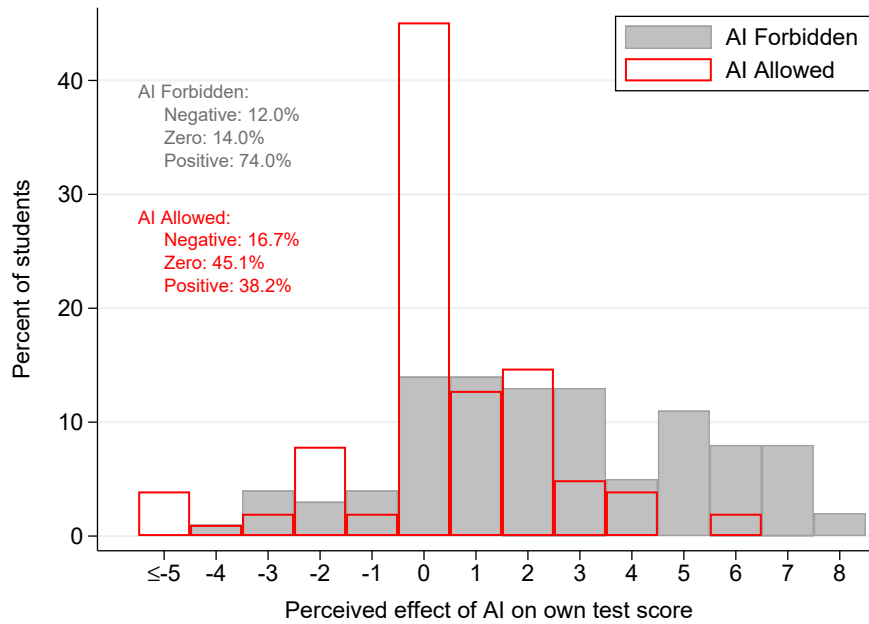


Automation users



Notes: This figure replicates the analysis from Figure 3, separately for augmentation and automation users as classified by Claude Opus (see Appendix B.6). Panels A and C present self-reported usage types from the exit survey; participants could select multiple categories. Panels B and D show the fraction of students with at least one prompt classified into each category from the actual ChatGPT conversation logs. Mixed users (with both augmentation and automation conversations) are included in both groups. Vertical bars denote 95 percent confidence intervals.

Figure A11: Distribution of Perceived Treatment Effects on Own Test Score



Notes: This figure plots the distribution of students' perceived treatment effect of AI access on their own Session Two test score, separately for AI-forbidden (control) and AI-allowed (treated) students. The perceived effect is each student's believed test score with AI access minus their believed score without it, expressed in number of questions answered correctly (for treated students, the believed own score minus the believed counterfactual; for control students, the believed counterfactual minus the believed own score). Values below -5 are bottom-coded at -5 . Bin width is 1 and bins are centered on integers. In-figure annotations report the share of each group reporting a negative, zero, or positive perceived effect.

Table A1: Determinants of AI Use Among AI-Allowed Students

	Bivariate regression		Multivariate regression	
	Used account (1)	Self- reported (2)	Used account (3)	Self- reported (4)
Panel A. Demographic characteristics				
Male	0.102 (0.102)	0.113 (0.110)	0.062 (0.110)	0.031 (0.109)
White	-0.202* (0.106)	-0.065 (0.112)	-0.183 (0.115)	0.051 (0.138)
International student	0.131 (0.114)	0.093 (0.127)	0.077 (0.145)	0.134 (0.146)
Private high school	0.007 (0.095)	-0.083 (0.106)	-0.102 (0.105)	-0.208** (0.100)
Panel B. Academic background				
Natural Sciences major	-0.063 (0.100)	-0.167 (0.106)	-0.103 (0.127)	-0.125 (0.129)
Social Sciences major	0.118 (0.096)	0.258** (0.101)	0.107 (0.119)	0.274** (0.125)
High GPA (at or above median)	-0.144 (0.090)	-0.114 (0.100)	0.007 (0.111)	0.010 (0.128)
Freshman	-0.143 (0.099)	-0.071 (0.109)	-0.226* (0.136)	-0.004 (0.147)
Sophomore	0.059 (0.105)	0.119 (0.125)	-0.119 (0.128)	0.034 (0.174)
Junior	-0.034 (0.124)	-0.016 (0.133)	-0.126 (0.140)	-0.085 (0.155)
Panel C. AI experience and beliefs				
Frequent AI user (at or above median)	0.357*** (0.113)	0.359*** (0.116)	0.300** (0.144)	0.349** (0.164)
Early AI adopter (before Fall 2024)	0.117 (0.096)	0.161 (0.103)	-0.038 (0.111)	-0.071 (0.131)
High AI proficiency (at or above median)	0.208** (0.096)	0.264** (0.103)	0.081 (0.112)	0.151 (0.146)
High perceived AI effect (at or above median)	-0.025 (0.121)	-0.171 (0.139)	0.079 (0.114)	-0.081 (0.128)
<i>N</i> (Students)	107	102	107	102

Notes: This table reports the association between student characteristics and AI use among treated students. Columns 1–2 are bivariate; columns 3–4 include all characteristics simultaneously. “Used account” (columns 1, 3) is based on ChatGPT server logs; “Self-reported” (columns 2, 4) is collected in Session Two. All variables are indicators. High GPA is at or above the sample median. Frequent AI user reports AI use during the semester at or above the median frequency (occasional or more frequent). Early AI adopter first used AI for academics before Fall 2024. High AI proficiency is self-assessed proficiency at or above the median. High perceived AI effect is a perceived treatment effect on test performance (believed score with AI minus believed counterfactual) at or above the median of 0 questions, so the reference group is the students who expect AI to lower their score. Missing characteristics are set to zero, and every specification includes an indicator for missingness. All specifications include session (strata) fixed effects. Heteroskedasticity-robust standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Table A2: Experimental Sample Versus the Middlebury Student Body

	Experimental sample (1)	Student body (2)
Female	0.635	0.533
Male	0.351	0.463
White	0.498	0.538
Black	0.085	0.052
Hispanic/Latino	0.242	0.124
Asian	0.284	0.073
International	0.265	–
Freshman	0.374	0.255
Sophomore	0.137	0.257
Junior	0.223	0.201
Senior	0.265	0.287
College GPA	3.683	3.670
Has not declared major	0.215	0.417
Natural-science major	0.308	0.244
Social-science major	0.365	0.243
Humanities/Arts/Languages major	0.104	0.145
<i>N</i> (Students)	211	2,760

Notes: Column 1 reports means among the Session One attendees; characteristics are self-reported at intake. Column 2 reports the Middlebury student-body figures compiled in Contractor and Reyes (2026), Table 1, from administrative enrollment records (Fall 2024) and the Middlebury Common Data Set 2024–25; the denominators there are 2,760 students for demographics, 2,738 for cohorts, and 3,158 declared degrees for fields of study; the Has not declared major row uses the 2,760 students as its denominator. The race rows are not strictly comparable: the sample indicators are self-reported and not mutually exclusive, while the administrative categories are exclusive and exclude international students (the sample is 26.5 percent international, and the administrative compilation reports no international share), so the Asian and Hispanic/Latino rows overstate the difference. College GPA is self-reported in column 1 and the registrar’s Spring 2024 grade point average in column 2. Fields of study in column 2 are classified by declared degree; the humanities row sums the compilation’s arts, humanities, languages, and literature categories.

Table A3: Effects of Generative AI Access on Academic Integrity

Outcome	Control mean (1)	ITT (2)	TOT (3)
Caught by proctor	0.029	0.057* (0.032)	0.081* (0.046)
Used unauthorized resources, Session One	0.010	0.096*** (0.032)	0.135*** (0.045)
Any integrity violation	0.039	0.136*** (0.043)	0.191*** (0.060)
<i>N</i>	102	205	205

Notes: This table reports treatment effects of AI access on rule violations during the Session One knowledge test. Each row reports the effect of being assigned to the AI-allowed group on the indicated outcome. Caught by proctor is an indicator for being flagged by a proctor during the Session One knowledge test. Used unauthorized resources, Session One is an indicator for answering “Yes, during Session 1 only” to the exit-survey question, asked at the end of Session Two, “Did you use any unauthorized resources or assistance (such as looking up answers online or using AI tools) while completing the multiple choice questions in either session?”; no student reported a violation in Session Two. Any integrity violation is an indicator for either. The knowledge tests prohibited all students from using external resources. Column 1 reports the control group mean. Column 2 reports intent-to-treat (ITT) estimates. Column 3 reports treatment-on-the-treated (TOT) estimates from two-stage least squares, instrumenting actual ChatGPT use with random treatment assignment. All specifications include controls selected through a double-lasso procedure (Belloni et al., 2014). Heteroskedasticity-robust standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Table A4: Effects of AI Access on Essay Writing Style

Outcome	Session One (with AI)			Session Two (without AI)		
	Control mean (1)	ITT (2)	TOT (3)	Control mean (4)	ITT (5)	TOT (6)
Length index	0.000	0.136 (0.119)	0.197 (0.173)	0.000	0.069 (0.114)	0.098 (0.164)
Readability index	0.000	0.061 (0.088)	0.087 (0.127)	0.000	0.045 (0.102)	0.064 (0.146)
Lex. diversity index	0.000	0.057 (0.126)	0.082 (0.181)	0.000	-0.092 (0.120)	-0.130 (0.170)
<i>N</i>	104	210	210	102	204	204

Notes: This table reports treatment effects of AI access on three z-scored writing style indices, standardized relative to the control group. Each row reports the effect of being assigned to the AI-allowed group on the indicated outcome. Length index averages the z-scores of tokens, words, and sentences. Readability index averages four z-scored measures (sentence length, syllables per word, Flesch-Kincaid grade level, and Flesch Reading Ease), with signs oriented so that higher values indicate easier readability. Lexical diversity index averages the z-scored type-token ratio and hapax proportion. Appendix Figure A7 reports results for the individual characteristics. Columns 1 and 4 report the control group mean. Columns 2 and 5 report intent-to-treat (ITT) estimates. Columns 3 and 6 report treatment-on-the-treated (TOT) estimates from two-stage least squares, instrumenting actual ChatGPT use with random treatment assignment. All specifications include controls selected through a double-lasso procedure (Belloni et al., 2014). Heteroskedasticity-robust standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Table A5: Effects of AI Access on Essay Quality, by Dimension

Outcome	Session One			Session Two		
	Control	ITT	TOT	Control	ITT	TOT
	mean (1)	(2)	(3)	mean (4)	(5)	(6)
Accuracy	6.745	0.240 (0.170)	0.343 (0.242)	5.204	0.261 (0.195)	0.380 (0.289)
Evidence	6.143	0.076 (0.223)	0.109 (0.320)	4.095	0.319 (0.199)	0.459 (0.292)
Relevance	6.957	0.112 (0.213)	0.162 (0.308)	6.073	0.325 (0.203)	0.454 (0.287)
Organization	5.930	0.279 (0.206)	0.402 (0.298)	5.310	0.204 (0.211)	0.292 (0.302)
Writing style	6.475	0.276 (0.182)	0.396 (0.263)	5.806	0.337* (0.176)	0.485* (0.260)
<i>N</i>	104	210	210	97	198	198

Notes: This table reports treatment effects of AI access on the five essay quality dimensions, averaging human and AI grades. Each row reports the effect of being assigned to the AI-allowed group on the indicated dimension. Each dimension is scored on a 0–10 scale. **Each row is one student-level regression on the average of the human graders' mean score and the AI grader's score for that dimension.** Columns 1–3 report Session One results (when AI-allowed students had access to ChatGPT); columns 4–6 report Session Two results (when all students wrote without AI access). Intent-to-treat (ITT) estimates report the effect of being assigned to the AI-allowed group; treatment-on-the-treated (TOT) estimates instrument ChatGPT use with random assignment. All specifications include controls selected through a double-lasso procedure (Belloni et al., 2014). Heteroskedasticity-robust standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Table A6: Baseline Characteristics and Unadjusted Outcomes by Type of AI Use

	Augmentation (1)	Automation (2)	Mixed (3)	Other uses (4)	Non-user (5)	AI-Forbidden (6)
Panel A. Baseline characteristics						
Age	20.4	20.2	20.5	22.1	19.9	19.9
Male (percent)	18.4	47.8	16.7	62.5	21.9	41.3
White (percent)	42.1	47.8	16.7	37.5	62.5	51.9
International (percent)	21.1	21.7	66.7	25.0	12.5	31.7
College GPA	3.64	3.60	3.78	3.81	3.67	3.71
SAT score	1325	1362	1420	1478	1434	1391
Study hours per week	16.1	16.2	18.2	19.6	16.7	16.7
Baseline self-assessed knowledge (0–10)	1.3	1.3	1.2	2.0	1.7	1.6
Baseline fraction correct (percent)	26.8	41.7	26.7	20.0	28.8	30.6
Panel B. Unadjusted outcome means						
Fraction correct, Session One (percent)	62.1	68.7	63.3	51.4	58.8	56.3
Test score, Session One (SD)	0.227	0.487	0.276	−0.194	0.095	0.000
Fraction correct, Session Two (percent)	49.4	53.5	48.3	51.3	56.0	50.0
Test score, Session Two (SD)	−0.030	0.190	−0.091	0.068	0.327	0.000
<i>N</i> (Students)	38	23	6	8	32	104

Notes: This table reports unadjusted means by how students in the AI-allowed condition used the provided ChatGPT account, with the AI-forbidden group in column 6 as the reference. Use types come from the LLM conversation classification (Appendix B.6): a student is an augmentation (automation) user if the account’s conversations include augmentation (automation) but not the other type; mixed if the conversations include a mixed conversation or both types; other if every conversation is classified other; and a non-user if the account has no conversation. Panel A reports characteristics measured at intake and the baseline test; Panel B reports unadjusted means of the test outcomes, with test scores standardized to the AI-forbidden group, and the Session Two rows average over the students in each group who attended Session Two. Use type is chosen by the student, so differences across columns reflect selection as well as any effect of the use type.

Table A7: Treatment Effects on Belief Accuracy

Outcome	Control mean (1)	ITT (2)	TOT (3)
Perceived AI gain on self (pp)	24.800	−21.512*** (3.616)	−30.350*** (5.416)
Perceived AI gain on others (pp)	14.060	−3.117 (3.102)	−4.410 (4.382)
Absolute deviation from estimated effect, self (pp)	28.221	−13.209*** (2.541)	−18.700*** (3.724)
Absolute deviation from estimated effect, others (pp)	21.368	−3.579 (2.188)	−5.050 (3.101)
<i>N</i>	100	202	202

Notes: This table reports treatment effects of AI access on belief accuracy. The sample is students who attended Session Two. Outcomes are in percentage points: “Perceived AI gain on self” is the signed predicted treatment effect on the student’s own Session Two test score; “Perceived AI gain on others” is the predicted effect on others’ scores; the “Absolute deviation from estimated effect” rows report $|\text{perceived gain} - \text{estimated ITT}|$, where the estimated ITT is the Session Two estimate from Table 4; these rows measure the distance of a student’s perceived gain from the average estimated effect, not an individual forecasting error. Column (1) reports the control-group mean of each outcome; columns (2) and (3) report ITT and TOT estimates. All specifications include controls selected through a double-lasso procedure (Belloni et al., 2014) with strata fixed effects. Heteroskedasticity-robust standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

B Empirical Appendix

B.1 Reading Materials

We wrote original texts on each of the three topics, designing them so that participants faced similar cognitive demands across topics. Each text follows a parallel structure, beginning with an overview of the technology, followed by its historical development, technical underpinnings, core applications, and limitations. The texts are information-rich but self-contained, requiring no prior knowledge of the topic. The full texts are available in the [Supplementary Materials](#).

The reading materials are 1,253–1,399 words long, or about 5.0–5.6 minutes of reading at an average silent reading speed of 250 words per minute for college students ([Carver, 1992](#); [Brysaert, 2019](#)). They are closely matched across several readability metrics (Appendix Table B1). Flesch-Kincaid grade levels of 15.1–15.9 place them at a late-undergraduate reading level. Measures of lexical and syntactic complexity, including sentence and word length, also indicate a consistent level of difficulty across topics.

Table B1: Readability Metrics for Topic Texts

Metric	Learning Topic:		
	Blockchain	Carbon Capture	CRISPR
Flesch-Kincaid Grade Level	15.90	15.76	15.11
SMOG Index	16.75	17.28	16.50
Automated Readability Index	17.75	16.26	15.86
Type-Token Ratio	0.51	0.49	0.51
Average Sentence Length	21.59	24.15	20.80
Average Word Length	6.43	6.09	6.07
Word Count	1,253	1,333	1,399
Estimated Reading Time (250 wpm)	5.0 min	5.3 min	5.6 min

Notes: This table presents readability metrics for the three reading materials. Estimated reading times assume an average silent reading speed of 250 words per minute for college students ([Carver, 1992](#); [Brysaert, 2019](#)).

B.2 Writing Assessment Prompts

This appendix presents the essay prompts we developed for each of the three topics. For each topic, we designed two complementary prompts of comparable difficulty. Each prompt requires participants to analyze relationships between concepts, evaluate the relative importance of different factors, and support their analysis with specific examples. Each

participant is randomly assigned to receive one prompt in Session One and the other in Session Two.

Blockchain Technology

Prompt 1. “Examine how blockchain has grown beyond cryptocurrency. Analyze which two application areas have the strongest potential for transformative impact and explain the specific factors that support your analysis. Draw on examples from the reading to develop your response.”

Prompt 2. “Examine key differences between blockchain technology and traditional databases. Analyze which specific features make blockchain unique, which types of applications benefit most from these differences, and why. Support your analysis with concrete evidence or examples.”

Carbon Capture

Prompt 1. “Analyze the three main barriers to scaling carbon capture technologies (technical limitations, economic costs, and policy challenges). Identify which barrier you believe is most critical to overcome first, how it impacts the other challenges, and what approaches might address it. Support your analysis with specific examples.”

Prompt 2. “Compare direct air capture with point-source carbon capture approaches. Analyze the specific advantages and limitations of each method, identify which contexts each approach is better suited for, and explain what factors most influence their effectiveness. Draw on evidence from the reading to support your analysis.”

CRISPR Gene Editing

Prompt 1. “Analyze how CRISPR differs from previous gene editing technologies in terms of precision, accessibility, and versatility. Identify which two differences have been most consequential for scientific advancement, explain why these particular differences matter, and provide specific examples that demonstrate their impact.”

Prompt 2. “Examine the three main technical challenges that limit CRISPR

today (delivery methods, off-target effects, and ethical considerations). Analyze which challenge is most important to address first, how it relates to the others, and what approaches show promise for overcoming it. Support your analysis with specific examples.”

B.3 Human Essay Grading

We recruit independent graders through Prolific to evaluate participants’ analytical essays, restricting eligibility to individuals holding a master’s or PhD degree. Graders are blind to treatment condition, participant identity, and all other experimental variables. Each grader evaluates five essays and is compensated with a \$15 fixed payment plus eligibility for a \$10 lottery bonus. To incentivize consistency, we tell graders that their chances of winning the bonus rise if their scores resemble those of other graders evaluating the same essays.

Before grading, each grader completes a comprehension check verifying their understanding of the task requirements and reviews two example essays with suggested scores and detailed rationales. The examples are chosen to illustrate different quality levels—one strong response (receiving scores of 6–9 across dimensions) and one weaker response (receiving scores of 4–8)—so that graders can calibrate their standards. Graders are instructed to read each essay carefully, consult external resources to verify factual claims if needed, and maintain consistent scoring standards across all five essays.

Graders evaluate each essay on five rubric dimensions plus an overall quality rating, each on a 0–10 scale. *Accuracy of Content* measures the factual correctness of the information presented in the essay. *Use of Evidence and Examples* assesses the integration of specific evidence, examples, and data from the reading to support arguments. *Relevance to the Prompt* evaluates how well the essay addresses the specific prompt and stays on topic throughout. *Organization and Structure* measures the logical structure, coherence, flow, and effectiveness of transitions. *Writing Style and Clarity* assesses clarity, readability, grammar, and precision of language. *Overall Essay Quality* captures the grader’s holistic assessment of the essay. Graders also report their prior familiarity with each of the three topics (blockchain, carbon capture, and CRISPR) on a 0–10 scale before beginning the grading task.

Each essay is evaluated by three to four independent graders.²⁹ We use the overall quality rating as our benchmark measure of essay quality and report results for each dimension

²⁹Two Session Two essays that were assigned to two grading batches received seven and eight ratings

separately.

B.4 AI Essay Grading

We complement our human grading with AI-generated scores to provide an independent assessment of essay quality. We use Anthropic’s Claude Opus 4.8 to grade each essay on the same five rubric dimensions plus the overall quality rating, using the same 0–10 scale as the Prolific graders.

For each essay, we make six separate API calls—one per grading dimension—to avoid cross-contamination between scores. Each call includes a system prompt establishing the grading context, the dimension-specific rubric, two calibration essays with suggested scores, and the student essay to be graded. The calibration essays are the same examples shown to human graders during training, ensuring that human and AI graders share a common scoring anchor. We constrain the model’s output to a JSON schema: a brief justification followed by an integer score on the 0–10 scale. Requiring the justification first lets the score reflect the model’s stated reasoning.

The system prompt is as follows:

You are an expert essay grader for a university research study. College students were given 35 minutes to learn about a topic and write a 300-500 word analytical essay. You must grade essays on a 0-10 scale following the rubric exactly. Be consistent and calibrated using the example essays and scores provided.

For each dimension, the user prompt includes the dimension name, a description of what the dimension measures, and the scoring scale. The dimensions and their scales are as follows:

Accuracy of Content. Evaluate the factual correctness and depth of understanding demonstrated in the essay. Consider whether key concepts are accurately described, whether technical details are correct, and whether the student shows genuine comprehension of the topic.

0-3: Major factual errors or fundamental misunderstanding of the topic. 4-6: Some general knowledge but oversimplified or vague; key technical distinctions are glossed over. 7-8: Strong factual understanding; correctly describes basic mechanics, benefits, and drawbacks with some nuance. 9-10: Demonstrates advanced, detailed, and precise understanding of the topic with no factual errors.

Use of Evidence & Examples. Assess whether the essay draws on evidence from the provided reading materials. Consider how specific and well-integrated the references are.

0-3: No evidence or examples from the reading. 4-6: Minimal or vague use of evidence; general claims with only vague references like "the reading says..." without integrating specific details. 7-8: Good use of evidence with some specific examples from the reading; references are relevant but could be more detailed. 9-10: Excellent integration of specific evidence and examples from the reading throughout the essay.

Relevance to the Prompt. Measure whether the essay directly addresses the prompt requirements. Consider whether all parts of the prompt are addressed and whether the essay stays focused throughout.

0-3: Does not address the prompt or is largely off-topic. 4-6: Partially addresses the prompt but misses key aspects or lacks depth. 7-8: Mostly answers the prompt and touches on each required area; stays focused throughout. 9-10: Directly and thoroughly addresses every aspect of the prompt with analytical depth.

Organization and Structure. Evaluate the essay's organization, clarity of argument flow, and coherence. Consider whether there is a clear introduction, body, and conclusion, and whether transitions between ideas are smooth.

0-3: No discernible structure; ideas are disorganized or incoherent. 4-6: Basic structure with intro and conclusion, but loosely organized; paragraphs may be short and somewhat repetitive. 7-8: Clear and mostly effective structure with introduction, comparison, context, and conclusion; transitions could be smoother. 9-10: Excellent organization with smooth transitions, logical progression, and well-developed paragraphs.

Writing Style and Clarity. Assess the clarity, readability, grammar, and language precision of the essay. Consider vocabulary, sentence variety, and overall polish of the writing.

0-3: Very poor writing with major grammar issues or incoherent prose. 4-6: Very basic style; sentences are short and repetitive, vocabulary is limited; reads more like notes or a summary than a full essay. 7-8: Mostly clear and readable with minor awkward phrasing; lacks some polish or precision in language but avoids major issues. 9-10: Excellent writing with varied sentence structure, precise vocabulary, and sophisticated prose.

Overall Essay Quality. Provide a holistic assessment of the essay's overall quality. Consider all previous dimensions together: accuracy, evidence, relevance, structure, and writing style.

0-3: Very poor quality; fails to demonstrate understanding or engage with the topic meaningfully. 4-6: Basic understanding demonstrated but lacks depth, sophistication, and engagement with the reading. 7-8: Solid response that answers the prompt with reasonable clarity and structure, though it may lack depth in evidence or style. 9-10: Exceptional quality; demonstrates deep understanding, excellent writing, strong evidence use, and thorough analysis.

The prompt also includes two calibration essays with suggested scores for the relevant dimension. After presenting the calibration material, the prompt provides the student's assigned topic, the exact prompt the student was shown in that session, and the essay text. Students saw a different prompt in each of the two sessions, so grading against the session-specific prompt ensures the relevance dimension reflects the task the student actually faced. Empty essays, or those with fewer than 20 characters, are skipped and recorded as missing without querying the model.

B.5 SAT Score Imputation

We lack admissions test scores (SAT or ACT) for approximately 26 percent of participants (Middlebury College has test-optional admissions). To include these students in our regressions, we impute SAT scores for students with missing scores using baseline covariates and assign them to decile bins alongside students with observed scores.

We estimate an OLS regression of SAT scores on baseline characteristics among the 189 students with observed scores. The predictors are college GPA, high school GPA, age, gender, race and ethnicity indicators (Black, Latino, Asian, with white as the omitted

category), international student status, public and private high school indicators, academic field indicators (natural sciences, social sciences, humanities and arts), weekly study hours, whether the student has declared a major, cohort fixed effects, baseline self-assessment of topic knowledge, and baseline number of test questions answered correctly (out of five).³⁰

We include a missing indicator (equal to one for imputed observations) in the pool of potential controls, so the double-lasso procedure can account for any residual differences between observed and imputed groups. The imputed SAT scores are used only for bin assignment; they do not enter the regression directly as a continuous covariate.

B.6 Classification of ChatGPT Conversations

During Session One, participants in the AI-allowed condition interacted with ChatGPT through monitored accounts, generating conversation logs that we classified using an LLM (Anthropic’s Claude Opus 4).

Prompt-Level Classification. We classify each student prompt into one of six categories that mirror the self-reported usage types from Figure 3: (1) *Explaining concepts*—asking ChatGPT to explain, clarify, or teach a concept from the reading; (2) *Summarizing readings*—asking ChatGPT to summarize or condense the reading material; (3) *Drafting responses*—asking ChatGPT to write, draft, or generate essay text; (4) *Proofreading*—asking ChatGPT to check grammar, spelling, or typos; (5) *Editing responses*—asking ChatGPT to revise, improve, rephrase, or restructure existing text; and (6) *Other*—anything that does not fit the above categories.

For each prompt, we provide the LLM with the full system prompt establishing the experimental context—that students were given a reading passage on a science topic and asked to write an essay—along with descriptions of each category. The classifier returns a single category label. Across all 435 student prompts, the distribution is: Explaining concepts (44 percent), Other (26 percent), Drafting responses (11 percent), Editing responses (10 percent), Summarizing readings (7 percent), and Proofreading (2 percent).

The exact system prompt provided to the LLM is:

³⁰Predictors with missing values enter as zero together with an indicator for missingness, so all 189 students with an observed score contribute to the regression and every student without a score receives a prediction.

You are classifying student prompts sent to ChatGPT during a learning experiment. Students were given a reading passage on a science topic (Blockchain, CRISPR, or Carbon Capture) and asked to write an essay. Some students had access to ChatGPT during this process.

Classify each student prompt into exactly ONE of these categories:

1. Explaining concepts -- Asking ChatGPT to explain, clarify, or teach a concept from the reading
2. Summarizing readings -- Asking ChatGPT to summarize or condense the reading material
3. Drafting responses -- Asking ChatGPT to write, draft, or generate essay text
4. Proofreading -- Asking ChatGPT to check grammar, spelling, or typos
5. Editing responses -- Asking ChatGPT to revise, improve, rephrase, or restructure existing text
6. Other -- Anything that does not fit the above (e.g., off-topic chat, testing the tool)

Respond with ONLY the category name, exactly as written above. Nothing else.

To illustrate, Appendix Table [B2](#) provides representative examples of student prompts classified under each category.

Table B2: Examples of Student Prompts by Classification Category

Category	Example prompt
Explaining concepts	“easy definition and understanding of crispr” “how might the cas9 attach to the wrong spot in crispr” “Give a brief overview on CRISPR technology and then explain why CRISPR differs from other gene editing technology...”
Summarizing readings	“summary with bullet points and main takeaways from this document...” “Can you summarize the difference between direct air capture and point source capture”
Drafting responses	“writte approximately 500 word response to the following prompt: analyze the three main barriers to scaling carbon capture technologies...” “Write a response paper of at least 500 words...”
Editing responses	“Make this more approachable and simpler for a regular reader” “rewrite the whole thing now”
Proofreading	“use appropriate grammer and capitalization”
Other	“keyboard shortcuts to copy and paste” “i think that policy challenges is most difficult what with trump in office and everything”

Notes: This table presents example ChatGPT prompts illustrating the classification categories. Each row shows a verbatim student prompt from the ChatGPT conversation logs. Prompts are reproduced exactly as typed, including spelling and grammatical errors.

Conversation-Level Classification. We also classify each full conversation—the complete sequence of student and ChatGPT messages—into one of four categories: (1) *Automation*, where the AI does the work for the student (generating essay text, producing draft paragraphs, or creating content that could be pasted directly into the essay); (2) *Augmentation*, where the AI works with the student (explaining concepts, answering clarifying questions, or providing feedback on the student’s own writing); (3) *Mixed*, where the conversation contains substantial elements of both; and (4) *Other*, where the conversation is off-topic, unrelated to the academic task, or consists of testing or experimenting with ChatGPT without engaging with the reading or essay.

For each conversation, we provide the LLM with the full exchange between the student and ChatGPT, along with descriptions of each category. Across all ChatGPT conversations, the distribution is: Augmentation (49 percent), Automation (31 percent), Mixed (5 percent), and Other (15 percent).

To illustrate the distinction, we reproduce excerpts from conversations classified under

each category. Appendix Figures [B1](#) and [B2](#) show selected exchanges; full transcripts are available via the hyperlinked URLs.

The exact system prompt provided to Claude Opus 4 is:

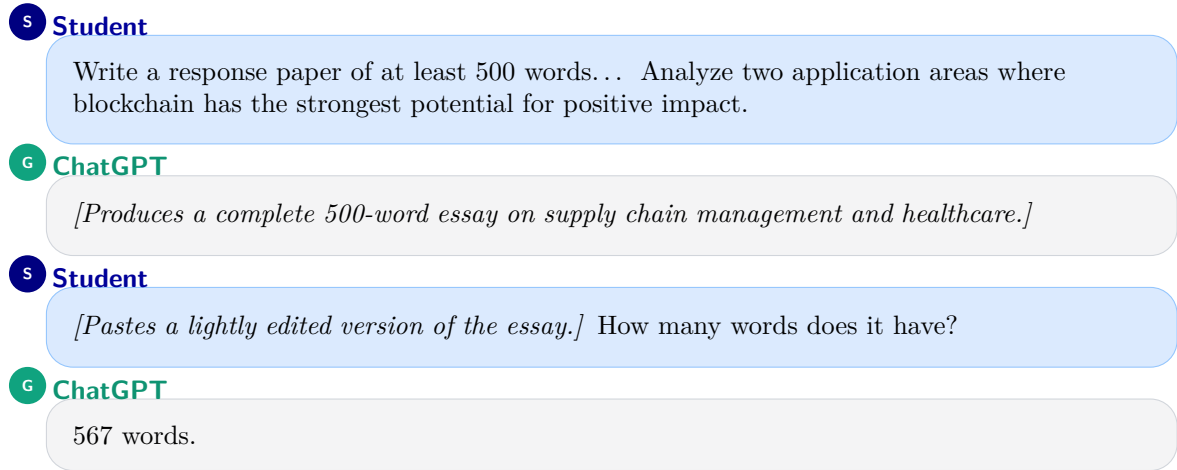
You are classifying a full ChatGPT conversation from a learning experiment. Students were given a reading passage on a science topic (Blockchain, CRISPR, or Carbon Capture) and asked to write an essay. Some students had access to ChatGPT during this process.

Classify the overall conversation into exactly ONE of these categories:

1. Automation -- The AI is doing the work FOR the student. The student asks ChatGPT to generate essay text, produce draft paragraphs, write responses, or create content that the student could paste directly into their essay. The student outsources the writing/thinking to the AI.
2. Augmentation -- The AI is working WITH the student. The student uses ChatGPT to understand concepts, get explanations, ask clarifying questions, get feedback on their own writing, or check their work. The student retains ownership of the thinking and writing process.
3. Mixed -- The conversation contains substantial elements of both automation and augmentation, making it hard to assign a single label.
4. Other -- The conversation is off-topic, unrelated to the academic task, or consists of testing/experimenting with ChatGPT without engaging with the essay or reading material.

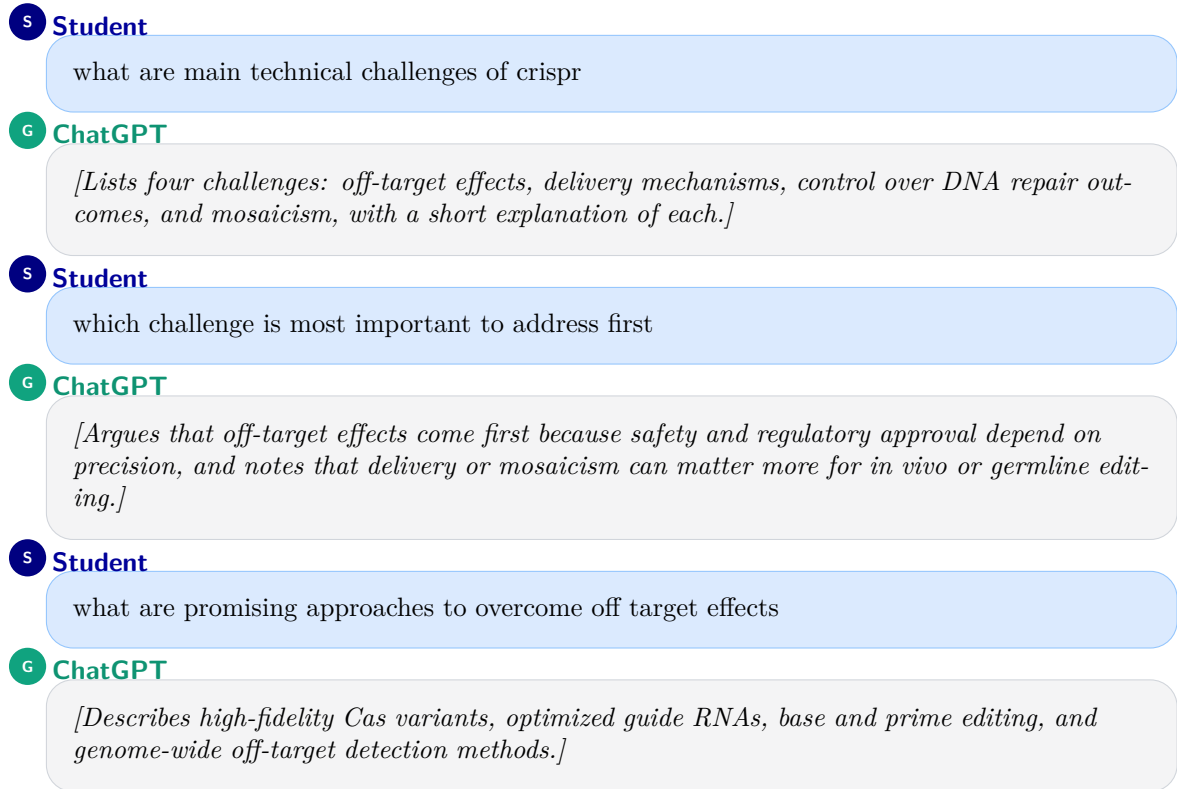
Respond with ONLY the category name (Automation, Augmentation, Mixed, or Other). Nothing else.

Figure B1: Automation Example: Student Delegates Writing to ChatGPT



Notes: This figure shows an excerpt of a ChatGPT conversation classified as automation. Full transcript: <https://chatgpt.com/share/681d3c21-9844-8012-b4f0-6a573222bd5b>.

Figure B2: Augmentation Example: Student Uses ChatGPT as an Explainer



Notes: This figure shows an excerpt of a ChatGPT conversation classified as augmentation (session 8, seat 17, conversation 1 in the study logs). The student's three prompts are verbatim; the bracketed text summarizes ChatGPT's replies.

Use in Heterogeneity Analysis. We use the conversation-level classification to construct student-level indicators of AI use type. A student is classified as an *automation user* if any of their conversations was labeled as Automation or Mixed, and as an *augmentation user* if any was labeled as Augmentation or Mixed. These groups are neither mutually exclusive nor collectively exhaustive: students with both types of conversations appear in both subsamples, and the “Other” category is excluded from the heterogeneity analysis. These indicators are defined only for treated students who used ChatGPT; control students are excluded from the classification by construction. In the heterogeneity analysis (Table 7), we compare each subgroup of treated students separately against the full control group.

B.7 Literature Comparison

We surveyed experimental papers on generative AI and learning outcomes published or circulated between 2023 and 2026. We required that each study satisfy seven criteria: (1) a randomized experimental design; (2) an effect size reported in standard deviations or convertible from reported statistics; (3) a standard error or confidence interval (reported or back-calculable); (4) an outcome measuring learning on an unassisted assessment, excluding tasks where AI was available during evaluation; (5) a clean no-AI control, rather than an active comparison such as human tutoring or structured hints; (6) a total sample size of at least 50 participants; and (7) a sample composed of students, rather than working professionals or general online participants.

Appendix Table B3 summarizes the 14 included studies; we describe each below.

Table B3: Included Studies in Literature Comparison

Study	Domain	N	Design	Outcome	Estimates
Ba et al. (2024)	Medicine	77	RCT	Unassisted exam	1
Bassner et al. (2026)	Coding	275	RCT	Unassisted post-test	2
Bastani et al. (2025)	Math	839	Cluster RCT	Unassisted exam	2
Dai et al. (2025)	Physics	349	RCT ($\times 2$)	Unassisted exam	3
De Simone et al. (2025)	Language	759	RCT	Unassisted post-test	2
Fischer et al. (2025)	Economics	334	Lab RCT	Unassisted exam	2
Franco et al. (2026)	Language	572	Lab RCT	Unassisted exam	2
Gan et al. (2024)	Medicine	110	RCT	Unassisted exam	1
Hou et al. (2026)	Engineering	95	RCT	Unassisted post-test	2
Huang et al. (2025)	Medicine	187	RCT	Unassisted skill test	1
Kavadella et al. (2024)	Medicine	70	RCT	Unassisted exam	1
Kazemitabaar et al. (2023)	Coding	69	RCT	Post-test + 1-wk retention	2
LearnLM Team (2026)	Math	1,423	Cluster RCT	Unassisted exam	1
Lehmann et al. (2025)	Coding	176	Lab RCT ($\times 2$)	Unassisted post-test	2

Notes: This table lists the studies included in our literature comparison. N is the total sample across all arms included in our comparison. “Estimates” indicates the number of point estimates included in Figure 5. All studies compare AI access to a no-AI control on unassisted assessments.

Ba et al. (2024) randomized 77 medical interns rotating through pediatric cardiology at Sun Yat-sen University to two weeks of ChatGPT-assisted instruction or standard bedside teaching with identical cases and instructors. Scores on a closed-book knowledge exam were statistically indistinguishable (-0.07 SD), with both groups near the test ceiling. We computed the effect size and standard error from reported means and standard deviations.

Bassner et al. (2026) randomized 452 introductory-programming students at TU Munich to a 90-minute Java exercise with a scaffolded LLM tutor (Iris), unrestricted Chat-

GPT, or no AI (275 analyzed after exclusions). Both AI arms scored higher on the exercise itself, but on the supervised, unassisted knowledge test neither arm learned more than the control: baseline-adjusted gains are -0.07 SD (Iris) and -0.01 SD (ChatGPT)—the paper’s titular dissociation between performance and learning. We standardized the difference in pre–post gains by the pooled pre-test standard deviation.

[Bastani et al. \(2025\)](#) conducted a cluster-randomized trial with 839 Turkish high school students learning math on an online platform. Students were assigned to base GPT-4 access, GPT-4 with a tutoring prompt that withheld direct answers, or a no-AI control, and took an unassisted math exam after the practice period. Although base GPT-4 students solved more practice problems during the AI-assisted phase, this advantage reversed on the unassisted exam: the base group scored -0.19 SD relative to the control, while the tutor-prompted group scored -0.01 SD. We converted the authors’ raw-scale standard errors to SD units by dividing by the control-arm standard deviation (0.277).

[Dai et al. \(2025\)](#) ran two randomized experiments with grade-10 physics students at a Chinese high school (Experiment 1: $N = 121$; Experiment 2: $N = 266$), providing LLM-generated feedback on regular homework over five weeks. In Experiment 1, treated students received recommended problems with AI heuristic hints; in Experiment 2, students could request AI help on demand while studying, either choosing the feedback type themselves or receiving the system’s choice. Effects on the unassisted end-of-term physics exam are small and insignificant in all three AI arms relative to no-intervention controls (0.03 – 0.21 SD). Exam scores are standardized, so regression coefficients are effect sizes directly.

[De Simone et al. \(2025\)](#) studied 759 secondary school students in Nigeria randomly assigned to practice English language skills with an AI chatbot or a no-AI control. Students showed gains on unassisted post-tests in English skills (0.24 SD) and on the school’s regular third-term curricular exam at the end of the intervention (0.21 SD). They also report a “total” estimate bundling AI and digital literacy skills directly taught by the intervention; we exclude it because it captures treatment-specific content rather than general learning.

[Fischer et al. \(2025\)](#) randomized 334 university students in Berlin to study two microeconomics textbook excerpts with a GPT-4-based tutor grounded in the readings—available either throughout the session or only after ten minutes of independent reading—or with the textbook alone. On an incentivized, unassisted 25-question exam, unrestricted AI tutoring raised performance by 0.34 SD, while restricted access produced a statistically insignificant 0.13 SD. Test scores are standardized, so the reported coefficients are effect sizes directly.

[Franco et al. \(2026\)](#) randomized 572 students at the University of Nottingham to study

Esperanto in a proctored lab session—a 15-minute lesson followed by about 20 minutes of practice questions—with a browsing-only control, unrestricted ChatGPT access, or ChatGPT access plus brief written guidance on learning-oriented use. On an incentivized, unassisted 15-question exam, neither AI arm differs from the control (0.04 SD for unrestricted access, -0.12 SD for guided access), and students with AI attempt fewer practice questions.³¹

Gan et al. (2024) randomized 125 third-year medical students at Jinan University to a one-week orthopedics review using ChatGPT-4 or conventional internet resources, 110 of whom were analyzed after 15 withdrawals. The ChatGPT group scored 0.40 SD higher on an unassisted 214-item orthopedics exam. We computed the effect size and standard error from reported means and standard deviations.

Hou et al. (2026) randomized 95 undergraduates with no prior construction-related coursework to review construction-engineering material with a generative-AI assistant—with or without a structured prompting framework—or with lecture slides only. On an unassisted post-test combining multiple-choice and open-ended items, guided AI use raised total scores by 0.86 SD, while unguided use produced a near-zero effect (0.09 SD). We computed effect sizes and standard errors from reported means and standard deviations.

Huang et al. (2025) randomized 187 dental students at Wuhan University to one week of operative-skills training with instructional videos plus ChatGPT-3.5 or the same videos alone. The ChatGPT group scored 0.67 SD higher on an unassisted virtual-reality skill assessment. We computed the effect size and standard error from reported means and standard deviations.

Kavadella et al. (2024) randomized 77 second-year dental students at European University Cyprus to complete a radiation-biology assignment using ChatGPT or conventional literature search. On an unannounced, unassisted ten-question exam, which 70 of the 77 sat, the ChatGPT group scored 0.52 SD higher. We computed the effect size and standard error from reported means and standard deviations.

Kazemitabaar et al. (2023) ran a 10-session randomized trial with 69 K-12 learners (ages 10–17, novice programmers) using OpenAI Codex to support Python authoring tasks. AI access produced large performance gains during training (1.67 SD) but no detectable difference on an immediate unassisted post-test (-0.05 SD; the baseline group scored

³¹The paper reports coefficients on the 0–15 exam scale but not the score’s standard deviation; we standardize by a standard deviation of 1.9, backed out from the residual standard deviation implied by the reported regression standard errors and consistent with the 1.7–2.0 range implied by the control group’s mean and tail shares under normality.

slightly higher, 62.9 versus 61.3 percent) and only a marginal difference on a one-week retention test of code modification (0.41 SD). Standard errors were back-calculated from the reported N and standardized effect size.

[LearnLM Team \(2026\)](#) conducted a preregistered trial randomizing 48 mathematics classrooms (1,763 grade 7–8 students enrolled, 1,423 in the analyzed balanced panel) in Sierra Leone to integrate Gemini’s Guided Learning feature into half of weekly lessons for eight weeks or to continue standard instruction. The intent-to-treat effect on an unassisted, IRT-scaled endline mathematics assessment is 0.26 SD, with standard errors clustered at the classroom level.

[Lehmann et al. \(2025\)](#) ran two laboratory experiments with German university students learning Python programming (Study 2: $N = 107$; Study 3: $N = 69$). In both, treated students used ChatGPT during practice and then completed a 20-question unassisted coding post-test. Effects are 0.25 SD (Study 2) and 0.42 SD (Study 3), neither individually significant at conventional levels. Study 2 included an unintended copy-paste restriction that may have affected the treatment.

The random-effects grand mean in Figure 5 uses [DerSimonian and Laird \(1986\)](#) weights, accounting for between-study heterogeneity through a variance component estimated from study-level effects. The pool is the 24 literature estimates, aggregated within each study by inverse-variance weighting so that each of the 14 studies enters once (our own estimates are excluded).

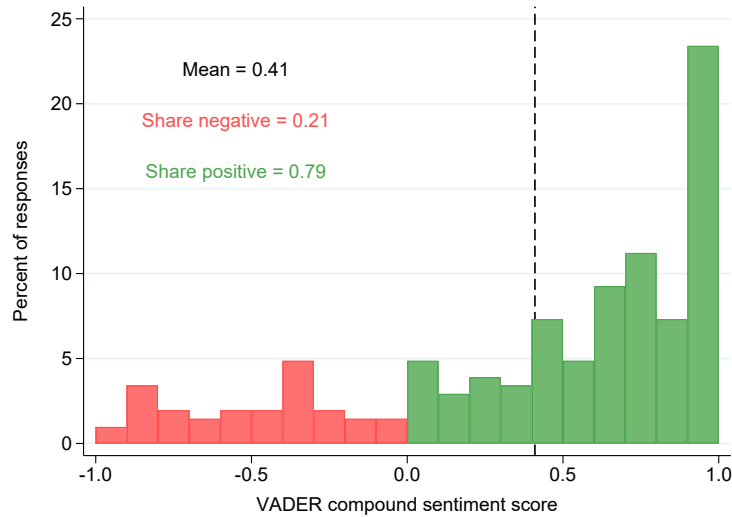
C Student Beliefs About AI and Learning

This section analyzes student responses to an open-ended question included in the Session Two survey: “In your opinion, how does generative AI (e.g., ChatGPT) affect student learning in college? Please explain your reasoning.” All 205 students who attended both sessions answered the question. We analyze these responses in two ways: sentiment analysis to show the measure carries signal rather than noise, and causal-graph coding, following [Andre et al. \(2026\)](#), to characterize how students reason about AI and learning.

C.1 Validating the Open-Ended Response Measure

We validate our open-ended response measure using VADER (Valence Aware Dictionary and sEntiment Reasoner), a lexicon-based sentiment analysis tool ([Hutto and Gilbert, 2014](#)). VADER assigns each response a compound sentiment score ranging from -1 (most negative) to $+1$ (most positive). Appendix Figure C1 plots the distribution of these scores. Responses skew positive, with most students expressing net-positive sentiment about AI’s effect on learning and a minority expressing negative sentiment.

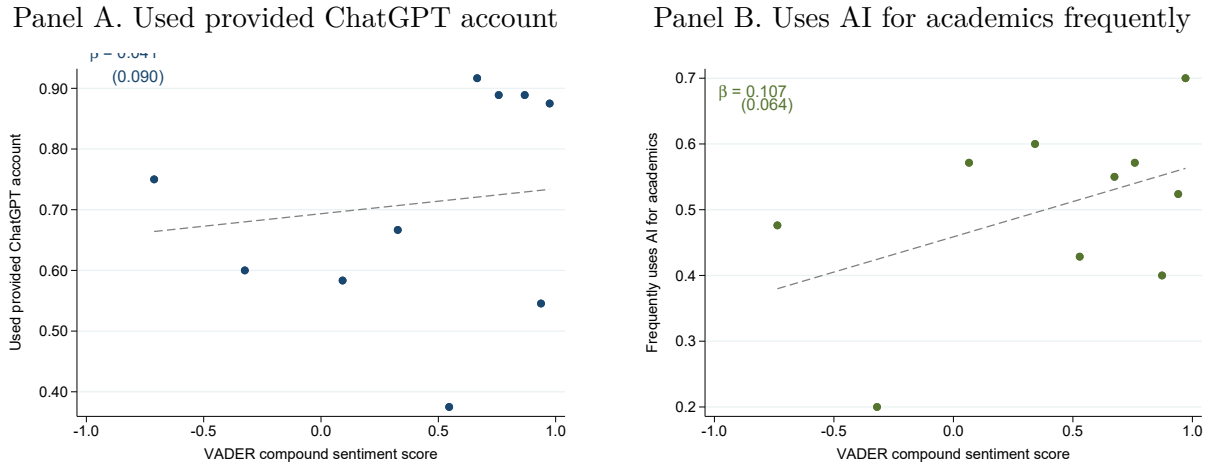
Figure C1: Distribution of AI Sentiment Scores



Notes: This figure plots the distribution of compound sentiment scores across students who responded to the open-ended question about how generative AI affects student learning. Scores are computed using [Hutto and Gilbert \(2014\)](#)’s VADER algorithm and range from -1 (most negative) to $+1$ (most positive). The dashed vertical line marks the mean.

To assess whether the open-ended responses carry a relevant signal, we test whether their sentiment tracks students’ actual AI use. Sentiment is weakly related to both measures. Appendix Figure C2 presents binned scatterplots relating compound sentiment to two measures of AI use. Panel A relates sentiment to whether an AI-allowed student used AI during Session One. A regression of AI use on the sentiment score yields a positive but statistically insignificant coefficient ($\hat{\beta} = 0.041$, $p = 0.647$). Panel B shows that students who express more positive views about AI’s effect on learning use AI somewhat more for academic work during the semester ($\hat{\beta} = 0.107$, $p = 0.096$). This pattern is consistent with the open-ended responses capturing a relevant signal.

Figure C2: Relationship Between AI Sentiment and AI Usage



Notes: This figure presents the relationship between AI sentiment and AI usage. Panel A restricts to students in the AI-allowed condition and shows the proportion who used AI during Session One (from usage logs). Panel B shows an indicator for using AI for academic purposes frequently or very frequently during the semester, for all respondents. Each point represents the mean outcome for respondents within sentiment score bins. Sentiment scores are compound scores computed using [Hutto and Gilbert \(2014\)](#)’s VADER algorithm applied to responses to the open-ended question about AI and learning. Positive values indicate positive sentiment and negative values indicate negative sentiment. The dashed lines show OLS best-fit lines estimated on the microdata.

C.2 Narratives as Causal Graphs

We characterize the content of the open-ended responses by coding each one as a causal graph, following [Andre et al. \(2026\)](#)’s analysis of how households and experts explain macroeconomic events. In their setting respondents explain why inflation rose, and the explanation is coded as a chain of cause-and-effect links running from a driving factor to inflation. We adapt their method in one substantive way: their links are unsigned, because

in the episode they study inflation only rose, whereas AI can raise or lower learning. Each link in our graphs therefore carries a sign. We represent each response as a signed causal graph from a source node, AI use, to a sink node, learning, passing through the mechanisms the student names. A link is positive if the student describes the mechanism as raising learning and negative if as lowering it.

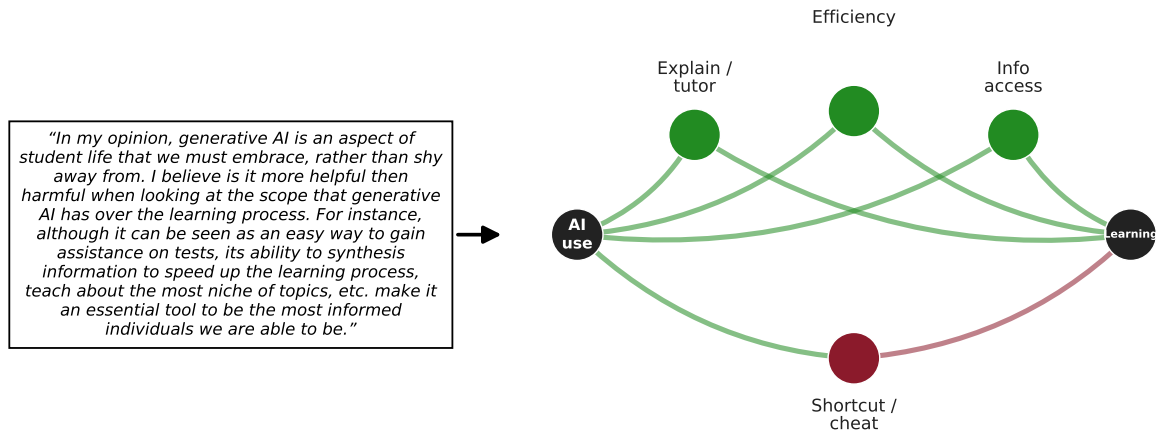
We proceed in two steps. First, we develop the codebook. Using the full set of student responses, we catalog the distinct mechanisms students raise, and consolidate near-synonyms into fifteen mechanisms. Second, we code every response against this fixed codebook with an LLM. We group the mechanisms by their effect on learning. Seven promote it: explanation and tutoring, efficiency, information access, brainstorming, personalized feedback, access and equity, and engagement. Eight harm it: shortcutting and cheating, cognitive offloading, skill atrophy, dependency, inaccuracy, reduced productive struggle, retention loss, and reduced human contact. A final node, *mode of use*, marks responses that make the effect explicitly conditional (“it depends on how you use it”). This node is a mediator between AI use and the channels rather than a channel itself: AI use feeds mode of use, which branches into channels that raise learning and channels that lower it, so conditionality is encoded by the node carrying both a promoting and a harming arm.

To illustrate how the coding works, Appendix Figure C3 shows the directed acyclic graphs of three specific students. The coder assigns each response an overall valence: the student’s bottom-line judgment of whether AI helps learning (Panel A), harms it (Panel B), or depends on how it is used (Panel C). The Panel A student concludes that AI helps learning yet flags the temptation to shortcut; the Panel B student concludes that AI harms learning yet credits AI with explaining difficult concepts. The pattern extends beyond these two students: 20 percent of net-positive narratives name at least one harming channel, and 27 percent of net-negative narratives name at least one promoting channel.

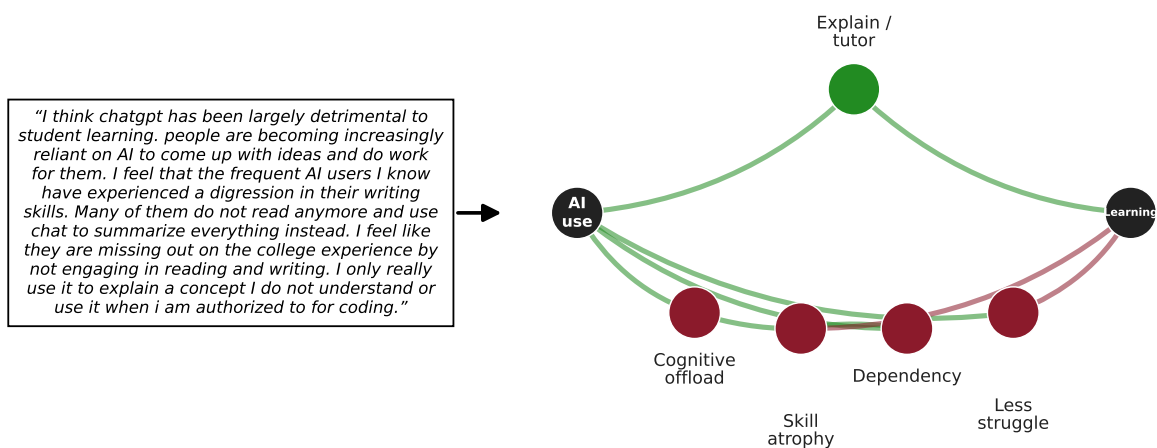
The taxonomy also maps onto the use classification in the main text. Explanation, feedback, and brainstorming are channels in which AI works *with* the student; shortcutting and offloading are channels in which AI works *for* the student. The coder accordingly classifies the manner of use each response describes: *augmentation* when the student describes working with AI (asking it to explain, give feedback, or check work), *automation* when the student describes AI doing the work in their place, both, or unspecified.

Figure C3: Three Example Student Narratives, Coded as Signed Causal Graphs

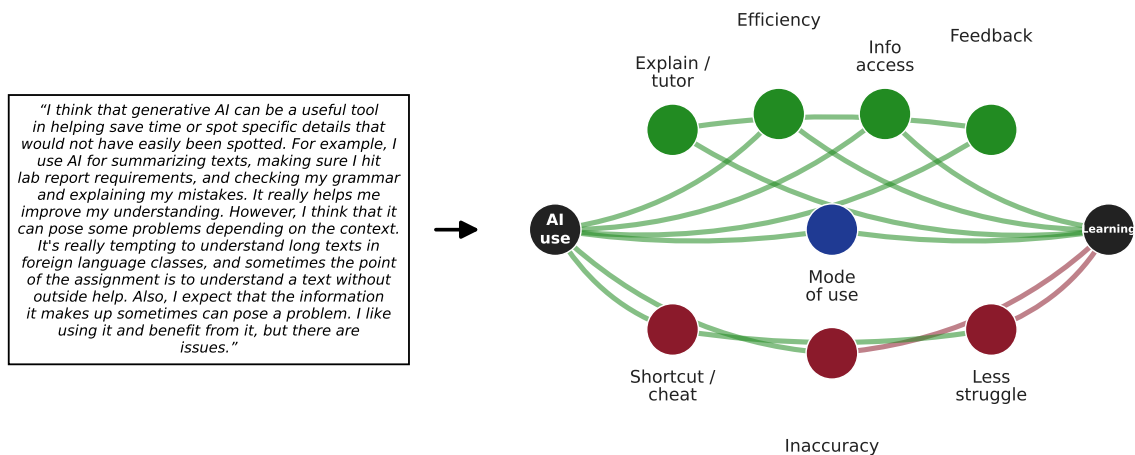
Panel A. A narrative in which AI helps learning



Panel B. A narrative in which AI harms learning



Panel C. A conditional narrative



Notes: Each panel pairs a student's verbatim open-ended response (left) with the response coded as a signed causal graph from AI use to learning (right), following Andre et al. (2026), so the mapping from text to graph can be assessed directly. Green links denote mechanisms the student describes as promoting learning, maroon links mechanisms that harm it, and navy the mode-of-use node.

We code each response with an LLM (Claude Opus 4.8). The majority of responses (98 percent) can be represented as a causal graph; the rest are non-substantive (e.g., “I don’t know”). As a validation, we code every response a second time with a model from a different developer (OpenAI’s GPT-5.5) and compute their inter-rater agreement statistic, the pooled probability that a code assigned by one rater is also assigned by the other. Agreement matches the trained human coders in [Andre et al. \(2026\)](#): 0.88 for the individual mechanisms (0.88 in their data), 0.97 and 0.96 for the coarser grouping into any promoting and any harming channel (0.94 and 0.93), and 0.73 for the signed links (0.77), with the mechanism and link statistics far above their random-coding benchmarks of 0.32 and 0.21, respectively. The two models agree on a response’s overall valence 94 percent of the time and on its augmentation-versus-automation framing 85 percent of the time.³²

C.3 The Narratives of College Students

The Average Narrative. The average narrative contains 3.1 mechanisms and 6.5 links, and 94 percent of coded responses mention more than one mechanism. Among students naming any mechanism, 69 percent name both a promoting and a harming channel, and 69 percent of coded responses frame AI’s effect as explicitly conditional. The most common single node is mode of use (54.7 percent), and the most common promoting and harming mechanisms are explanation and tutoring (40 percent) and shortcutting and cheating (42 percent).

Variation Across Students. We ask how narratives vary with student characteristics. Appendix Table [C1](#) regresses each of six narrative features—three *sophistication* measures (the number of mechanisms, the number of signed links, and an indicator for naming more than one mechanism) and the three *framing* measures (augmentation, automation, and conditional-on-use)—on the randomized treatment and the full set of student characteristics from Appendix Table [A1](#).

Among the AI-experience measures, only self-assessed proficiency predicts how students frame AI. Students with above-median proficiency are 17.1 percentage points more likely

³²Two checks confirm that the coding is not model-specific. Coding every response a third time with a different model from the primary coder’s developer (Anthropic’s Claude Sonnet 4.6) gives nearly identical agreement with the primary coder (0.86 for mechanisms, 0.66 for links, and 0.92 for valence). Agreement rises with model capability: within the OpenAI line, GPT-5.5 reaches mechanism and link agreement of 0.88 and 0.73, whereas two smaller models reach 0.76 and 0.52, and 0.70 and 0.47. All remain far above the random-coding benchmarks, but we use frontier models for both coders.

than less proficient students to frame AI as augmentation ($p = 0.019$) and 12.0 pp less likely to frame it as automation ($p = 0.062$). Early adopters—those who first used AI for academics before Fall 2024—do not differ from later adopters on any narrative feature, and randomized AI access shifts none of the six at the 5 percent level: the largest treatment difference, a 7.2 pp lower automation share among treated students, is imprecisely estimated ($p = 0.202$), and treated students are 6.8 pp less likely to name more than one mechanism ($p = 0.099$). Framing thus tracks how well students use AI, not how often they use it, how early they began, or whether they received a single session of access.

Demographics, academic background, and beliefs, by contrast, do predict narratives. Men name 0.38 fewer mechanisms and 0.71 fewer links than women (both marginally significant; $p = 0.059$ and $p = 0.087$), the same gender difference Andre et al. (2026) document in narratives about inflation. Students with above-median GPAs are 7.1 pp more likely to name more than one mechanism ($p = 0.133$). Framing varies instead with field of study and beliefs about AI. Natural-science majors are 29 pp more likely to condition AI’s effect on how it is used ($p = 0.003$) and 20.8 pp less likely to frame it as automation ($p = 0.002$). Students who expect large AI effects on their own test scores frame AI the same way: they are 12.6 pp less likely to frame it as automation ($p = 0.031$) and 14.2 pp more likely to condition its effect on use, though the latter is marginally significant ($p = 0.077$).

Table C1: Correlates of Students' Narratives

	Narrative sophistication			Use-mode framing		
	# mech. (1)	# links (2)	>1 mech. (3)	Aug- ment. (4)	Au- tomat. (5)	Condit. (6)
AI access (randomized)	0.079 (0.204)	0.239 (0.449)	-0.068* (0.041)	-0.006 (0.059)	-0.072 (0.056)	0.048 (0.075)
Panel A. Demographic characteristics						
Male	-0.378* (0.199)	-0.706* (0.410)	-0.038 (0.040)	-0.063 (0.056)	0.007 (0.058)	-0.032 (0.073)
White	0.214 (0.197)	0.360 (0.407)	0.002 (0.041)	0.089 (0.069)	-0.017 (0.062)	-0.114 (0.075)
International	0.040 (0.262)	0.055 (0.559)	0.010 (0.044)	0.075 (0.089)	-0.038 (0.079)	-0.085 (0.102)
Private high school	0.110 (0.196)	0.107 (0.441)	0.010 (0.041)	0.012 (0.066)	-0.022 (0.071)	0.014 (0.084)
Panel B. Academic background						
Natural Sciences	-0.286 (0.233)	0.059 (0.491)	-0.017 (0.062)	-0.092 (0.072)	-0.208*** (0.068)	0.287*** (0.094)
Social Sciences	0.016 (0.231)	0.205 (0.468)	0.012 (0.045)	0.062 (0.069)	-0.080 (0.073)	0.053 (0.091)
High GPA	0.471** (0.200)	0.676 (0.430)	0.071 (0.047)	-0.012 (0.069)	-0.009 (0.063)	0.014 (0.079)
Freshman	0.451* (0.244)	0.999** (0.504)	-0.005 (0.057)	-0.109 (0.074)	-0.059 (0.078)	0.145 (0.092)
Sophomore	0.263 (0.311)	0.646 (0.687)	0.034 (0.064)	0.036 (0.114)	-0.056 (0.084)	0.076 (0.114)
Junior	0.510** (0.255)	0.806 (0.537)	0.056 (0.049)	-0.109 (0.076)	0.019 (0.078)	0.031 (0.096)
Panel C. AI experience and beliefs						
Frequent AI user	0.281 (0.249)	0.625 (0.497)	-0.031 (0.041)	0.007 (0.073)	-0.079 (0.059)	0.083 (0.078)
Early AI adopter	-0.254 (0.214)	-0.255 (0.434)	-0.035 (0.041)	0.028 (0.059)	0.002 (0.055)	0.009 (0.068)
High AI proficiency	0.143 (0.264)	0.320 (0.520)	0.036 (0.049)	0.171** (0.072)	-0.120* (0.064)	-0.027 (0.084)
High perceived AI effect	0.273 (0.214)	0.680 (0.476)	-0.020 (0.048)	-0.043 (0.066)	-0.126** (0.058)	0.142* (0.080)
N (Students)	201	201	201	201	201	201

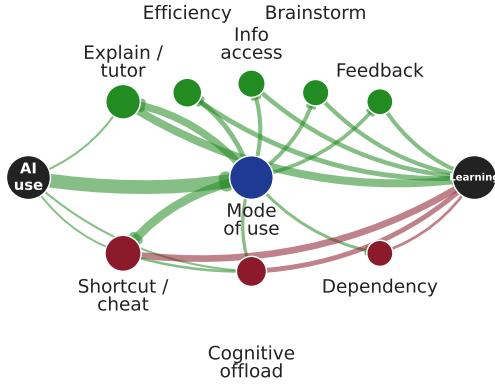
Notes: Each column reports an OLS regression of the indicated narrative feature on all rows jointly, estimated on the codable open-ended responses. Outcomes come from each student's coded causal graph: the number of mechanisms, the number of signed links, an indicator for naming more than one mechanism, and indicators for the augmentation, automation, and conditional-on-use framings. All characteristics are indicators, defined as in Appendix Table A1 with the medians recomputed in this sample: frequent AI user is use during the semester at or above this sample's median (frequently or more often), and high perceived AI effect is a perceived gain of at least one question; missing values are set to zero, with missing indicators included. Heteroskedasticity-robust standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Narrative Clusters. We next ask whether students’ narratives come in recurring types or “clusters.” Following [Andre et al. \(2026\)](#), we measure the distance between two narratives as the fraction of their combined signed links that the two do not have in common (the Jaccard distance), group narratives with agglomerative hierarchical clustering with average linkage, and choose the number of clusters to maximize the average silhouette, a standard measure of how well each narrative fits its own cluster relative to the nearest alternative.

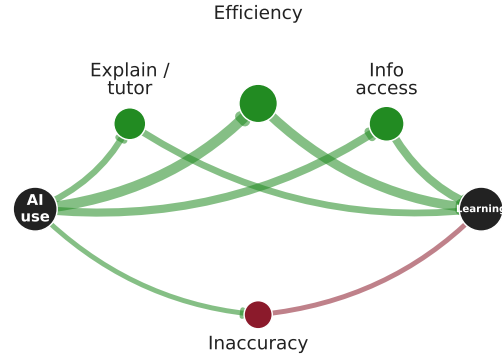
Four narrative types cover 94.0 percent of coded responses, while the rest scatter across small clusters (Appendix Figure [C4](#)). The modal type (47 percent) is the double-edged narrative, in which mode of use feeds both an explanation channel and a shortcutting channel. An augmentation type (28 percent) chains efficiency, information access, and explanation to higher learning, though many of these narratives also flag accuracy concerns, and an automation type (10 percent) chains shortcutting and offloading to lower learning. The remaining type is smaller and one-sided: a deskilling narrative (8.0 percent), in which skill atrophy and offloading erode critical thinking.

Figure C4: Narrative Clusters

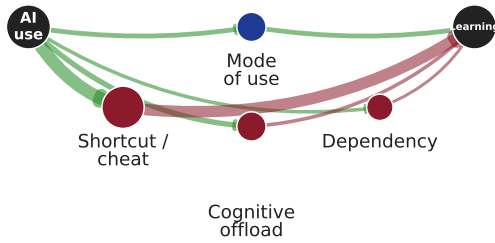
Panel A. Double-edged (47 percent)



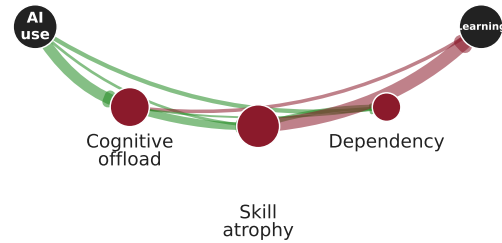
Panel B. Augmentation (28 percent)



Panel C. Automation (10 percent)



Panel D. Deskilling (8.0 percent)



Notes: This figure shows the average causal graph within each narrative cluster; panel headings report each cluster's share of coded narratives. We form the clusters with agglomerative hierarchical clustering, using average linkage on the Jaccard distance between signed-link sets and choosing the number of clusters to maximize the average silhouette. We show the four clusters with at least eight responses and, within each panel, omit mechanisms below 20 percent and links below 6 percent.

D Conceptual Framework

This appendix presents the complete model of Section 5.2. Students do research and write to produce an essay and acquire knowledge. We distinguish AI that helps students understand the material from AI that writes the essay in their place, and we show how the two capabilities affect effort, knowledge, and the incentive to delegate. We first present a baseline model without AI and then introduce the two AI capabilities and derive how each changes effort and knowledge relative to the baseline.

D.1 Setup

A student chooses cognitive effort in research, $r \geq 0$ (e.g., reading and searching), and writing, $w \geq 0$ (e.g., drafting, editing, note-taking, and organizing). Both activities produce knowledge (denoted by K), that is, lead to learning. Writing also produces the essay the student submits, and the quality of that essay reflects the knowledge the student has acquired. Without AI, knowledge and Session One essay quality are

$$\begin{aligned} K &= \theta_r r + \theta_w w, \\ Q_1 &= w + \phi K, \end{aligned} \tag{D1}$$

where $\theta_r, \theta_w > 0$ are the returns to effort in each activity and $\phi \geq 0$ measures how much of the student's knowledge shows in the essay. We focus on the case $\phi > 0$. Writing raises essay quality directly, and both activities raise it through the knowledge they produce, so the reward for essay quality enters both effort decisions.

The student values knowledge at $\omega_K > 0$ per unit and essay quality at $\omega_Q > 0$ per unit. These valuations combine intrinsic interest with external rewards. In our experiment, the lottery tickets paid for correct test answers and for essay quality contribute to them. In ordinary coursework, grades and the future returns to skill are external incentives, while interest in the topic are intrinsic incentives. Effort x in each activity costs $x^2/2\mu$, with $\mu > 0$. The cost is convex, so each additional unit of effort costs more than the last, and a higher μ lowers the marginal cost of effort.

D.2 The Student's Problem Without AI

We begin by studying the student problem without access to AI. The student solves

$$\max_{r,w \geq 0} \underbrace{\omega_K(\theta_r r + \theta_w w)}_{\text{Value of knowledge}} + \underbrace{\omega_Q[w + \phi(\theta_r r + \theta_w w)]}_{\text{Reward for essay quality}} - \underbrace{\frac{r^2 + w^2}{2\mu}}_{\text{Cost of effort}}. \quad (\text{D2})$$

The objective is strictly concave in (r, w) , so the first-order conditions give the unique solution

$$r_0 = \mu(\omega_K + \phi\omega_Q)\theta_r \quad \text{and} \quad w_0 = \mu[(\omega_K + \phi\omega_Q)\theta_w + \omega_Q]. \quad (\text{D3})$$

Substituting these choices into equation (D1) gives an expression for learning without AI:

$$K_0 = \underbrace{\mu\omega_K(\theta_r^2 + \theta_w^2)}_{\text{Value of knowledge}} + \underbrace{\mu\omega_Q[\theta_w + \phi(\theta_r^2 + \theta_w^2)]}_{\text{Reward for essay quality}}. \quad (\text{D4})$$

Equation (D4) makes explicit the two sources of learning. The first term is the knowledge the student acquires through research and writing because they value knowledge. The second term is the knowledge that the reward for essay quality generates: producing the essay requires writing, and writing produces knowledge; and because the essay shows what the student knows, the reward also induces research and writing that add to knowledge.

D.3 Introducing AI

We extend the baseline model by giving the student access to AI, which enters through two non-mutually exclusive capabilities. First, used for augmentation, AI raises the return to research effort by $\tau \geq 0$. The parameter τ is a reduced form for the many channels through which AI makes each unit of research effort more productive. AI can act as a tutor that explains confusing material, as a research assistant that finds relevant and more rigorous sources faster, as a summarizer that condenses reading to what matters most, as an illustrator that produces visualizations on demand, and as an examiner that quizzes the student on what they have read. For concreteness, we refer to this capability as tutoring. Second, the student may automate the process of writing the essay by delegating this task

to AI. Let $m \in \{0, 1\}$ indicate delegation.³³ Let $\bar{Q} \geq 0$ denote the quality of the AI-written essay.³⁴

With AI, knowledge and Session One essay quality are

$$\begin{aligned} K &= (\theta_r + \tau) r + \theta_w w, \\ Q_1 &= (1 - m) (w + \phi K) + m \bar{Q}. \end{aligned}$$

Delegation saves the student the time that writing the essay would have taken. Let $v \geq 0$ denote the student's valuation of this time. It is distinct from the cognitive effort that delegation saves, whose cost the cost-of-effort term already captures. Given AI access and the tutoring gain, the student solves

$$\max_{r, w \geq 0, m \in \{0, 1\}} \underbrace{\omega_K K}_{\text{Value of knowledge}} + \underbrace{\omega_Q [(1 - m) (w + \phi K) + m \bar{Q}]}_{\text{Reward for essay quality}} + \underbrace{\frac{mv}{2\mu}}_{\text{Value of recovered time}} - \underbrace{\frac{r^2 + w^2}{2\mu}}_{\text{Cost of effort}}, \quad (\text{D5})$$

where $K = (\theta_r + \tau) r + \theta_w w$.

For a given delegation choice, the first-order conditions yield:

$$r^*(\tau, m) = \mu \tilde{\omega}_m (\theta_r + \tau) \quad \text{and} \quad w^*(\tau, m) = \mu [\tilde{\omega}_m \theta_w + \omega_Q (1 - m)], \quad (\text{D6})$$

where $\tilde{\omega}_m \equiv \omega_K + \phi \omega_Q (1 - m)$ is the value of a unit of knowledge given the delegation choice: $\omega_K + \phi \omega_Q$ for a student who writes the essay, as in the baseline, and ω_K for a student who delegates, because the student's knowledge no longer raises the quality of the submitted essay.

Equation (D6) nests the baseline effort levels of equation (D3) and shows how each

³³We model delegation as a binary decision for simplicity. In practice there are gradations: students can have AI write parts of the essay, or use it for narrower purposes such as proofreading or paraphrasing. The model also abstracts from the combination of an AI draft with the student's own revisions. It treats delegation as the extreme case in which the submitted essay is AI-written, whereas the empirical automation category of Section 5.3 includes partial delegation. Allowing m to vary continuously between 0 and 1 would not change the solution: the student's utility, maximized over effort, is convex in m , so the student would still choose one of the two corners. Interior delegation would require, for example, diminishing returns to delegating a larger share of the essay.

³⁴Both τ and $\bar{Q}$ depend on the capabilities of the model, which may explain why students made little use of language models before ChatGPT. They may also depend on the student's academic preparation, knowledge of the topic, and prompting skill, which would make them vary across students and add a second source of selection into use types. For simplicity, we take them as given parameters when solving for effort and delegation.

capability changes effort, holding the other fixed. Tutoring raises research effort by $\mu\tilde{\omega}_m\tau$, because the return to research effort is higher, and leaves writing effort unchanged. Delegation removes the reward for essay quality from the research and writing decisions. Thus, both types of effort decrease. In particular, writing effort falls by $\mu\omega_Q(1 + \phi\theta_w)$ because the student's writing no longer determines the submitted essay and research effort falls by $\mu\phi\omega_Q(\theta_r + \tau)$, because the knowledge research produces no longer improves the essay. In proportional terms, research effort falls by the fraction $\phi\omega_Q/(\omega_K + \phi\omega_Q)$ and writing effort by the larger fraction $\omega_Q(1 + \phi\theta_w)/[(\omega_K + \phi\omega_Q)\theta_w + \omega_Q]$, so delegation lowers the share of effort devoted to writing.³⁵ Notably, writing effort remains positive even with delegation ($m = 1$) because students still get the learning benefit of writing, for example when they take notes or organize their ideas.

Substituting the effort choices into the knowledge equation yields the level of learning with AI:

$$K^*(\tau, m) = \mu\tilde{\omega}_m[(\theta_r + \tau)^2 + \theta_w^2] + \mu\omega_Q\theta_w(1 - m). \quad (\text{D7})$$

The effect of AI access on knowledge, the model's counterpart of the treatment effect on test scores, is

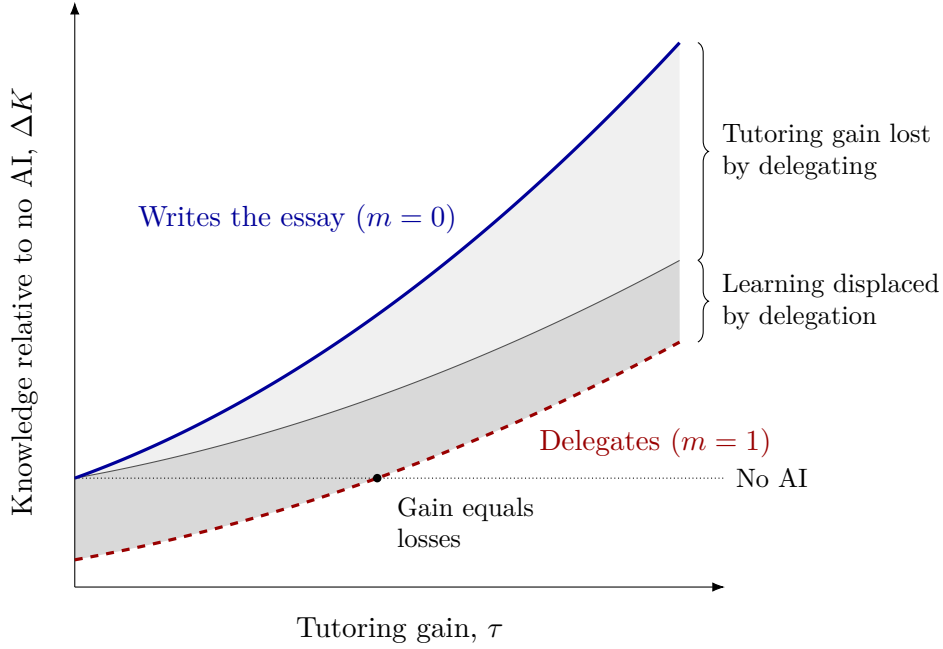
$$\Delta K = \underbrace{\mu(\omega_K + \phi\omega_Q)[(\theta_r + \tau)^2 - \theta_r^2]}_{\text{Gain from tutoring}} - \underbrace{\mu\omega_Q[\theta_w + \phi(\theta_r^2 + \theta_w^2)]m}_{\text{Learning displaced by delegation}} - \underbrace{\mu\phi\omega_Q[(\theta_r + \tau)^2 - \theta_r^2]m}_{\text{Tutoring gain lost by delegating}}. \quad (\text{D8})$$

Equation (D8) shows that the impact of AI on learning is ex-ante ambiguous and it depends on the relative size of three forces. The first one is the gain from tutoring for a student who writes the essay. It combines the increase in the knowledge that each unit of research effort produces with the increase in research effort that the higher return induces. The second force is the learning displaced by delegation: the knowledge that the reward for essay quality generates when the student writes the essay, through writing directly and through the research and writing that the essay value of knowledge motivates. The third force is the knowledge that tutoring produces no longer improves the submitted essay, so research effort responds less to the higher return. The effect of AI access on knowledge

³⁵The difference between the two fractions is $\omega_K\omega_Q/\{(\omega_K + \phi\omega_Q)[(\omega_K + \phi\omega_Q)\theta_w + \omega_Q]\} > 0$. Delegation removes the direct reward for writing, which is a larger part of the motive for writing than the essay value of knowledge is of the motive for research.

therefore depends on whether the student delegates and, if so, on whether the gain from tutoring exceeds the two losses from delegation. Figure D1 plots the change in knowledge against the tutoring gain for a student who writes the essay and for the same student when delegating, and shows the two losses from delegation, whose sum widens with the tutoring gain. We now turn to the choice of whether to delegate.

Figure D1: The Effect of AI Access on Knowledge With and Without Delegation



Notes: The figure plots the change in knowledge relative to no AI, $K^*(\tau, m) - K_0$, against the tutoring gain, for a student who writes the essay ($m = 0$) and for the same student when delegating ($m = 1$), with research and writing effort chosen optimally in each case. The curves are drawn for $\mu = \omega_K = \omega_Q = \theta_r = \theta_w = \phi = 1$. The values are illustrative, so the axes show no scale. The two curves are two counterfactual choices for the same student, holding everything else fixed, not average outcomes for two groups of students. The shaded area between the curves is the loss from delegation, the sum of the last two terms of equation (D8). The darker band is the learning displaced by delegation, which does not depend on the tutoring gain. The lighter band is the tutoring gain lost by delegating, which grows with it, so the delegation curve rises more slowly and the gap between the curves widens. The braces show the two parts at the largest tutoring gain drawn. At the marked point on the no-AI line, the gain from tutoring equals the two losses from delegation.

D.4 The Decision to Delegate Work to the AI

The student compares the utility obtained with and without delegation, each evaluated at the effort levels of equation (D6). Substituting those effort levels into the objective

function gives the utility under each choice,

$$V_m = \underbrace{\frac{\mu}{2} \tilde{\omega}_m^2 (\theta_r + \tau)^2}_{\text{Net benefit of research effort}} + \underbrace{\frac{\mu}{2} [\tilde{\omega}_m \theta_w + \omega_Q (1 - m)]^2}_{\text{Net benefit of writing effort}} + \underbrace{m(\omega_Q \bar{Q} + v)}_{\text{AI-written essay and recovered time}}.$$

The student delegates when

$$\begin{aligned} V_1 - V_0 &= \omega_Q \bar{Q} + v - \frac{\mu}{2} \left[(\tilde{\omega}_0^2 - \omega_K^2) (\theta_r + \tau)^2 + (\tilde{\omega}_0 \theta_w + \omega_Q)^2 - \omega_K^2 \theta_w^2 \right] \geq 0 \\ &\iff \underbrace{\omega_Q \bar{Q} + v}_{\substack{\text{Benefit of delegation} \\ \text{Reward for the AI-written essay} + \text{Value of recovered time}}} \\ &\geq \underbrace{\mu \omega_K \omega_Q [\theta_w + \phi((\theta_r + \tau)^2 + \theta_w^2)]}_{\substack{\text{Cost of delegation} \\ \text{Value of the forgone knowledge}}} + \underbrace{\frac{\mu \omega_Q^2}{2} [(1 + \phi \theta_w)^2 + \phi^2 (\theta_r + \tau)^2]}_{\substack{\text{Own-essay reward} \\ \text{net of extra effort}}}. \quad (\text{D9}) \end{aligned}$$

Equation (D9) clarifies the sources that determine the decision to delegate. On the benefit side, a higher-quality AI-written essay ($\bar{Q}$) and a higher value of recovered time (v) make delegation more attractive. The cost of delegation is the value of the knowledge that a student with AI gives up by delegating, $\omega_K[K^*(\tau, 0) - K^*(\tau, 1)]$, plus the reward the student's own essay would have earned, net of the extra effort it would have taken. A higher value of knowledge makes delegation less attractive, since $\partial(V_1 - V_0)/\partial\omega_K = -\mu\omega_Q[\theta_w + \phi((\theta_r + \tau)^2 + \theta_w^2)] < 0$: students who value learning more are less willing to give up the knowledge that the essay reward induces. Less costly effort (a higher μ), a higher return to writing effort (θ_w), a larger role of knowledge in the essay (ϕ), and a larger tutoring gain (τ) also make delegation less attractive, the last because a student who writes the essay captures the essay value of the knowledge that tutoring produces: $\partial(V_1 - V_0)/\partial\tau = -\mu\phi\omega_Q(2\omega_K + \phi\omega_Q)(\theta_r + \tau) < 0$.

Whether delegation hurts or helps essay quality is ambiguous. A student who delegates submits an essay of quality $\bar{Q}$ in place of the essay they would have written, whose quality is $w^*(\tau, 0) + \phi K^*(\tau, 0)$ by equations (D6) and (D7). Solving equation (D9) at equality for $\bar{Q}$ gives the quality of the AI-written essay at which the student is indifferent between delegating and writing. This threshold is below the quality of the student's own essay: the AI-written essay can be worse than the student's own by $(\mu\omega_Q/2)[(1 + \phi\theta_w)^2 + \phi^2(\theta_r + \tau)^2] +$

v/ω_Q and the student still delegates. The first term is the effort that delegation saves, net of the knowledge that effort would have produced, and the second is the value of recovered time, both in units of essay quality. A student whose AI-written essay falls in that range delegates to a worse essay, trading essay quality for the effort and time that delegation saves. A student whose AI-written essay is better than their own delegates to a better one. The model therefore allows a gain in essay quality without a gain in knowledge, and it does not require an essay-quality gain whenever the student delegates. For a student who writes the essay, tutoring raises its quality through the knowledge it produces, by ϕ times the gain from tutoring in equation (D8), so a gain in essay quality can also come with a gain in knowledge.

Equation (D9) also makes clear that delegation is endogenous to the incentives the student faces. The value of knowledge, ω_K , combines the student’s intrinsic motivation to learn with the external rewards for learning: the lottery tickets for correct test answers in our experiment, or grades and the returns to skill in a course. A student with a higher ω_K is less willing to delegate and also supplies more effort in both activities, so students who delegate and students who do not already differ in motivation and incentives before AI enters. The reward for essay quality and the value of recovered time, which outside the laboratory come from grades and from competing demands on the student’s time, shift the decision as well. A higher reward for essay quality discourages delegation: it raises the quality the AI-written essay must reach, since the student at the threshold delegates to an essay worse than their own.

D.5 Relationship Between the Model and the Experiment

The experiment measures the model’s outputs and fixes several of its parameters. Both tests measure K . The Session One essay measures Q_1 . The Session Two essay is written without AI, so its quality reflects the student’s writing effort during the assessment and, through ϕ , the knowledge the student retains. The time-use survey reports minutes, which relate to effort loosely, as discussed below, and the learning-experience measures relate to the cost of effort. We take each result in turn and end with three implications.

Test scores. AI access raised test scores in both sessions (Section 4.2). In the model, treated students’ scores measure $K^*(\tau, m)$ and control students’ scores measure K_0 . Random assignment equalizes the two groups’ preferences and returns to effort in expectation, so the Session One effect is the average of ΔK in equation (D8) across treated students,

each with their own tutoring gain and delegation choice, including students who did not use AI, for whom $\tau = 0$ and $m = 0$, so $\Delta K = 0$. Through the lens of the model, a positive average effect means that, averaged across treated students, the tutoring gain exceeded the two losses from delegation.

Essays. The Session One essay measures Q_1 . For a student who writes the essay, tutoring raises its quality through the knowledge it produces. For a student who delegates (and the rise in the share of text flagged as AI-generated shows that many treated students did) the submitted essay can be better or worse than the student’s own. The model therefore does not sign the average Session One essay effect. The unaided Session Two essay reflects retained knowledge and writing effort during the assessment. The direct contribution of AI-written text to the first essay, $m\bar{Q}$, does not enter it. Knowledge gains can appear in both the follow-up test and the unaided essay, whereas the features of AI-written text cannot.

Time allocation. Treated students shifted time from writing toward research (Section 5.1). In the model, both AI capabilities lower the share of effort devoted to writing: a higher τ raises research effort without changing writing effort, and delegation lowers effort in both activities, writing by proportionally more.³⁶

The learning experience. Treated students enjoyed the task more (Section 5.1). One interpretation is that AI changes the cost of effort, a channel that the decomposition in equation (D8) holds fixed.³⁷ Holding constant the delegation choice, equation (D7) implies

$$\frac{\partial K^*}{\partial \mu} = \tilde{\omega}_m [(\theta_r + \tau)^2 + \theta_w^2] + \omega_Q \theta_w (1 - m) > 0.$$

Less costly effort induces more research and writing effort and raises knowledge.

Additional implications. The model delivers three implications that the experiment can examine descriptively by use type.

³⁶The time-use estimates measure minutes, whereas r and w in equation (D6) measure cognitive effort, which also depends on how intensively a student works during those minutes. The two move in the same direction here, but not necessarily one for one.

³⁷We note that other interpretations are possible: students may enjoy the task more because they are learning more or because less writing is required.

First, a delegating student’s Session One essay advantage comes from $\bar{Q}$ and is absent from unaided work, whereas knowledge, whether from tutoring or from the student’s own research and writing, persists into the unaided assessments, including the unaided essay, which expresses knowledge through ϕ . The essay advantage of automation users should therefore be concentrated in Session One, any test-score advantage should appear in both sessions, and an essay advantage of augmentation users, which can arise only through knowledge, should appear in both sessions.

Second, the two capabilities move effort in different directions. Relative to a student without AI, tutoring raises research effort and leaves writing effort unchanged, whereas delegation lowers writing effort and frees time, which in the laboratory students could use only by stopping early and browsing. Augmentation users should therefore do more research than control students, and automation users should write less than control students and stop earlier. The model does not sign automation users’ research effort relative to control students, since tutoring raises it and delegation lowers it. Holding the tutoring gain fixed, however, automation users differ from augmentation users only by the delegation effect in equation (D6), which lowers effort in both activities. Section 5.3 reports that automation users spend less time on research and less time overall than augmentation users.

Third, the choice to delegate reveals the student’s incentives and type. By the delegation condition, students who delegate place a lower value on knowledge, a higher value on recovered time, expect a better AI-written essay, or gain less from tutoring, so pre-determined characteristics that correlate with these parameters should predict use type. Appendix Table A6 shows that they do: automation users enter with higher baseline test scores than augmentation users, which the model can rationalize through a higher $\bar{Q}$ for better-prepared students.